Near-Earth Asteroid Orbit Determination & Yarkovsky Effect Detection

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Near-Earth Asteroid Orbit Determination & Yarkovsky Effect Detection

by

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Cover: High-resolution mosaic of Dimorphos, which forms a binary asteroid system with 65803 Didymos, captured by the DRACO instrument on NASA's DART mission shortly before impact with the Dimorphos. The north pole of Dimorphos is on the left side of the image. Credit: NASA/Johns Hopkins APL (2023).
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Preface

I would first of all like to thank my supervisor, Dominic Dirkx, for his patience, expertise, and guidance throughout this project. His advice and answers to my questions during our meetings were always clear, insightful, and thoughtful, and I have learned a lot from them. His commitment to his work whilst balancing research, teaching, and developing and maintaining Tudat is exemplary, and is something I admire greatly. I am also glad to have had the opportunity to work on a topic that I find genuinely interesting, which definitely made the long journey of the thesis that little bit easier.

I want to thank my parents, brother, and sister for their continuous support at all times in life and for nagging me just the right amount about my thesis progress during our weekly family calls. I would not have been able to do it without them. I'm also grateful to my friends for their support throughout this time and for providing the fun and social counterpoint to the sometimes heavy workload of a thesis. In particular, I would like to thank Matthias for welcoming me to work in his office at the Statistics department of EEMCS. Working alongside him and his colleagues made the solo work of a thesis feel a lot less solitary than it may have otherwise been. Discussions with them about certain statistical aspects of this thesis were invaluable to my understanding and analysis. I am looking forward to finally showing my friends and family what I have been working on all this time in detail during my graduation presentation.

Although this thesis has been a longer journey than I initially expected, it has also been a valuable learning experience. I am proud that I am finally reaching the finish line and I'm grateful to the people who supported me along the way. I hope the work presented in this thesis reflects not only what I have learned about the subject, but also the perseverance that completing it has required.

Rónán McIntyre
Co. Kildare, 30th June 2026

Abstract

The Yarkovsky effect is a non-gravitational acceleration caused by the anisotropic thermal emission of rotating asteroids. The primary manifestation of the Yarkovsky effect is a gradual drift in the semi-major axis of the asteroid's orbit. The Yarkovsky effect is equally relevant to understanding asteroid evolution as gravitational forces and collision events, and it is key in describing the mechanisms that feed the supply of near-Earth asteroids (NEAs) from the main asteroid belt. Accurate measurements of this effect are important for planetary defence and for assessing the risk of potential Earth impacts.

Much of the research on small-body orbit determination and the Yarkovsky effect uses closed-source, proprietary software, such as JPL's Monte and ESA's Aegis. Research publications which make use of such software typically omit many precise details of how results were produced. This limits scientific transparency, impedes independent verification of results, and makes it difficult to assess discrepancies in results from different sources. The selection of optimal settings can be a time-consuming process. Without the free availability of the settings that have been used to produce successful results, researchers wishing to build upon these results are forced to re-conduct the selection of settings from scratch.

The problems identified can be addressed by using free, open-source software for astrodynamics and orbit determination, and the transparent documentation of the process of selecting settings. This thesis aims to reproduce results from literature relating to the Yarkovsky effect using Tudat, the free open-source astrodynamics toolbox developed at Delft University of Technology. This thesis also aims to be thorough and transparent about documenting the methods, settings, and inputs used.

A detailed selection process was carried out and documented regarding the selection of numerical integration settings, a robust dynamic model, and observation outlier treatment. Using these settings, orbit estimations were performed using a weighted least-squares method, including the estimation of a parameter (A_2) describing a non-gravitational acceleration in the transverse direction. The resulting orbits are assessed on their signal-to-noise ratio, the quality of their fit to the data, and are checked for convergence.

Successful Yarkovsky detections were found for 281 asteroids (81%) out of 347 asteroids for which estimations were performed. The remaining estimations had low signal-to-noise ratios of the A_2 parameter and thus were not considered valid detections. The determined A_2 values generally provide a good match to values found in literature, being compatible with values determined by ESA's Near-Earth Object Coordination Centre (NEOCC) in 93.2% of cases, and being compatible with values from JPL's Small-Body Database (SBDB) in 97.7% of cases. All estimations provide good fits to the observation data, in line with existing high-fidelity research that uses similar methods.

Contents

Nomenclature

Abbreviations

Abbreviation	Definition
1b1	One-by-one
AB	Additional bodies
AB	Adams-Bashforth (numerical integration method)
ABM	Adams-Bashforth-Moulton (numerical integration method)
ANR	Absolute normalised residual
arcmin	Arcminute or $'$ (one-sixtieth of one degree)
arcsec	Arcsecond or $''$ (one-sixtieth of one arcminute, or $\frac{1}{3600}$ of one degree)
AU	Astronomical unit (approximate average Earth-Sun distance)
BL	Baseline accelerations
BS	Bulirsch-Stoer (numerical integration method)
CCD	Charge-coupled device
CP	Close pass(er)(s)
DEC	Declination
DP	Dormand-Prince (numerical integration method)
EIH	Einstein-Infeld-Hoffman
ESA	European Space Agency
FE	Formal error
GTE	Global truncation error
IAU	International Astronomical Union
J2000	January 1 2000, 12:00 TT
JPL	Jet Propulsion Laboratory (NASA)
LB	Large bodies (Sun, eight planets, Moon)
LMBA	Large main-belt asteroid
LTE	Local truncation error
MANR	Mean of the absolute normalised residuals
MBA	Main-belt asteroid
MOID	Minimum orbit intersection distance
MONTE	Mission analysis, Operations, and Navigation Toolkit Environment (JPL tool)
MPC	Minor Planet Center
NASA	National Aeronautics and Space Administration (United States)
NEA	Near-Earth asteroid
NEO	Near-Earth object
NEOCC	Near-Earth Object Coordination Centre (ESA)
NR	Normalised residual
O-C	Observation minus calculated (i.e. residual)
PDS	Planetary Data System (NASA)
PHO	Potentially hazardous object
PHA	Potentially hazardous asteroid
RA	Right ascension
RK	Runge-Kutta (type of numerical integration method)
RKF	Runge-Kutta-Fehlberg (type of numerical integration method)
RMS	Root mean square
RMSNR	RMS of the normalised residuals
SBDB	Small-Body Database (JPL)
SiMDA	Size, Mass and Density of Asteroids (catalogue)
SNR	Signal-to-noise ratio
SRP	Solar radiation pressure
TE	True error

Abbreviation	Definition
TT	Terrestrial Time
UTC	Coordinated Universal Time
WLS	Weighted least-squares
WSL	Windows subsystem for Linux
YORP	Yarkovsky-O'Keefe-Radzievskii-Paddack

Symbols

Symbol	Definition	Unit*
A	(Cross-sectional) area	$\mathrm{m^2}$
A_2	Yarkovsky parameter	$\mathrm{m\,s^{-2}}$
$\mathbf{a}$, $\ddot{\mathbf{r}}$	Acceleration (vector)	$\mathrm{m\,s^{-2}}$
a	Semi-major axis	m
C	Heat capacity	$\mathrm{J\,kg^{-1}\,K^{-1}}$
C_R	Radiation pressure coefficient	-
c	Speed of light	$\mathrm{m\,s^{-1}}$
D	Diameter	m
e	Eccentricity	-
$\mathbf{F}$	Force (vector)	N
F	Magnitude of force vector	N
f	Function of	-
G	Gravitational constant	$\mathrm{m^3\,kg^{-1}\,s^{-2}}$
$\mathbf{H}$	Jacobian matrix	-
H	Specific angular momentum	$\mathrm{m^2\,s^{-1}}$
H	Absolute magnitude (brightness)	-
$\mathbf{h}$	Modelled observation vector	-
i	Inclination	° or rad
J	Loss function	-
k_1, k_2	Relative errors between estimations	-
$L_\odot$	Solar luminosity	W
M, m	Mass	kg
m	Spherical harmonic order	-
n	Spherical harmonic degree	-
n	Mean motion	$\mathrm{rad\,s^{-1}}$
n	Number of bodies in a system	-
n	Number of time steps	-
N	Order of integrator	-
P	Rotation period	s
p	Momentum	$\mathrm{kg\,m\,s^{-1}}$
$\mathbf{P}$	Covariance matrix	situation dependent
$\mathbf{p}$	Vector of free parameters of force model	situation dependent
$\mathbf{q}$	Vector of free parameters of measurement model	situation dependent
R	Mean (equatorial) radius of body	m
$\mathbf{r}$	Position (vector)	m
$\hat{\mathbf{r}}$	Unit position vector	-
r	Magnitude of position vector / distance between bodies	m
S_r, S_v	Measures of total formal error in position, velocity	m, $\mathrm{m\,s^{-1}}$
t	Time	s
U	Gravitational potential	$\mathrm{m^2\,s^{-2}}$
$\mathbf{v}$, $\dot{\mathbf{r}}$	Velocity (vector)	$\mathrm{m\,s^{-1}}$
$\hat{\mathbf{v}}$	Unit velocity vector	-
v, $\dot{r}$	Magnitude of velocity vector	$\mathrm{m\,s^{-1}}$
$\mathbf{W}$	Weight matrix	$\mathrm{rad^{-2}}$
W	Radiation flux	$\mathrm{W\,m^{-2}}$
$\mathbf{x}$	State vector	situation dependent
x, y, z	Cartesian coordinates/axes/directions	situation dependent

Symbol	Definition	Unit*
$\mathbf{z}$	Observation data vector	situation dependent
α	Right ascension	° or rad or hour
α	Absorption coefficient	-
Γ	Thermal inertia	$\mathrm{J\,m^{-2}\,K^{-1}\,s^{-1/2}}$
γ	Obliquity	° or rad
γ	Lorentz factor	-
Δ	Discrete step or change	situation dependent
δ	Declination	° or rad
ε	Error	situation dependent
ε	Emissivity	-
Θ	Occultation function	-
θ	True anomaly	° or rad
Λ	Body-centric longitude	° or rad
μ	Gravitational parameter, $\mu = GM$	$\mathrm{m^3\,s^{-2}}$
μ	Proper motion	mas/year
ξ	Yarkovsky efficiency factor	-
$\boldsymbol{\rho}$	Residual vector	situation dependent
ρ	Density	$\mathrm{kg\,m^{-3}}$
σ	Standard deviation	situation dependent
$\boldsymbol{\Phi}$	State transition matrix	-
Φ	Body-centric latitude	° or rad
φ	Phase lag (Yarkovsky)	° or rad
Ω	Right ascension of the ascending node	° or rad
ω	Argument of periapsis	° or rad
Υ	Vernal equinox	
$\odot$	Sun / solar	

**Note that units of length other than metres may also be used, such as kilometres and AU.*

1 Introduction

1.1. Background & Motivation

Gaining a better understanding of asteroids, their physical properties, and their orbital evolution is scientifically important for three primary reasons. The first is that this knowledge can lead to a better understanding of Solar System evolution. According to Bottke et al. (2006), non-gravitational phenomena such as the Yarkovsky effect are as relevant to understanding asteroid evolution as gravitational forces and collision events, and this effect in particular is key in describing the mechanisms that feed the supply of near-Earth asteroids (NEAs) from the main asteroid belt (Granvik et al. 2017), as well as the dispersion of asteroid families (Bottke et al. 2002). A clearer comprehension of the dynamical properties of specific (groups of) asteroids can allow us to derive information about their origins, and can contribute improvements to the models that study long-term orbital evolution of the Solar System (Morbidelli et al. 2015).

The second primary reason is related to Solar System exploration. Certain physical properties of asteroids can be derived from measurements of the Yarkovsky effect, such as bulk density (e.g. Chesley et al. 2014), internal structure (e.g. Rozitis et al. 2014), spin direction (e.g. Farnocchia and Chesley 2014), and surface characteristics and thermal conductivity (e.g. Fenucci et al. 2021; Fenucci et al. 2023). Such information is important to know when it comes to making informed decisions about selecting objects of interest and designing space missions such as lander and sample return missions (Murdoch et al. 2021) and impactor-deflection missions (Bruck Syal et al. 2016; Cheng et al. 2023), hopefully leading to more useful scientific outcomes. The interpretation of the results of such missions can also be aided by this information (Murdoch et al. 2021). A good understanding of an asteroid's orbit and physical properties is also key when designing mission trajectories whose destinations are asteroids (Ferrari and Lavagna 2018), as well as missions that simply pass near asteroids.

The third and perhaps most salient reason is related to planetary protection. The primary manifestation of the Yarkovsky effect is a gradual drift in the semi-major axis of the asteroid. Uncertainties in this effect are large contributors to uncertainties in the predicted paths of NEAs which may potentially impact Earth (Chesley et al. 2014; Spoto et al. 2014). Furthermore, in the event that an asteroid is discovered to be on a collision course with Earth, a clearer understanding of its physical and orbital properties would be critical for studying the effectiveness of various measures for deflecting the asteroid (Cheng et al. 2018; Cheng et al. 2023).

By using astrometric observations of an asteroid over a reasonably long observational arc, its orbit can be estimated. By analysing if and how this orbit changes over time, or whether a better orbital fit is achieved by including certain perturbations, some physical properties of the asteroid can be derived. For example, the Yarkovsky effect is a non-gravitational acceleration caused by the anisotropic thermal emission of some rotating asteroids, producing a small propulsive force. The resulting drift in the semi-major axis of an asteroid's orbit can be observed if sufficient observation data is available, from which the magnitude of the effect can be derived.

The Yarkovsky effect has been detected for many objects, using several different methods. Typically, the approach is to use observation data and a dynamical model to produce an orbit estimation using a weighted least-squares (WLS) method (e.g. Fenucci et al. 2024; Dziadura et al. 2023; Hung et al. 2023; Li et al. 2023; Reddy et al. 2022; Kretlow 2022). While fundamental physical models of the Yarkovsky effect exist (e.g. Vokrouhlický 2001), they depend on specific physical parameters of the asteroid which are typically not known. For this reason, simplified parametrised models are often used instead (e.g. Chesley et al. 2014; Fenucci et al. 2024), such that a parameter related to the Yarkovsky effect is estimated alongside other orbital parameters, which accounts for how the semi-major axis of the body has drifted over the observational arc. The strength of the effect is typically reported as the drift rate of the body's semi-major axis, $\mathrm{d}a/\mathrm{d}t$ (often in AU/My), or as an acceleration term (often in AU/day^2).

For example, Dziadura et al. (2023) used the above approach with astrometric observations from the Gaia mission and reliably detected the Yarkovsky effect for 49 NEAs (and also calculated their bulk densities). Fenucci et al. (2024) used a similar approach with astrometric and radar observations to detect the Yarkovsky effect for 348 NEAs, whilst corroborating the detections against a physical model to assess plausibility. Farnocchia et al. (2013) and Greenberg et al. (2020) also use this method, detecting the Yarkovsky effect for 21 and 247 NEAs respectively.

However, much of this research uses proprietary software for orbit propagation and determination, such as JPL's Monte and ESA's Aegis software, which are not freely available for all researchers to use. The use of proprietary software in research limits methodological transparency and the independent reproducibility of results, as well as the ability to build upon existing research.

Furthermore, precise details about the methods and settings used to produce the published results are frequently omitted from research papers, further curtailing the ability of other researchers to independently verify results. Any differences in results are also harder to explain when it is uncertain which settings are common and which differ between two analyses. An example of this will be shown in Subsection 4.3.4 (see Figure 4.32), where there are some notable discrepancies between the results obtained by two different sources (JPL's Small-Body Database and ESA's Near-Earth Object Coordination Centre) in the determination of a parameter relating to the Yarkovsky effect, including two instances where the two sources disagree on the direction of the Yarkovsky acceleration. Since neither source provides comprehensive explanations of the methodologies and inputs used to make these determinations, it is difficult to know how the discrepancies arise and which source should be considered more reliable.

The omission of precise aspects of the methodology can also result in time being wasted on the process of selecting optimal settings. While the choice of optimal settings is highly dependent on the research being undertaken, closely related studies can often employ very similar settings. Without the free availability of which settings have been used to produce successful results, researchers wishing to extend existing analyses or perform closely related analyses are forced to re-conduct the selection of settings from scratch. This aspect can affect both researchers with and without access to the proprietary software tools mentioned.

The problems identified could be remedied through the use of free open-source software for astrodynamics and orbit determination, as well as the transparent documentation of the processes and results of selecting the settings relating to a particular kind of orbit determination problem. Open-source software is freely available to use and allows for independent verification that the methods available have been implemented correctly. Thorough documentation of settings selection processes and the corresponding results can be used by future researchers to inform the settings they will use in their own work.

The research of this thesis will therefore focus on reproducing existing results from literature relating to the Yarkovsky effect using free open-source software whilst being transparent and explicit about documenting the methods and settings used, as well as the reasons for choosing these. Reproducing results from literature will allow for an assessment of the quality of the selected settings by comparing the results with those from literature. The open-source software used is Tudat, the TU Delft Astrodynamics Toolbox, a set of libraries developed at the Faculty of Aerospace Engineering in the Delft University of Technology for the purposes of astrodynamics research (Dirkx et al. 2022; Gisolfi et al. 2025). Some more details about Tudat can be found in Subsection 3.1.2.

1.2. Research Questions

Based on the problems identified above, this thesis will address the following research questions:

RQ1: To what extent can results relating to the Yarkovsky effect reported in literature be reproduced via orbit determinations using open-source software (Tudat)?

RQ2: Which specific methodological processes and settings are required to achieve good orbit determinations for the purposes of detecting the Yarkovsky effect?

1.3. Report Structure

Chapter 2 provides the theoretical background of the topics that are relevant to understand in order to follow the methodology used and the results presented. This includes discussions of dynamical modelling, the Yarkovsky effect, astrometric data, and the orbit estimation process. Throughout this chapter, references are made to the current practices used in literature with comparable goals. Chapter 3 presents the methodology that was implemented to achieve the goals of the research. This includes detailing the selection of the asteroid set on which estimations were performed, the selection of numerical integration settings, and the selection of accelerations to use in the dynamic model, as well as documenting the inputs to the estimation process and the outlier rejection strategy. The results of the research and the corresponding analyses and discussions are given in Chapter 4, including comparisons of results with values published in literature, and an overview of results for a few selected bodies. The conclusions arising from the analyses of the results, as well as recommendations for future work are presented in Chapter 5.

2
Background

The purpose of this chapter is to introduce and explain the topics which are relevant to understanding the methodology, results, and conclusions of this research. First, some general background information about asteroids is presented in Section 2.1. Section 2.2 (starting on page 7) details the calculation of a celestial body's orbital motion, as influenced by gravity, relativity, and orbital perturbations, as well as a discussion of perturbation models that have been used in literature. Section 2.3 (starting on page 15) goes into more detail about the mechanisms of the Yarkovsky effect, followed by a discussion of Yarkovsky models that have been used in literature and an overview of some existing Yarkovsky detections in literature. The observational data of asteroids is treated in Section 2.4 (starting on page 21), including how observations are made, the errors that affect them, and the schemes that can be used to process them. Finally, the process of performing an orbit estimation using a weighted least-squares method is outlined in Section 2.5 (starting on page 29), as well as the process of covariance and uncertainty analysis, and details regarding the numerical integration of the equations of motion.

2.1. Asteroid Background

This section contains some basic background information about asteroids including a brief history of their discovery and classification (Subsection 2.1.1), an overview of modern terminology (Subsection 2.1.2), notable groups of asteroids (Subsection 2.1.3), some statistics on the rate of asteroid discovery over time (Subsection 2.1.4), and naming conventions (Subsection 2.1.5). Readers knowledgable about these matters already may prefer to skip (parts of) this section.

2.1.1. History

Ceres, the first body to be discovered which is now categorised as an asteroid (designated as 1 Ceres), was discovered in 1801, closely followed by the discovery of Pallas in 1802 (Hilton 2001). Although William Herschel argued that these objects should be not be classified as planets but instead as asteroids (a term he coined, meaning "star-like", because observations appeared as points of light rather than disks due to their small size), most other astronomers disagreed and considered them as newly discovered planets (Metzger et al. 2019). As more and more small bodies with orbits between Mars and Jupiter were discovered, it became convenient to number them, as an alternative to prescribing a new astronomical symbol for each one (Hilton 2001).

Contrary to what is sometimes reported, these bodies continued to be considered planets by most scientists and astronomers, as the large number and commonality of orbits was not considered disqualifying for being part of the planet taxon. They were considered to be a subset of the planets, variously referred to as "minor planets", "small planets", and "asteroids". This changed in the 1950s when new scientific reporting showed that asteroids to be geophysically different than the large planets. Around this time, asteroids began to be considered taxonomically different to planets. To this day, however, it is often argued that Ceres — the largest body in the main asteroid belt — should be considered a planet on the basis of its geophysical properties (Metzger et al. 2019).

2.1.2. Modern Terminology

A number of terms are used when referring to asteroids and other Solar System bodies. Since 2006, the International Astronomical Union (IAU) uses three main classifications: "planet", "dwarf planet", and "small Solar System body" (IAU 2018). Planets are bodies which 1) orbit the Sun, 2) have a mostly round shape, and 3) are gravitationally influential enough to have cleared their orbital neighbourhoods of other large bodies; there are eight such bodies in the Solar System. Dwarf planets are bodies that meet the first two planetary criteria but not the third; examples are Ceres and Pluto.

Small Solar System bodies are those which meet only the first planetary criterion: they orbit the Sun but are too small to have an almost round shape. The terms "asteroid", "minor planet", and "minor body" are also commonly used in a similar way to "small Solar System body", though these terms typically include dwarf planets and exclude comets. The term "meteoroid" may be used to refer to small asteroids. Comets are icy bodies characterised by their outgassing in the presence of solar radiation, forming a tail of gas and dust (IAU Office of Astronomy for Education n.d.(b)). Asteroids are typically irregular lumps of rock, metal, and/or ice and come in a wide range of sizes, orbits, origins, compositions, and characteristics (IAU Office of Astronomy for Education n.d.(a)). The number of identified asteroids is about 330 times larger than the number of identified comets (Minor Planet Center 2026). Any body that enters Earth's atmosphere and incandesces due to atmospheric friction is called a meteor. Any piece of material that survives the trip from space to Earth's surface without entirely burning up is called a meteorite (Ostro 1997).

2.1.3. Groups

Most known asteroids are in the main asteroid belt, which is a region that lies between the orbits of Mars and Jupiter. Other notable populations of asteroids include the near-Earth objects and the trojans. The trojans are groups of asteroids that exist at the stable Lagrange points (L4 and L5) of the planets — the Jupiter trojans are the most numerous of these (NASA n.d.). Despite asteroids existing in very large numbers, the total mass of the asteroid belt is only 3.3% of the mass of Earth's Moon (Pitjeva and Pitjev 2018).

A near-Earth object (NEO) is an asteroid or comet whose perihelion distance (closest approach to the Sun) is less than 1.3 astronomical units (AU). Most NEOs are asteroids — near-Earth asteroids (NEAs) — and a small number are comets (Chapman 2004; CNEOS n.d.). If a NEO meets the following two criteria, it is categorised as a potentially hazardous object (PHO): the closest possible distance between the NEO and Earth (called the Earth Minimum Orbit Intersection Distance or Earth MOID) must be 0.05 AU (equal to $\sim$19.5 times the average Earth-Moon distance) or less and its absolute magnitude H must be 22 or less. The absolute magnitude limit corresponds to an asteroid size of around 140 m, assuming an albedo of 0.14. If such an object is an asteroid, as opposed to a comet, it is called a potentially hazardous asteroid (PHA). An NEA is further classified into one of four groups — the Amors, Apollos, Atens, and Atiras — based on its orbital characteristics, namely its semi-major axis and its perihelion or aphelion distance (CNEOS n.d.).

2.1.4. Discoveries Over Time

According to Figure 1 in DeMeo et al. (2015) only $\sim$100 asteroids had been discovered by 1870, $\sim 10^3$ by 1920, and $\sim 10^4$ by 1980. The rate of discovery then sped up significantly, reaching 10^5 known asteroids in 2000 (DeMeo et al. 2015; MPC 2019) and 10^6 in 2020 (IAWN 2020). The total now stands at over 1.5×10^6 (Minor Planet Center 2026). The increasing rate of discovery is similar for NEOs: before 1980, fewer than 50 NEOs had been discovered. By 2000, that number rose to almost 10^3. As of June 2026, approximately 4.2×10^4 are known (Minor Planet Center 2026), which includes more than 90% of all NEOs larger than one kilometre (CNEOS 2024).

2.1.5. Naming[1]

If observations reported to the MPC of an asteroid (see Subsection 2.4.2 for more information about the MPC and observations) cannot be linked to any known asteroids' orbits, the asteroid is designated a new discovery and is assigned a *provisional designation* (MPC n.d.(a)). This is formatted as the year in which the discovery was made, a letter which identifies the half-month of the discovery[2], and a letter which marks the order of the discovery within that half-month[3]. If more than 25 discoveries are made within that half-month, the second letters are reused in order and the number '1' is appended to the end of the designation. If more than 50 discoveries are made, the number '2' is appended, and so on (MPC n.d.(b)). The appended numeral(s) are preferably subscripted. For example, the asteroid 2026 EB_1 was the 27th asteroid to be discovered in the first half of March 2026 (the fifth half-month of that year).

When observations are available for four or more oppositions[4], the asteroid's orbit is generally known

[1] Note that historical exceptions exist to modern processes and rules described in this subsection.
[2] The capital letters of the Latin alphabet are used for this, excluding I and Z.
[3] The capital letters of the Latin alphabet are used for this, excluding I.
[4] For a body with an orbit that exists (partially) outside of Earth's orbit, opposition occurs when the body appears on the opposite

with enough certainty that it may be given a *permanent designation*, which is an integer number. Such an asteroid may be described as *numbered*. NEAs may receive permanent designations after only two or three oppositions (MPC n.d.(a)). The numbers are generally designated sequentially, in order of orbital confirmation, as opposed to the order of first discovery (IAU n.d.). As of June 2026, ~887 000 asteroids have been numbered (JPL 2026), which accounts for ~57% of discovered asteroids.

The discoverer of the asteroid can then suggest a name by submitting a proposal to the MPC with the name and a short explanation. The names are subject to certain rules, such as a character limit of 16 and restrictions on using the existing name of another Solar System body, the discoverer's own name, numbers, generic words, commercial names, and naming after political or military persons. Name proposals are judged by the Working Group Small Body Nomenclature (WGSBN) within the IAU. If accepted, the suggested name becomes official when published in a bulletin by the WGSBN (WGSBN 2025). As of June 2026, ~26 000 asteroids have been named (WGSBN 2026), which accounts for ~1.7% of discovered asteroids and ~2.9% of numbered asteroids.

Asteroid name formats vary by their status and also by convention. The various formats for are shown in Subsection 2.1.5.

Table 2.1: Possible formatting of the names of unnumbered, numbered & unnamed, and named asteroids.

Unnumbered	Numbered & unnamed	Named
* 2026 EB_1	* 612197	* 1566 Icarus
* 2026 EB1	* 612197 (2000 WP19)	* Icarus
(2026 EB_1)	(612197)	(1566) Icarus
(2026 EB1)	* 612197 (2000 WP_{19})	1566 Icarus (1949 MA)

* Indicates naming formats used in this document.

side of the sky compared to the Sun. At this point, the full illuminated face of the body is facing Earth, so it is typically when the body is the brightest and most visible. For example, the Moon at opposition corresponds to a full moon (IAU Office of Astronomy for Education n.d.(c)).

2.2. Dynamical Modelling

Celestial mechanics is a branch of astronomy in which the motion of bodies in space is studied and described. In order to model and predict the motion of any given celestial body, the forces which affect its motion — gravitational and otherwise — must be considered carefully. This section provides the mathematical expressions relevant to modelling the motion of the bodies relevant to this research. Subsection 2.2.1 gives an overview of the fundamentals of motion, Subsection 2.2.2 discusses the main phenomena which 'perturb' orbital motion as compared to the simplest 2-body formulation, and Subsection 2.2.4 describes a few dynamic models that have been used in literature for similar research.

2.2.1. Equations of Motion

This subsection details the fundamentals of motion, starting from Newton's laws. Then the limitations of these laws in relation to relativistic effects are discussed, followed by a presentation of the general n-body problem.

Newton's Laws

The fundamentals of classical celestial mechanics arise from Isaac Newton's three laws of motion and his law of gravitation, published in his *Principia* in 1687. These laws apply to 'particles' or 'point-masses', whose motion is described relative to an inertial (non-rotating and non-accelerating) frame of reference. The three laws of motion can be expressed as follows (Wakker 2015, Sec 1.1):

1. A particle at rest or in uniform rectilinear motion will remain in that state unless/until a force acts upon that particle.
2. The rate at which a particle's linear momentum changes with time is proportional to the resultant of all external forces acting upon the particle. The direction of this momentum change is the same as the direction of the resultant force.
3. When two particles exert forces on each other, these forces are equal in magnitude and opposite in direction.

Newton's second law of motion can be written symbolically as:

$$\mathbf{F} = \frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}) = \frac{\mathrm{d}}{\mathrm{d}t}\left(m\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right) \tag{2.1}$$

where $\mathbf{F}$ is the net external force acting on the particle, m is the mass of the particle, $\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$ is the velocity of the particle in the inertial frame, and $\mathbf{r}$ is the position vector of the particle in the inertial frame. Variables in **bold** are vectors: for example, $\mathbf{v}$ is a vector whose magnitude is $|\mathbf{v}| = v$ and unit vector is $\mathbf{v}/v = \widehat{\mathbf{v}}$. A rigid body (a body with zero flexibility) can be thought of as a set of small, discrete, constant-mass elements with a constant spatial relation to each other. The assumption that celestial bodies are rigid bodies is a good first-order approximation, and for such a body, the second law of motion can be expressed as (Wakker 2015, Sec. 1.2):

$$\mathbf{F} = M\frac{\mathrm{d}\mathbf{v}_{\mathrm{cm}}}{\mathrm{d}t} \tag{2.2}$$

where M is the total instantaneous mass of the body, and $\mathbf{v}_{\mathrm{cm}} = \frac{\mathrm{d}\mathbf{r}_{\mathrm{cm}}}{\mathrm{d}t}$ is the velocity of the body's centre of mass. Regarding the modelling of translational dynamics, a rigid body can be assumed to be a point mass with mass M, located at $\mathbf{r}_{\mathrm{cm}}$.

Newton's law of gravitation can be expressed as:

$$F = G\frac{m_1 m_2}{r^2} \tag{2.3}$$

where m_1 and m_2 are the masses of two bodies, r is the distance between the centres of mass of these two bodies, G is the gravitational constant (equal to $\sim 6.6742 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$), and F is the magnitude of the gravitational force attracting the two bodies together (Curtis 2014, Ch. 1).

Because celestial mechanics tends to deal with bodies that are quite far apart, roughly spherical in shape, and with density distributions that are approximately radially symmetric, it is typical to treat each mass as a point mass at the centre of its respective body (Wakker 2015, Sec. 1.4). This assumption does not necessarily hold when bodies are relatively close to each other and when one or more of these bodies does not have a radially symmetric density distribution. In these cases, a more detailed gravity field can be modelled, for example using a spherical harmonic model (discussed further in Subsection 2.2.2). Examples of this in literature include modelling the orbits of artificial satellites in low Earth orbit (e.g.

Pedroso and Batista (2023)) and the orbits of the Galilean moons around Jupiter (e.g. Fayolle et al. (2023)).

G can be determined by measuring the gravitational force between two precisely-known masses. However, because of the weakness of gravitational forces, G is difficult to directly determine with extreme precision. For this reason, the known masses of the Sun and planets also have limited precision. Therefore, the gravitational parameter $\mu = GM$, where M is the mass of the body, is often used where possible because μ is typically known to a much higher degree of precision than either G or M (Wakker 2015, Sec. 1.4).

The force acting upon mass m_2 due to the mutual gravitation between m_1 and m_2 is (Idema 2018):

$$\mathbf{F}_2 = -G\frac{m_1 m_2}{{r_2}^3}\mathbf{r}_2 = -G\frac{m_1 m_2}{{r_2}^2}\widehat{\mathbf{r}}_2 \tag{2.4}$$

where $\mathbf{r}_2$ is the position vector from m_1 to m_2 and $\widehat{\mathbf{r}}_2$ is the corresponding unit vector. The minus sign is necessary because the direction of the force is opposite to the direction of the position vector. Dividing by m_2, we get the acceleration $\mathbf{a}_2$ of mass m_2 due to mutual gravitation between m_1 and m_2:

$$\mathbf{a}_2 = -G\frac{m_1}{{r_2}^2}\widehat{\mathbf{r}}_2 \tag{2.5}$$

Relativistic Considerations

Newton's laws accurately describe the motion of bodies in a wide range of scenarios and they form the basis of the classical approach to celestial mechanics. In contrast to this classical approach are Einstein's theories of special and general relativity. Special relativity challenges the underlying assumption of classical mechanics that time runs at equal rates for all inertial reference frames, by postulating that the speed of light (denoted $c = 299\,792\,458\ \mathrm{m\,s^{-1}}$) has the same value in all inertial reference frames. A consequence of this is that c is the universal speed limit and nothing can travel faster than it. Newton's third law implies that equal and opposite forces act instantaneously and simultaneously. But according to special relativity, if two inertial reference frames are moving relative to each other at a constant velocity v, events which appear to be simultaneous in one reference frame will not necessarily be simultaneous in the other reference frame (Idema 2018, Ch. 10). If two events separated by a distance Δx occur simultaneously for an observer in a stationary reference frame S, they will appear sequentially with a time difference of $\Delta t'$ for an observer in S', a reference frame moving at velocity v in the x direction relative to S (Idema 2018, Ch. 11):

$$\Delta t' = \gamma(v)\left(\frac{v}{c^2}\right)\Delta x \tag{2.6}$$

where $\gamma(v)$ is the Lorentz factor:

$$\gamma(v) = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \tag{2.7}$$

Clearly, the differences are very small when the velocities involved are much lower than the speed of light ($v \ll c$).

General relativity posits that gravity is not a true force but a *fictitious* force (also called an apparent force), and that bodies in free fall are in fact following straight paths in a curved 4-dimensional space-time continuum — such paths are called geodesics. The curvature of space-time is caused by objects with mass, with more massive objects causing more extreme curvature. Additionally, any changes to space-time's curvature (due to a change in a body's position, for example) propagate at only a finite speed: c. For the purposes of computing typical stellar system orbits however, it is acceptable to use Newtonian gravitation and assume that it is a force that acts instantaneously (Wakker 2015, Sec. 1.2, 1.4). That is, it acts along the "instantaneous" straight line between the bodies, as opposed to considering the line between where one body is "now" and where the other body "was" according to the light-time delay of the distance between the two bodies. This is because the expected effect of the delay due to the propagation of gravity at the finite speed of light is almost exactly cancelled by velocity-dependent terms in the formulation of the Einstein field equations. The mismatch can therefore assumed to be negligible. This assumption loses validity in more extreme gravitational situations, such as binary pulsars, whose orbits measurably decay over time due to the loss of energy via gravitational radiation, which is a result of the aforementioned mismatch (Carlip 2000).

***n*-Body Systems**

So far, we have described gravitation between two bodies. A general case of a gravitational problem is a system with many bodies, say n bodies, which can be considered to be point masses; see Figure 2.1. The following explanation of n-body systems and the accompanying equations are from Chapter 2 of Wakker (2015).

The body of interest, i, has mass m_i and Cartesian coordinates (x_i, y_i, z_i) in an inertial reference frame. Any other body can be called j, with mass m_j and Cartesian coordinates (x_j, y_j, z_j). The position vector

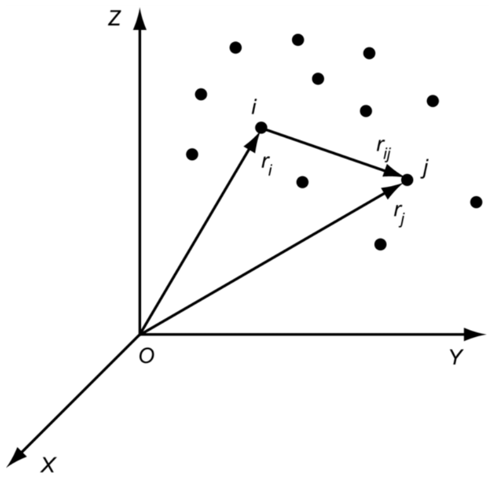


Figure 2.1: An n-body system of point masses in an inertial reference frame XYZ (Wakker 2015, p. 25)

from body i to body j is:

$$\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i \tag{2.8}$$

Assuming the only relevant forces to be gravitational, the motion of body i can be described with:

$$\sum \mathbf{F}_i = m_i \frac{\mathrm{d}^2\mathbf{r}_i}{\mathrm{d}t^2} = \sum_{j \neq i} G \frac{m_i m_j}{r_i^3} \mathbf{r}_{ij} \tag{2.9}$$

Where the summation takes place over all bodies from $j = 1$ to $j = n$, except for $j = i$. The acceleration of body i is:

$$\mathbf{a}_i = \frac{\mathrm{d}^2\mathbf{r}_i}{\mathrm{d}t^2} = \sum_{j \neq i}^{n} G \frac{m_j}{r_i^3} \mathbf{r}_{ij} \tag{2.10}$$

The n-body problem is to determine the position of each of the bodies at any/every point in time, when the positions and velocities of all bodies are known at a particular moment in time. A general solution exists for $n = 2$, and while solutions do exist for some restricted cases when $n > 2$, there are no practical general solutions for $n > 2$. Therefore, to analyse such systems, numerical integration techniques are typically required.

Some properties of an isolated n-body system as described above include:

- The total linear momentum of the system is constant
- The total angular momentum of the system is constant
- The total energy (kinetic plus potential) of the system is constant
- The barycentre (centre of mass) of the system is stationary or moves with constant velocity

The last of these properties means that the barycentre can be used as the origin of an inertial reference frame, relative to which the motion of the bodies can be described. Doing this, we can describe the motion of body i with:

$$\begin{aligned} \mathbf{a}_i = \frac{\mathrm{d}^2\mathbf{r}_i}{\mathrm{d}t^2} &= -G\frac{M}{r_i^3}\mathbf{r}_i + G\sum_{j \neq i}^{n} m_j \left(\frac{1}{r_{ij}^3} - \frac{1}{r_i^3} \right) \mathbf{r}_{ij} \\ &= \mathbf{a}_{\text{prim. grav.}} + \sum_{j \neq i}^{n} \mathbf{a}_{\text{3b pert. } j} \end{aligned} \tag{2.11}$$

where M is the total mass of the n bodies and $\mathbf{r}_i$ is the position vector of body i in the barycentric reference frame. With this expression, body i's motion can be seen as the sum of two main components: the first term on the right-hand side ($\mathbf{a}_{\text{prim. grav.}}$) represents the 2-body motion about the barycentre (see

Equation (2.5) for comparison), and the second term ($\sum \mathbf{a}_{3\text{b pert. } j}$) represents the set of 'perturbing' accelerations exerted by each body j on body i. This interpretation (and the use of Equation (2.11) in general) is most useful when the perturbing accelerations are much smaller than the first acceleration term; for example, when modelling the motion of an asteroid around the Sun, with the planets and other large bodies acting as the perturbers.

2.2.2. Perturbations

In reality, point-mass gravitational forces are not the only forces acting on a celestial body. Although many orbits can be well-approximated by a 2-body Kepler orbit, there are generally other forces present which cause deviations from this ideal Kepler orbit, called perturbations. We have already encountered one source of these perturbations in Equation (2.11): third-body perturbations. Further perturbations arise from relativistic effects, non-radially symmetric mass distributions, and non-gravitational forces such as radiation pressure, atmospheric drag, electromagnetic force, outgassing, and the Yarkovsky effect. The perturbations relevant to this research are discussed in more detail in this subsection.

Third-Body Perturbations

Third-body perturbations apply in scenarios where a body's motion can approximately be described as a 2-body problem, but when other bodies in the vicinity also exert some gravitational effect. An example is Earth's orbit around the Sun: considering only the mutual gravitation of these two bodies will give a good approximation of their motion. For a more accurate result, however, the 'perturbing' effects of the gravitation of other Solar System bodies may be included, such as those of the Moon, Venus, Mars, and Jupiter, and perhaps more.

In the barycentric reference frame, the acceleration due to body j's gravitational perturbation on body i is given by (Curtis 2014, Ch. 12):

$$\mathbf{a}_{i3\text{b pert. } j} = \mu_j \left(\frac{1}{{r_{ij}}^3} - \frac{1}{{r_i}^3} \right) \mathbf{r}_{ij} \tag{2.12}$$

where $\mu_j = Gm_j$ is the gravitational parameter of body j and $\mathbf{r}_{ij}$ is defined as in Equation (2.8). The acceleration $\mathbf{a}_{i3\text{b pert. } j}$ is akin to one of the terms summated on the right-hand side of Equation (2.11).

Because the distances to third bodies are typically large, and thereby the accelerations they cause are much smaller than the acceleration due to the primary body, they are typically treated as point-masses, as any gravity field variations are usually negligible from such distances (Wakker 2015, Sec. 1.4).

The most relevant third-body perturbations to include will be those that have significant mass and are not too distant from the body whose motion is being modelled. The cut-off point for including more bodies will depend on the accuracy requirements and the available computation time, among other factors. This cut-off may, for example, be defined as an absolute value of acceleration such that any perturbing accelerations with magnitudes below the cut-off are considered negligible and thus excluded from the dynamical model.

Spherical Harmonics

The gravity field of a celestial body is typically described with a scalar potential function, which is a function of the position of a test mass relative to that body. The gradient of this potential is equal to the field strength (force per unit of mass) at the location of the test mass. By convention, the point of zero potential is chosen to be at infinity, so the gravitational potential of a point-mass body with gravitational parameter μ is (Wakker 2015, Sec. 1.4):

$$U = -\frac{\mu}{r} \tag{2.13}$$

where r is the distance to the body. Thus, U is negative for any finite r.

Real celestial bodies have non-radially symmetric mass distributions, which lead to more complex gravity fields. A body with an arbitrary shape and mass distribution (which are assumed to be constant over time) can be described using spherical harmonics. The gravitational potential of such a body at a point given in spherical coordinates as (r, Φ, Λ) is (Wakker 2015, Sec. 20.1):

$$U = -\frac{\mu}{r} \left[1 + \sum_{n=2}^{\infty} \sum_{m=0}^{n} \left(\frac{R}{r} \right)^n P_{n,m}(\sin \Phi) \left(C_{n,m} \cos(m\Lambda) + S_{n,m} \sin(m\Lambda) \right) \right] \tag{2.14}$$

where r is the distance from the centre of mass of the body, Φ is the body-centric latitude, Λ is the body-centric longitude (these coordinates are for a reference frame with origin at the body's centre of mass and which rotates with the body), R is the mean equatorial radius of the body, $C_{n,m}$ and $S_{n,m}$ are the parameters that define the spherical harmonic model of the gravity field, and $P_{n,m}(\sin \Phi)$ are the Legendre functions that correspond to degree n and order m. Equation (2.14) is valid for values of r greater than the maximum radius of the body.

The degree n relates to latitudinal zones and the order m relates to longitudinal sectors. The model parameters corresponding to these values of n and m define the influence of these zones on the total gravity field (Wakker 2015, Sec. 20.1). All together, these components account for the full effect of the exact mass distribution of the body on the gravitational field. For example, the parameters determined for Earth's gravitational field account for the planet's pole-to-pole flattening, its slight pear shape, and any other significant deviations from the idealised radially symmetric mass distribution. The parameters describing Earth's gravitational model are known quite precisely, with the EGM2008 model being complete up to degree and order 2159 (Pavlis et al. 2012), while those of other celestial bodies are known less well.

In terms of the accelerations experienced by a test mass in a body's gravitational field, the term relating to the acceleration due to a point mass at the body's centre of mass is proportional to r^{-2} (see Equation (2.11)). Other terms, relating to the acceleration due to the body's asymmetric mass distribution, are proportional to $r^{-(n+2)}$, where the degree $n \geq 2$. This shows that deviations due to asymmetric mass distributions are most significant when orbiting close to the body, and that these deviations diminish quickly with distance, meaning bodies are generally well approximated as point-masses when far away.

Relativistic Effects

Another source of perturbations when working with a Newtonian dynamic model is the correction for relativistic effects. In many cases, a Schwarzschild correction acceleration $\mathbf{a}_{\text{rel.}}$ to account for the relativistic effect of the primary body p on body i, modelled as the following is sufficient (Farnocchia et al. 2015b):

$$\mathbf{a}_{\text{rel.}} = \frac{\mu_p}{c^2 {r_{pi}}^3} \left[\left(\frac{4\mu_p}{r_{pi}} - {v_{pi}}^2 \right) \mathbf{r}_{pi} + 4(\mathbf{r}_{pi} \cdot \mathbf{v}_{pi})\mathbf{v}_{pi} \right] \tag{2.15}$$

The vectors $\mathbf{r}_{pi}$ and $\mathbf{v}_{pi}$ are the position and velocity of body i with respect to the primary body p, and μ_p is the gravitational parameter of the primary body p.

Farnocchia et al. (2015b) note that the more Einstein-Infeld-Hoffmann (EIH) formulation of relativity may be necessary if body i experiences close encounters with other large bodies, such as planets. The EIH equations allow for gravitation to be modelled for an n-body system, including the mutual relativistic effects of all n bodies. The formulation for the EIH acceleration $\mathbf{a}_{\text{EIH}}$ exerted on body i is (Will 2018):

$$\begin{aligned}\mathbf{a}_{\text{EIH}} = \mathbf{a}_i = &- G \sum_{\substack{j=1 \\ j\neq i}}^{n} m_j \frac{\mathbf{r}_{ji}}{{r_{ji}}^3} \left[1 - \frac{1}{c^2} \left((1+2\gamma) \sum_{k=1}^{n} \frac{Gm_k}{r_{kj}} + (1+2\beta) \sum_{k=1}^{n} \frac{Gm_k}{r_{ki}} \right) \right. \\ &\left. + \frac{1}{2c^2} \left(v_i^2 + v_j^2 - 4\gamma \mathbf{v}_i \cdot \mathbf{v}_j - \frac{3}{2} (\widehat{\mathbf{r}}_{ji} \cdot \mathbf{v}_j)^2 \right) \right] \\ &+ \frac{G}{c^2} \sum_{\substack{j=1 \\ j\neq a}}^{n} m_j \left[\frac{\mathbf{r}_{ji}}{{r_{ji}}^3} (\mathbf{r}_{ji} \cdot (\mathbf{a}_i - \mathbf{a}_j)) + \frac{7}{2} \frac{\mathbf{a}_j}{r_{ji}} \right]\end{aligned} \tag{2.16}$$

where:

- j and k are other bodies in the n-body system
- m_a is the mass of a body a
- $\mathbf{r}_{ba} = \mathbf{r}_a - \mathbf{r}_b$ is the position vector from body b to body a
- $r_{ba} = |\mathbf{r}_{ba}|$
- $\widehat{\mathbf{r}}_{ba} = \mathbf{r}_{ba}/r_{ba}$ is the unit vector from body b to body a
- $\mathbf{v}_a = \dot{\mathbf{r}}_a$ is the velocity vector of body a
- $v_a = |\mathbf{v}_a|$
- $\mathbf{a}_a = \dot{\mathbf{v}}_a$ is the acceleration vector of body a
- γ and β are the Parametrized Post-Newtonian (PPN) parameters

Note that this expression is implicit, since $\mathbf{a}_i$ appears on both the left- and right-hand side.

Non-Gravitational Perturbations

There are a number of other perturbative forces which are unrelated to gravitation that can apply to the motion of a body, depending on the conditions. The non-gravitational Yarkovsky effect ($\mathbf{a}_{\text{Yark.}}$) is discussed in much more detail in Section 2.3. The rest will not be discussed in detail here because they are not directly relevant to this research, but short descriptions of them are provided below.

- **Solar radiation pressure $\mathbf{a}_{\text{SRP}}$**: Bodies are often exposed to electromagnetic radiation: directly from the light of the nearest star, reflected off a third body, or emitted from a third body as infrared. Each photon colliding with the body in question has momentum, and thus momentum is imparted onto the body when these photons are reflected or absorbed by the body. The magnitude of the perturbing force depends on the incoming radiation flux, the cross-sectional area exposed to the radiation, the body's mass, and the optical/thermal properties of the body's material. It is most relevant for small bodies whose area to mass ratios are higher. It causes a small drag-effect which causes the orbit to decay slowly (Wakker 2015, Sec. 5.11, 20.4).
- **Outgassing $\mathbf{a}_{\text{outgas}}$**: For bodies with icy compositions (comets), the sublimation of ices due to the warming effect of solar radiation causes gases and dust to escape from the surface. This is typically not uniform in all directions and so the gases and dust released can cause a small thrust force (Whipple 1950; Yeomans and Chodas 1989) or even a torque (Jewitt 2021).
- **Empirical accelerations $\mathbf{a}_{\text{empirical}}$**: Assuming the overall acceleration model to be quite accurate, small remaining accelerations which cannot be modelled (accurately) individually using a physical model can be grouped together and modelled as empirical accelerations in order to generate a better fit to observation data. The precise parametrisation of the empirical accelerations will depend on the type of motion being modelled and on the observed deviation from the observation data. For example, an Earth-orbiting satellite may use an acceleration model with a constant component and components proportional to the sine and cosine of the true anomaly (Montenbruck and Gill 2001, Subsec. 3.7.4). When Micheli et al. (2018) investigated non-gravitational accelerations affecting the hyperbolic trajectory of 1I/'Oumuamua (an interstellar object that passed through the Solar System in 2017), they tried a variety of candidate empirical acceleration models to explain the deviation (see Subsection 2.2.4 for more details on this). The parameter(s) of the empirical model are fitted during an orbit estimation (see Section 2.5).

2.2.3. Final Equation of Motion

As demonstrated in the previous subsections, the motion of a celestial body is affected by many different phenomena. To be able to model the motion of a particular body, we would like to combine all of these phenomena into a single acceleration model. Along with values describing the initial state of the body, this acceleration model can then be numerically integrated over a series of time steps to model the body's motion over time (see Subsection 2.5.3).

Taking into account all of the accelerations described above, the motion of a body i around a primary body p can be described by the following equation of translational motion:

$$\mathbf{a}_i = \ddot{\mathbf{r}}_i = \mathbf{a}_{\text{prim. grav.}} + \sum_{j=1}^{N} \mathbf{a}_{\text{3b pert. }j} + \mathbf{a}_{\text{rel.}} + \mathbf{a}_{\text{Yark.}} + \mathbf{a}_{\text{SRP}} + \mathbf{a}_{\text{outgas}} + \mathbf{a}_{\text{empirical}} \tag{2.17}$$

where N is the number of perturbing bodies included. The reference frame for the vectors in this equation is the barycentric reference frame (some, such as atmospheric drag, may have to be converted from frames that are local to a particular body). Note that the gravitational terms may include higher spherical harmonic terms if relevant. Any of the accelerations in Equation (2.17) may be set to zero or emitted if they are deemed irrelevant to the motion of body i.

The equations of motion discussed thus far have all been equations of translational motion. Other types of motion and dynamics do exist and can be relevant as well. For example, rotational dynamics, mass variation, and body deformation can all have significant effects. To model these, they would each need their own equations of motion. The different forms of dynamics can also be coupled together, adding further complexity and requiring concurrent propagation (Dirkx and Mooij 2019). While mass variation and body deformation are mostly applicable to spacecraft, the rotational state of an asteroid is important for modelling spherical harmonics, and can also be relevant for the Yarkovsky and YORP effects (see Subsection 2.3.1). Modelling and propagating the rotational dynamics will not be considered here because the models used for the Yarkovsky effect in this research are simplified, parametrised models that do not require information about the rotational state of the asteroid.

2.2.4. Use of Perturbation Models in Literature

It is important to understand which acceleration models are used in practice, especially in cases of the modelling of asteroid orbits. For this, it is useful to look at a few examples from literature to see which models researchers used in their investigations, how these were implemented, and the reasons for these decisions (if provided by the authors).

In **Micheli et al. (2018)**, the authors investigated a notable non-gravitational acceleration that acted on 1I/'Oumuamua, an interstellar object that passed through the Solar System in 2017. The authors found that a purely gravitational model was not sufficient to reproduce the observed motion of the object. That is, an orbit estimation (see Section 2.5) with only gravitational accelerations included in its dynamic model was not able to converge to a solution that was statistically compatible with the observations, i.e. the observation residuals were unreasonably large. Consequently, they included a series of different empirical acceleration models (one at a time) and compared the resultant trajectories to the observations to find the best match.

The purely gravitational model they used included the effects of "the Sun, the eight planets, the Moon, Pluto, the 16 largest bodies in the asteroid main belt and relativistic effects". The bodies included in this model stem from a recommendation provided in Farnocchia et al. (2015b). The exact formulation of the relativistic effects is not provided. In the estimation using this model, 6 parameters are fitted.

Because the purely gravitational model was found to have a poor fit to the observational data, several different empirical acceleration models were devised (where $k \in 0, 1, 2, 3$ in all cases):

- a constant, inertially fixed acceleration — 9 parameters;
- a pure radial acceleration proportional to r^{-k} — 7 parameters;
- a pure along-track acceleration proportional to r^{-k} — 7 parameters;
- an acceleration proportional to r^{-k} decomposed in the radial, transverse, and normal directions (also called the RTN or RSW directions) — 9 parameters;
- an acceleration proportional to r^{-k} decomposed in the along-track, cross-track, and normal directions (also called the ACN or TNW directions) — 9 parameters;
- an impulsive velocity change at a particular moment — 10 parameters;
- a pure radial acceleration based on one of two cometary outgassing models ($g_{H_2O}(r)$ and $g_{CO}(r)$) — 7 parameters each
- acceleration based on $g_{H_2O}(r)$ and $g_{CO}(r)$, decomposed in the RTN directions — 9 parameters each
- the same as above but including a time delay parameter with respect to the time of perihelion passage — 10 parameters

The free fit parameters of each model (e.g. the proportionality constants of the accelerations) were determined by fitting the model to the observation data, and the quality of the fits was used to determine the empirical acceleration model which, when included, best describes 'Oumuamua's motion. Other non-gravitational accelerations (e.g. solar radiation pressure, Yarkovsky effect) were not included in the acceleration model. The authors found that the most plausible model in this case was a positive radial acceleration proportional to r^{-2} or r^{-1}, with outgassing being the likely physical cause.

The orbit determination code used in this research is proprietary to JPL and is therefore not publicly available. The authors note that the key results of the research can be replicated using free software such as Find_Orb.

In **Fenucci et al. (2024)**, the authors devised a procedure for identifying near-Earth asteroids (NEAs) for which the Yarkovsky effect may be detectable, estimating a non-gravitational acceleration in the transverse direction, and checking if the acceleration is statistically significant and whether it fits with the Yarkovsky model. This procedure identified 348 NEAs with a detectable Yarkovsky effect, and was adopted by the ESA NEO Coordination Centre (NEOCC) for updating the full catalogue of NEAs.

The dynamical model used in this research, described by the authors as "high precision", uses the same set of celestial bodies for gravitation as Micheli et al. (2018). The masses and positions of these bodies are supplied by JPL's DE441 planetary and lunar ephemerides. Relativistic corrections for the Sun, Moon, and planets were included, seemingly using the Einstein–Infeld–Hoffmann equations.

For some NEAs, an acceleration due to solar radiation pressure (SRP) was included in the dynamic model. This was modelled as an acceleration in the radial direction $\hat{\mathbf{r}}$:

$$\mathbf{a}_{\mathrm{SRP}} = \frac{A}{m} \cdot (1 - \Theta) \cdot \frac{W}{c} \cdot \left(\frac{1\mathrm{AU}}{r}\right)^2 \hat{\mathbf{r}} \tag{2.18}$$

where W is the solar radiation flux at 1 AU ($W = 1.361\,\mathrm{kW\,m^{-2}}$), c is the speed of light, A/m is the area-to-mass ratio, and Θ is a function whose value is 1 when the body is fully occulted by another body, 0 when in full sunlight, and between 0 and 1 when in penumbra. A/m if the free parameter fitted in the estimation in the cases when SRP is included in the dynamic model. The authors cite Montenbruck and Gill (2001) for this equation, which apparently comes from Equation 3.75 of that source. Montenbruck and Gill (2001) include the radiation pressure coefficient C_R in their equation, but Fenucci et al. (2024) do not mention how they treat or define the C_R. Implicitly, it seems that the body is assumed to be a black body and so reflectivity ($\mathbb{R}$ or ε) is zero.

Of course, a Yarkovsky acceleration was also included in the dynamical model used in this research. This is discussed further in Subsection 2.3.2. The orbital fits for this research were carried out using the ESA's proprietary Aegis software for orbit determination and impact monitoring. The methodology of this paper is outlined in more detail in Subsection 3.2.1.

In **Chesley et al. (2014)**, the authors use optical astrometry and radar measurements to very precisely define the orbit of the asteroid 101955 Bennu, including determining its semi-major axis drift rate due to the Yarkovsky effect and predicting potential future impacts with Earth. They also estimated Bennu's bulk density.

The orbit determination was performed with the use of a high fidelity dynamical model. This model includes the Newtonian gravitation of the Sun, the eight planets, the Moon, Pluto, and 25 main-belt asteroids. Earth's gravitational model also includes the terms related to its oblateness when Bennu is within a distance of 1 AU. The perturbing relativistic effects of the Sun, Moon, and eight planets are also included in the model. The non-gravitational accelerations included are direct solar radiation pressure and the acceleration due to the Yarkovsky effect. The Yarkovsky effect was modelled in three different ways as part of this research. Explanations of these models are provided in section Subsection 2.3.2. Other perturbations were not included as they were deemed insignificant over the time interval being considered.

Although not explicitly stated, it seems that the model used for SRP is very similar to the one shown in Equation (2.18). The known shape model of Bennu was used to find the area-to-mass ratio, A/m. The authors note that the pressure due to reflected radiation is negligible due to Bennu's low Bond albedo. They also note that although the magnitude of the SRP acceleration is much larger than that of the Yarkovsky acceleration, it has only a minor effect on the predicted trajectory, due to its direction being purely radial (the effect is therefore equivalent to a slight reduction in the Sun's mass).

Several model variations (such as changing which gravitational bodies were included, which bodies were included in the relativistic corrections, and the integration tolerances, for example) were considered and their effects on the eventual solutions were assessed. This helps to demonstrate the sensitivity of the dynamical model and the relative importance of the different facets of the model. The software used in this research was the proprietary JPL orbit determination and propagation package. The authors cross-referenced their results using OrbFit, a publicly available and free software package for orbit determination, and found very close agreement between the programs ($< 0.2\sigma$ in the calculations of the semi-major axis drift rate $\mathrm{d}a/\mathrm{d}t$).

2.3. Yarkovsky Effect

Typically, the largest accelerations experienced by a celestial body are gravitational accelerations. However, there are several other accelerations that also often act upon these bodies which can have a significant effect on their motion and orbital evolution over time, despite being much smaller in magnitude than the gravitational accelerations. The primary perturbations that are relevant to the motion of asteroids in the Solar System are discussed in Subsection 2.2.2; one important such perturbation is the Yarkovsky effect, which will be discussed in more detail in this section.

The acceleration caused by the Yarkovsky effect is one of the most important non-gravitational accelerations in terms of its effect on the evolution of asteroid orbits. The semi-major axis drift rate da/dt varies between approximately 10^{-4} and 10^{-3} AU/My for asteroids with diameters between 10 m and 1 km and for various values of surface conductivity (Bottke et al. 2006). Main-belt asteroids and fragments that undergo this motion often drift into chaotic mean-motion or secular resonance zones after which they can be injected into orbits that overlap with Earth's orbit, and it seems that this is the main mechanism supplying the population of NEOs (Vokrouhlický and Farinella 2000). The Yarkovsky effect is also important for understanding how asteroid families evolve and spread out over time, and has been used to explain several properties of such families that could not be explained using purely collisional models (Bottke et al. 2001).

Explanations of the physical mechanisms behind the Yarkovsky effect are given in Subsection 2.3.1, along with a list of factors that influence the effects. Subsection 2.3.2 describes a number of acceleration models that have been used in literature to model the Yarkovsky effect. Subsection 2.3.3 then describes how the Yarkovsky effect has been measured in literature.

2.3.1. Physical Mechanism

The Yarkovsky effect is an acceleration that acts upon a celestial body if its thermal radiation is non-isotropic. In fact, there are two types of Yarkovsky effect: the diurnal effect, which is related to the rotational period of the body about its own axis, and the seasonal effect, which is related to the orbital period of the body around its primary (typically a star). The following discussions on the mechanisms of these two effects and the factors that influence them are based on Bottke et al. (2006) unless otherwise specified.

Diurnal Yarkovsky Effect

The diurnal effect can be described by a simplified scenario of a uniform spherical body in a circular orbit around its star, with its spin axis perpendicular to the orbital plane. Such a scenario is depicted in Figure 2.2(a). The side facing the star is illuminated and warmed by the star's radiation. Ultimately, any energy absorbed by the body will be re-radiated, typically at infrared wavelengths. The material of the body is not a perfect conductor, so the material on the star-facing side will generally be warmer than on the dark side. However, due to the rotation of the body about its own axis and the thermal inertia of the body's material, this heat distribution will not be aligned with the visible light distribution on the body's surface. The material on the morning side will be relatively cold after spending half a rotation period pointed away from the star, while the part that has just experienced sunset will still be relatively warm after spending half a rotation period absorbing radiation from the star. Therefore, the heat distribution is rotated somewhat as compared to the visible light distribution, with the warmest point being somewhere on the afternoon side. We are familiar with this on Earth, as afternoons are often warmer than midday.

Each infrared photon that is emitted from the body as thermal radiation has a momentum proportional to its frequency (or energy). Because the warmer side emits more photons and higher frequency photons than the cooler side, there is a net push that acts on the body, directed away from the warmer side. If the body is rotating in the prograde sense — that is, in the same sense as its orbital motion around the star, as depicted in Figure 2.2(a) — then there is an acceleration component in the (outward) radial direction and a component in the (positive) along-track direction. The along-track component results in a gradual increase in the semi-major axis of the orbit, which can lead to significant changes over long periods. If the body rotates in the opposite sense (retrograde rotation), there would again be an outward radial acceleration component, but the along-track component would be in the negative direction, meaning the size of the orbit would shrink over time.

Seasonal Yarkovsky Effect

The seasonal effect can be explained with another simplified scenario, as depicted in Figure 2.2(b): the spin axis of the body now lies in the orbital plane, meaning there is no diurnal effect. Although the northern hemisphere receives the greatest solar irradiance at point A, thermal inertia means that there is a delay between this moment and the time at which the northern hemisphere is the warmest, which

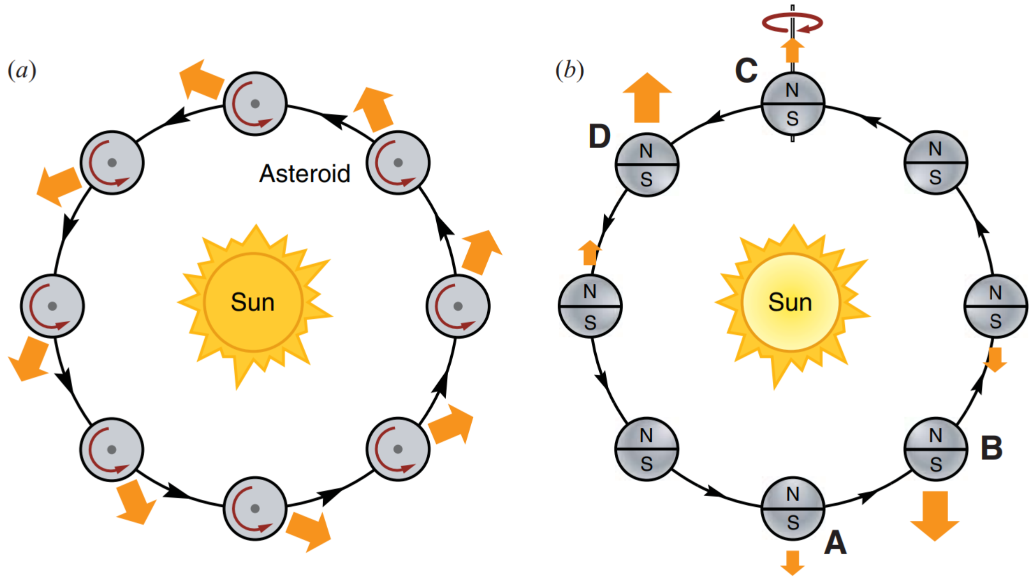


Figure 2.2: (a) Diurnal Yarkovsky effect: greater thermal emissions on the evening side than the morning side of the body means that a body in prograde rotation experiences a net acceleration with positive along-track and outward radial components
(b) Seasonal Yarkovsky effect: maximum thermal emissions at points B and D (rather than A and C) cause net accelerations in the negative along-track direction (Bottke et al. 2006)

occurs at point B. The same happens for the southern hemisphere at points C (maximum irradiance) and D (maximum temperature). The resulting uneven temperature distributions result in net accelerations with negative along-track components, causing orbital decay over time (at least for relatively small orbital eccentricities). Without the lag from the thermal inertia, these accelerations would be symmetric in both directions and would cancel each other out over the course of an orbit. This effect is analogous to seasonal temperature variations on Earth, where the warmest temperatures typically occur in the month or two after the summer solstice.

Influencing Factors

There are quite a number of factors that influence the diurnal and seasonal Yarkovsky effects:

- The **spin rate** about the body's own axis: for the diurnal effect, a very fast spin rate would result in an even heat distribution in the longitudinal direction, and thus no net effect. Zero spin would result in zero diurnal effect and a nonzero seasonal effect.
- The **spin direction** about the body's own axis: as discussed above, for the diurnal effect, prograde rotation leads to a positive along-track acceleration (which increases the size of the orbit), while a retrograde rotation results in a negative along-track acceleration (which decreases the size of the orbit). For the seasonal effect, the direction of rotation does not matter.
- The body's **axial tilt**, or **obliquity**: if the spin axis lies in the orbital plane, there will be zero diurnal effect and maximum seasonal effect. If the spin axis is perpendicular to the orbital plane, the diurnal effect is maximum and the seasonal effect is zero.
- The body's **thermal and radiative characteristics**: both Yarkovsky effects are due to the thermal inertia of the body's material(s), i.e. the time lag involved in heating up and cooling down. Thus, characteristics such as emissivity, albedo, specific heat capacity, and conductivity of the material at and below the surface of the body significantly effect both Yarkovsky effects.
- The body's **orbital period**: while this has no influence on the diurnal effect, it does influence the seasonal effect in the same way the spin rate influences the diurnal effect.
- The **size and mass** of the body: in general, the Yarkovsky effect is most relevant for bodies with diameters between $\sim$0.1 m and $\sim$40 km. A very small body (e.g. a speck of dust) will essentially have a uniform temperature distribution, as heat from the star-facing side can quickly permeate throughout the body, and thus no Yarkovsky effect will be present. Conversely, a very large body (e.g. Earth) has a high mass-to-surface-area ratio, meaning the net force is very weak compared to the mass of the body, making the Yarkovsky effect negligible. For asteroids in the Solar System above 40 km diameter, the gravitational effects of close encounters with large asteroids dominate over the Yarkovsky effect in terms of semi-major axis drift (Delisle and Laskar 2012). For typical Solar

System objects, the diurnal effect is maximum for bodies with sizes on the order of centimetres to metres, and the seasonal effect is maximum for bodies with diameters ∼10 metres (Farinella et al. 1998).

- The body's specific **shape and surface topography**: bodies with non-spherical shapes can have further influences on the Yarkovsky effect, for example if the area exposed to the star is not constant, or if the shape is asymmetric, or if there are craters or lumps that cast shadows and change how energy is absorbed and radiated. Furthermore, the material distribution on and under the surface may not be uniform (e.g. one side may have more reflective material).
- **Solar irradiance**: the amount of energy incident on the surface influences the magnitude of the effects; bodies in the inner Solar System experience a more significant Yarkovsky effect than comparable bodies in the outer Solar System. This is particularly true for the diurnal effect.
- The **eccentricity** of the orbit: related to the influence of solar irradiance, a significant orbital eccentricity will influence the magnitude of the Yarkovsky effect throughout the orbit. The direction of the Yarkovsky accelerations will also be influenced due to the body's motion being non-perpendicular to the radial vector from the star.
- The **interplay with the YORP effect**: the Yarkovsky-O'Keefe-Radzievskii-Paddack (YORP) effect is similar to the Yarkovsky effect as already discussed, but relates to net torques that can be produced by thermal emissions from irregularly shaped bodies. This effect will not be discussed in detail here, but over time, these torques can change the spin rate and axial tilt of small bodies. As mentioned above, the rate, direction, and orientation of the bodies spin vector can all influence the Yarkovsky effect.

2.3.2. Yarkovsky Models in Literature

The Yarkovsky effect can be modelled in many different ways. Similarly to the approach in Subsection 2.2.4, this section will outline some Yarkovsky acceleration models that have been used in literature.

Vokrouhlický (2001) presents quite a fundamental approach to modelling the Yarkovsky effect, with some simplifications that lead to a relatively simple linearised derivation and result. The assumptions include: the body is spherical, the orbit is circular, the body acts as a grey body. The general approach will be outlined here, followed by the eventual acceleration model.

The first step is to find the surface temperature distribution, which is done using heat diffusion equations for the flow of energy inside the body and across its surface. The parameters involved include the material's thermal conductivity, specific heat at constant pressure, density, surface thermal emissivity, and Bond albedo. For each surface element of the body, the solar radiation flux (insolation) must be known; this information follows from the body's shape, orientation, and distance from the Sun. The equations can then be solved numerically to find the temperature of each element.

Following from the temperature calculation and some derivations, the acceleration due to thermal emission can be formulated as an integral over the surface of the whole body:

$$\mathbf{a}_{\text{Yark}}(\zeta) = \frac{2\sqrt{2}}{3\pi}\alpha\Phi \int d\Omega \Delta T'(R', \theta, \varphi, \zeta)\mathbf{n} \tag{2.19}$$

where:

- ζ is a complex variable that replaces time t, defined $\zeta = e^{i\omega_{\text{rev}}(t-t_0)}$
- ω_{rev} is the mean motion of the orbit
- α is one minus the Bond albedo
- Φ is the radiation force factor, defined $\Phi = (\varepsilon_\star \pi R^2/mc)$
- $\varepsilon_\star$ is the radiation flux at the body's distance from the Sun
- R is the radius of the body
- m is the mass of the body
- c is the speed of light
- $d\Omega$ is a surface element
- $\Delta T'$ is a function describing the temperature anywhere in the body at any time ζ, in polar coordinates (r', θ, φ), with $r' \leq R$
- $\mathbf{n}$ is the unit vector normal to the surface

This acceleration can be broken down into two parts: a diurnal component that is mostly dependent on the rotation frequency and a seasonal component that is mostly dependent on the mean motion of the orbit.

The primary effect of the Yarkovsky effect on an asteroid's orbit is a drift in the semi-major axis. The diurnal and seasonal components of the rate of change of the semi-major axis can be expressed separately, if the body is assumed to be spherical and in a circular orbit. Since these expressions are functions of all of the parameters mentioned so far, they can be used to show how the strength of the Yarkovsky effect (manifested as semi-major axis drift) is affected by variations in any of these parameters. One finding that is apparent from these equations is that the seasonal component always results in orbital decay (decreasing semi-major axis), while the diurnal component can result in decay or growth, depending on the rotational sense.

While these equations can be used to model different types of theoretical asteroids and their orbital changes over time, it is often not straightforward to apply them to real asteroids. This is because properties such as thermal conductivity or density are not always precisely known, and the assumptions of spherical shape and circular orbit may not always be suitable either.

Greenberg et al. (2017) presents a simpler and more direct formulation for the Yarkovsky acceleration vector (also used by Greenberg et al. (2020)). The assumptions behind this model include: the body is spherical and acts as a black body (absorbing and eventually re-emitting all incident radiation). The absorbed radiation flux $\dot{E}$ is therefore dependent on solar luminosity $L_\odot$, heliocentric distance r, and the diameter of the body D, but not on any thermal characteristics such as albedo or emissivity:

$$\dot{E} = \frac{L_\odot}{4\pi r^2}\pi\left(\frac{D}{2}\right)^2 \tag{2.20}$$

The total change in momentum $\dot{p}$ of the re-emitted photons is:

$$\dot{p} = \frac{\dot{E}}{c} \tag{2.21}$$

And the acceleration $\ddot{r}$ acting on the body is:

$$\ddot{r} = \frac{\dot{p}}{m} \tag{2.22}$$

Modelling the asteroid as a sphere with diameter D and uniform density ρ, the acceleration can be written as:

$$\ddot{r} = \frac{3}{8\pi}\frac{1}{D\rho}\frac{L_\odot}{c}\frac{1}{r^2} \tag{2.23}$$

However, this does not yet take into account the direction of the re-emitted photons. Rather than modelling a temperature distribution using the asteroid material's thermal conductivity, this model simply assumes that a majority of the photons are re-emitted at a rotational phase that is offset from the subsolar longitude by a fixed amount, called the phase lag φ. The direction of the resultant acceleration is then opposite to the location of this majority re-emission and is expressed using $X_{\hat{\mathbf{z}}}(\varphi)$, a rotation matrix of angle φ about the spin axis $\hat{\mathbf{z}}$, together with the Sun-body vector $\mathbf{r}$:

$$\mathbf{a}_{\text{Yark.}} = \xi\frac{3}{8\pi}\frac{1}{D\rho}\frac{L_\odot}{c}\frac{X_{\hat{\mathbf{z}}}(\varphi)\mathbf{r}}{r^3} \tag{2.24}$$

An efficiency factor ξ is also introduced to account for the fact that not all photons are actually re-emitted with phase lag φ. ξ may be positive or negative, allowing for the modelling of the Yarkovsky effect on asteroids with retrograde and prograde rotations (respectively).

When trying to apply this model to real asteroids, fewer assumptions need to be made: only the diameter, density, and rotation state, if they are not already known. Greenberg et al. (2017) made use of this model to help in the detection of a drift in semi-major axis of asteroid 1566 Icarus due to the Yarkovsky effect, and Greenberg et al. (2020) used it in their detection of Yarkovsky drift in 247 NEAs. It is important to note that, when modelling the semi-major axis drift, ξ is an estimated parameter and thus any inaccuracies in the assumptions made about the asteroid's properties will be 'absorbed' by this parameter. This also obfuscates the physical interpretation of ξ. Because of this, the phase lag φ is set to a constant value of -90°, the position that maximises the transverse component of the acceleration. This does not have an affect on the resulting calculated value of semi-major axis drift.

Fenucci et al. (2024) (whose dynamical model was also discussed in Subsection 2.2.4) uses a very simple model for the Yarkovsky effect: simply a distance-dependent acceleration in the transverse direction $\hat{\mathbf{t}}$:

$$\mathbf{a}_{\text{Yark.}} = A_2\left(\frac{1\text{AU}}{r}\right)^2\hat{\mathbf{t}} \tag{2.25}$$

where r is the heliocentric distance (in AU) and A_2 is an estimated parameter that is determined using a weighted least-squares method to fit the model to the observation data. This model requires no information about the the physical and thermal properties of the body, since all of these are lumped together in the estimated term A_2. This is useful because such properties are often not known. Only the acceleration component in the transverse direction is modelled because this has the greatest effect on the semi-major axis drift of the orbit; the radial component does comparatively little to affect the orbit (Greenberg et al. 2017). However, this model assumes that the magnitude of the acceleration depends only on the heliocentric distance r, being proportional to $1/r^2$, and that the direction is purely transverse. This captures the dependence on semi-major axis and eccentricity, but it neglects seasonal variations, components of the effect in the radial and normal directions, the interaction with the YORP effect, and any thermal characteristics that would cause the transverse acceleration to be proportional to something other than $1/r^2$.

This model was used in the development of a procedure for identifying NEAs for which the Yarkovsky effect may be detectable, estimating a non-gravitational acceleration in the transverse direction, and checking if the acceleration is statistically significant and whether it fits with a Yarkovsky model based on (partially assumed) physical characteristics. The methodology of this paper is outlined in more detail in Subsection 3.2.1.

Chesley et al. (2014) (whose dynamical model was also discussed in Subsection 2.2.4) used three different Yarkovsky models in their research into asteroid 101955 Bennu's orbit. The first is the simplest and is the same as the model used by Fenucci et al. (2024) (described above), with one difference: the acceleration is modelled as proportional to $1/r^2.25$ instead of $1/r^2$, as this was found to provide a better fit given Bennu's physical and thermal properties. The second model is similar to what was formulated by Vokrouhlický (2001) (described above): it solves linearised heat transfer equations for a homogeneous spherical body.

The third and highest fidelity model divides the asteroid into a finite-element mesh of plates and solves non-linear heat transfer equations for the specific radiative scenario that applies to Bennu, until the temperature profiles over orbital revolutions converge. The force (as a function of orbital anomaly) is then calculated from the vector sum of the forces due to the thermal emission on each plate. Acceleration is then calculated using the volume of the asteroid and an assumed bulk density. This approach is only possible when a shape model of the asteroid is available. The authors compared the results from each of the models and believe that the third model provided the most accurate results.

2.3.3. Measurement of the Yarkovsky Effect

The Yarkovsky effect is typically detected by using astrometric data and a dynamical model to produce an orbit estimation (more details about orbit estimation, especially relating to the weighted least-squares method, are provided in Subsection 2.5.1). A parameter related to the Yarkovsky effect is estimated, which accounts for how the semi-major axis of the body has drifted over the observational arc. The strength of the effect is typically reported as the drift rate of the body's semi-major axis, $\mathrm{d}a/\mathrm{d}t$ (often in AU/My), or as the acceleration term A_2, (often in AU/day^2). Some more details from specific examples from literature are provided below.

Greenberg et al. (2017) investigated a number of characteristics of the asteroid 1566 Icarus, including a detection of a semi-major axis drift rate of $\mathrm{d}a/\mathrm{d}t = (-4.62 \pm 0.48) \times 10^{-4}$ AU/My, equivalent to a change of around 60 m per orbit. The authors used radar measurements of Icarus together with optical astrometry spanning an observation arc of 65 years to estimate its orbit.

A subset of the authors wrote a new software program for ephemeris generation, based on Monte (Mission analysis, Operations, and Navigation Toolkit Environment), a proprietary tool developed by JPL with many applications, including trajectory design and spacecraft tracking. The dynamical model used a variable time step integrator (see Subsection 2.5.3) and considered the gravitational effects of the Sun, the major planets, and 24 of the most massive minor planets (including using more detailed gravitational field models during close approaches to these bodies, though details about these gravitational fields are omitted), as well as general relativistic effects. The non-gravitational Yarkovsky acceleration (Equation (2.24)) is also applied. The efficiency factor ξ is estimated alongside Icarus' state (position and velocity). The residuals between the observations and the simulated observations (from the dynamical model) are minimised using Monte's residual minimisation algorithm, which is based on a Kalman filter. The estimated value of ξ is then used to convert to an averaged semi-major axis draft rate, $\mathrm{d}a/\mathrm{d}t$. A number of tests were performed to verify the Yarkovsky detection, including a demonstration that the dynamical model including the Yarkovsky acceleration was significantly better at making orbital predictions than the model excluding the the Yarkovsky acceleration.

As mentioned in the previous section, **Fenucci et al. (2024)** developed a procedure for identifying the Yarkovsky effect in NEAs whilst using a simple Yarkovsky acceleration model (Equation (2.25)). The A_2 term in this model is treated as an estimable parameter, similar to ξ in Greenberg et al. (2017), along with the parameters defining the body's state. In this research, this was done using a weighted least-squares procedure, by way of the ESA Aegis orbit determination software. This procedure detected 348 instances of the Yarkovsky effect on NEAs. The methodology of this paper is outlined in more detail in Subsection 3.2.1.

Dziadura et al. (2023) used astrometric data from the Gaia mission to estimate the Yarkovsky effect for 446 NEAs and $\sim$54000 inner main-belt asteroids, eventually reliably detecting a Yarkovsky effect for 49 NEAs, as well as calculating their bulk densities. Again, the authors of this research estimated the A_2 parameter for the Yarkovsky effect (although the acceleration model dependent on A_2 is not explicitly stated) along with the six parameters describing the body's state. The estimation was done using a linearised least-squares method with differential corrections, using the OrbFit software.

2.4. Data & Observations

Astrometry is the practice of taking measurements of the positions and motions of celestial bodies. There are several different methods by which such measurements can be made, including using telescopes on the ground, observation from space, radar signals, and several other less common or more outdated methods. Before these methods are discussed in Subsection 2.4.2, the primary coordinate system used for angular measurements is explained in Subsection 2.4.1. Next, Subsection 2.4.3 discusses the errors that can arise when making astrometric measurements, and how some of these errors can be circumvented. Finally, Subsection 2.4.4 describes the procedures for processing astrometric data before use, including adjusting for systematic errors and applying weighting schemes to the observations.

2.4.1. Equatorial Coordinate System

The explanations the of coordinate systems in this subsection are based on Wakker (2015, Ch. 11) unless otherwise specified. A common method of taking astrometric observations is to describe the position of a celestial body using two angular measurements, i.e. using spherical coordinates with the radial distance coordinate omitted, as this is not usually directly measurable. These angular measurements are typically made using the celestial sphere as a reference, with the angles called the the right ascension (RA) and declination (DEC). The celestial sphere is an imaginary sphere which is centred on Earth and which has an infinitely large radius. It is divided into northern and southern hemispheres by the celestial equator, which is co-planar with Earth's equator. The celestial poles are the intersection points between Earth's rotation axis and the celestial sphere. Specifically, Earth's rotation axis at the J2000 epoch[5] is used as the standard reference for this in modern data[6]. Each star or other celestial body can be considered to have a position on the 'inner surface' of this sphere. In the RA-DEC system, distant stars have (almost) fixed coordinates. If these coordinates are well-known, RA-DEC measurements of other bodies can be obtained by measuring the positions relative to these "fixed" stars.

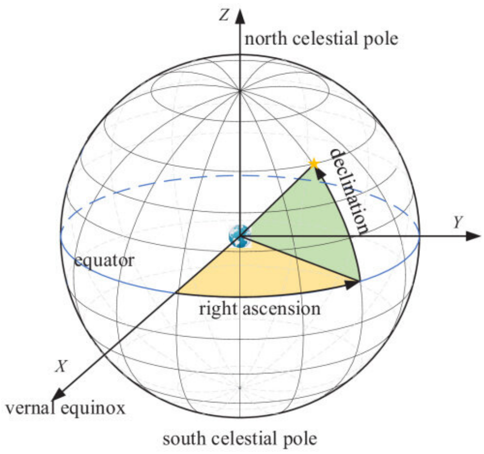


Figure 2.3: A diagram of the equatorial coordinate system using spherical coordinates to describe the location of an object on the celestial sphere (Sun et al. 2023)

Declination is analogous to latitude on Earth's surface, while right ascension is analogous to longitude. Declination (also denoted as δ) is the angular distance between the celestial equator and the object in question, as measured along the great circle which is perpendicular to the celestial equator and coincident with the object. Such a great circle is also called the object's hour circle. Declination values are always between -90° and 90°; values are positive for objects in the celestial northern hemisphere and negative for the southern hemisphere. See the angle of the green sector in Figure 2.3.

Right ascension (also denoted as α) is the angular distance between the vernal equinox and the intersection of the celestial equator with the object's hour circle (measured eastward). See the angle of the yellow sector in Figure 2.3. The vernal equinox (often denoted with the symbol ♈) is an arbitrarily chosen reference direction, analogous to the choice of the Greenwich meridian as the prime meridian in the definition of longitude. The equinox line is the intersection of the planes of the celestial equator and the ecliptic (the plane of Earth's orbit around the Sun). The angle between these planes is $\sim$23.4°. The Earth-Sun vector is coincident with the equinox line twice per year: once in spring and once in autumn[7].

[5]This is a standard reference epoch used in astronomy, defined as January 1 2000, 12:00 Terrestrial Time (TT).

[6]This is necessary because of the motion of Earth's rotation axis over time due to precession, nutation, and other shorter-period motions.

[7]At these times, the Earth's rotation axis is perpendicular to the Earth-Sun vector, meaning the duration of day and night is

The reference direction of the vernal equinox is defined as the direction from the Earth to the Sun during the spring (vernal) equinox in the northern hemisphere, which occurs in March. Right ascension can be measured as an angle from 0° to 360°, but can also be measured in hours, minutes, and seconds, where the full circle is divided into 24 hours, each hour into 60 minutes, and each minute into 60 seconds (Voisey 2018). Angular measurements can also be referred to with units of arcminutes and arcseconds (abbreviated as arcmin and arcsec), respectively.

A complete spherical coordinate system would include the radial distance to the object, but this is typically omitted in astrometric measurements because it is not directly measurable using standard optical methods. Note that this reference frame does not rotate along with Earth's rotation. Distant stars' coordinates are almost constant within the RA-DEC system, but not exactly due to the relative motion of the stars within the Milky Way galaxy. Thanks to the Gaia mission, the positions and proper motions of around 1.5 billion stars are known with very high precision (Gaia Collaboration et al. 2021).

In reality, astrometric observations are taken in a reference frame that is centred on the observer. If a body is relatively close to Earth, two observers on different sides of the Earth taking measurements of the body at the same time will observe the body to be near different 'fixed' stars, resulting in difference RA-DEC measurements. Similarly, the motion of Earth around the Sun will result in different measurements of a body's position at different times of the year. For these reasons, it is important to know the precise time and location at which an observation was taken.

2.4.2. Observational Methods

Astrometric data can be recorded in a number of different ways. This section provides background information on each of the main methods of making astrometric observations (ground- and space-based optical astrometry and radar measurements) and how they work. A number of other methods (e.g. stellar occultations, measurements from probes that have visited asteroids, filar micrometers) can also be used to derive astrometric information, but such data are relatively few in number (Desmars et al. 2013) and will not be discussed here. The Minor Planet Center plays a major role when it comes to collecting and processing astrometric data; this context is explained in the first subsection below.

The Minor Planet Center

The Minor Planet Center (MPC), a division of the International Astronomical Union (IAU), is the worldwide central hub for receiving, processing, and publishing of positional observations of small bodies in the Solar System (asteroids, NEAs, comets, and small satellites of planets). The MPC is based at the Smithsonian Astrophysical Observatory (affiliated with Harvard University), and funding comes from NASA's Near-Earth Object (NEO) Observations program (Minor Planet Center 2026), which itself is a part of the NASA Planetary Defense program.

Any observatory can apply for an MPC observatory code, as long as a certain set of requirements are met. Then any observations made can be reported to MPC, with requirements on the format and accuracy of the data (measured using the post-fit residuals of MPC's own orbit determinations) (Minor Planet Center 2023). Using astrometric measurements sent in from observatories around the world, as well as observations from space telescopes, the MPC performs orbit determinations of the observed bodies. Discoveries that are potentially new objects are placed on a list and given a temporary designation. If the observations match where a known object should be at that time (according to previous observations of that object and propagation to the appropriate time), then they are tagged as such. If it is confirmed that the observations do not correlate with any known object, then the object will be given an official designation and credit for discovery will assigned to the first person(s) to observe the object (Ostro 1997). See Subsection 2.1.5 for more details about this process.

After processing, the MPC publishes the observation data along with the object designations they have determined. As of June 2026, the MPC's database contains over 540 million observations, the earliest of which date back to 1802, with more than a million more added each month (Minor Planet Center 2026). This database and other related data is freely and publicly available for use, and can be accessed online.

Ground-Based Optical Astrometry

Ground-based optical astrometry refers to astrometric measurements which are taken by observatories on Earth's surface using (near-)visible wavelengths of light. These observatories range from large institutions with many full-time researchers to small operations run by amateurs. Typically, the operators of an observatory will have goals for the survey they are currently conducting. These goals will feed into the strategies employed that determine where to point the telescope at a given time. Large areas of interest

(approximately) equal everywhere on Earth, hence *equi-nox*, deriving from the Latin for *equal* and *night* (Online Etymology Dictionary n.d.).

in the sky may be divided into smaller areas, which are searched through sequentially. Directions towards the (centre of the) Milky Way are typically avoided, due to the star fields being too dense and bright for the reliable identification of minor bodies. The information in this paragraph is based on Langbroek (2024) (a personal communication) unless otherwise specified.

Most observatories involved in observing asteroids and other minor bodies make use of reflecting telescopes, with diameters of approximately 25-40 cm for amateur observatories and up to a few metres for larger observatories. Images are most commonly taken using charge-coupled devices (CCDs), which are electronic light-sensing devices which are well-suited to astrometry due to their high sensitivity and dynamic range, as well as the simple process of extracting and processing the astrometric measurements (Desmars et al. 2013). Before CCDs, photographic plates were common, and even earlier, filar micrometers were used to make measurements.

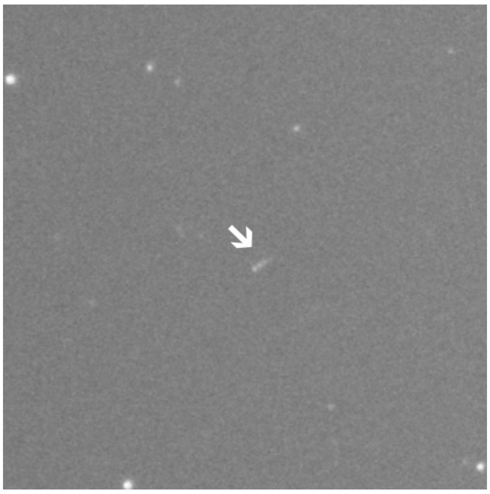


Figure 2.4: Four overlaid images of the minor body 483 Seppina, taken by an amateur astronomer over a span of 60 minutes. If the frames are viewed in sequence, one sees that the motion of the body is towards the bottom-left corner of the frame. (Sagatowski 2023)

Typically, the same area of the sky will be imaged at least 3 times in a row, with exposure times of approximately 1-3 minutes, and with time between exposures up to 20 minutes. The exact exposure times and intervals between exposures depend on the (expected) rate of motion of the target bodies across the sky, the (expected) brightness of the bodies and background stars, the instrument sensitivity, et cetera. By comparing the images taken, points of light that move relative to the background stars can be identified — see Figure 2.4 for an example of the motion of a small body over a period of one hour. The identification of points of light that move across the frame can be done automatically (using image-processing algorithms) or manually. Automatic detection algorithms can save a lot of time but they are susceptible to producing false positives and also to missing detections (especially faint ones), so at least some human judgement is typically needed.

The RA-DEC coordinates of any identified moving objects is then calculated for each exposure. This calculation uses a number of values as input to convert to RA-DEC: the x-y pixel coordinates of the centre of the moving object in the images, the reference/background stars present in the images, a star catalogue to look up the precise coordinates of these background stars, and the plate constants of the image (which take into account any distortions produced by the telescope). The data is then compiled according to the MPC format, which includes a Coordinated Universal Time (UTC) time stamp, RA-DEC coordinates, and brightness for each observation, along with the observatory/station code and the star catalogue used. This data can then be sent to the MPC, where object identification takes place.

For the set of objects that will be considered in this research (see Section 3.2), ground-based optical measurements account for 89.5% of optical observations available via the MPC database (as of June 2026).

Space-Based Astrometry

Another method of collecting astrometric data is via spacecraft. This is primarily done using space telescopes, which operate in similar ways to ground-based telescopes and observatories (in terms of sky-searching strategies, imaging processes, observation processing, and reporting). These space telescopes (including Hipparcos, Hubble Space Telescope, and the James Webb Space Telescope, for example) also have MPC observatory codes, and their observations of small bodies are reported to the MPC. The information in this subsection is based on National Academies of Sciences, Engineering, and Medicine (2019, Ch. 4) unless otherwise specified.

Space-based telescopes have a number of advantages over their terrestrial counterparts. The lack of an atmosphere in space means that observations can be free from atmospheric effects, such as distortion, light pollution, clouds, inclement weather, and the scattering of sunlight during the day, all of which can degrade the quality of astrometric measurements or prevent them entirely. This means that space telescopes can spend much more time in active operation (though much of this time may be dedicated to scientific objectives unrelated to asteroid observation). Furthermore, because ground-based telescopes can only observe at night, they typically can't easily view the region inside Earth's orbit, while space telescopes can.

Space telescopes are also more suited to operating in the infrared range than ground-based telescopes. NEOs have similar temperatures to Earth due to their similar distance from the Sun, meaning ground-based infrared telescopes have a much worse signal-to-noise ratio (SNR) due to the infrared emissions from the atmosphere and the telescope itself. Cooled infrared telescopes in space, however, have proven to be very productive thanks to their high sensitivity. Infrared observations can also provide estimations of asteroid diameters, and are more suited to finding low-albedo objects, which visible light surveys can miss.

There are also a number of disadvantages when it comes to space telescopes. Compared to terrestrial telescopes, they are very expensive, with costs increasing significantly along with aperture size. Space telescopes also have a potentially more limited lifetime, in part because it can be difficult and expensive (or impossible) to maintain or upgrade the hardware. Additionally, whilst ground observatories have fixed positions that can be defined relative to a shape model of the Earth, space telescopes constantly move in their orbits. This adds some complexity to the implementation of space-based observations in orbit estimations, and requires an accurate ephemeris for the space telescope(s).

For the set of objects that will be considered in this research (see Section 3.2), space-based optical measurements account for 10.5% of optical observations available via the MPC database (as of June 2026).

Radar Astronomy

Radar astronomy is a method of observing celestial bodies by means of actively transmitting radio waves in the direction of a body and analysing the received reflected signal (sometimes called the echo). Radar astronomy is carried out using large radio telescopes, such as the 70 m diameter telescope at the Goldstone Observatory or the (now defunct) 305 m diameter Arecibo telescope — these facilities act as both signal transmitters and receivers (Ostro 1997). While other observation methods rely passively on reflected light, radar astronomers can precisely choose the frequency, polarisation, and timing of their signals to perform controlled experiments on the target body (Ostro n.d.).

Radar astronomy can be used to measure the distance (or range) to an object by measuring the round-trip time of the radar pulse. A component of the velocity of the object can also be measured by analysing the Doppler shift of the echo; according to Ostro (1997), these measurements can be accurate to a tenth of a millimetre per second at 20 lunar distances. Radar observations can also be used to construct 2-dimensional models of asteroids, with spatial resolutions down to 10 m, provided the asteroid is close enough for the echoes to be sufficiently strong. With several such observations over a span of time, 3-dimensional models of an asteroid's shape can be constructed, its rotational state can be defined, and information can be derived about its density distribution (Ostro n.d.). This is not possible even for high quality optical telescope images, as the points of reflected visible light are simply too small (Ostro 1997).

The highly accurate measurements of range (accurate to around 3.3 km (Desmars et al. 2013)) and velocity that radar astronomy can provide can be very useful additions to optical observations of asteroids for the purpose of orbit determination. They can greatly reduce the uncertainty in orbit predictions, even with relatively few radar observations (Ostro 1997; Ostro n.d.). However, radar astronomy is heavily limited by range because the power of the reflected echo signal is proportional to $1/4r^2$, where r is the distance to the target object. Furthermore, one must already have a good estimate of where a target object is in order to transmit a radar signal to it, making it less suitable for the discovery of new objects. These factors mean that very few asteroids have been studied or even detected using radar: out of a total of 1.5 million main-belt asteroids, NEAs, and comets that have been designated by the MPC (Minor Planet Center 2026), only 1310 have been observed using radar (JPL 2024), as of June 2026.

2.4.3. Astrometric Errors & Uncertainties

The quality of astrometric observations is not the sole determiner of the precision of the resultant asteroid orbit estimation (Desmars et al. 2013), but observation precision and how the data is processed does have a significant impact on the quality of asteroid orbit estimations (Vereš et al. 2017). The accuracy of asteroid ephemerides can be broken down into internal and external precision (Desmars et al. 2013). Internal precision comes from the numerical models being used (how well the equations of motion represent the true motion), the precision of the numerical integration process (truncation and rounding errors (see Subsection 2.5.3)), and the precision of any methods used to compress the representation of the ephemerides. External precision stems from the observational accuracy and how the observational data is processed, and is thus affected by random and systematic errors that may be introduced in the observation process. Internal precision can be made to be very good through the selection of appropriate acceleration models, numerical integration settings, and other inputs, so most of the remaining uncertainty in asteroid ephemerides is the result of external errors (Desmars et al. 2013).

The quality of optical astrometric observations is dependent on the size, focal length, and part quality of the telescopes used, the pixel resolution of the image-capturing technique, and the local visibility conditions. These aspects can never be fully optimised, however: telescope size and quality is limited by cost (and by spatial and mass requirements for space telescopes), and visibility conditions are often a limiting factor for ground-based astrometry (Vereš et al. 2017). Thus, some amount of error and uncertainty is inherent in any kind of astrometric measurement. While some of these errors are random and unpredictable, there are also some systematic errors and biases which can sometimes be corrected for. The sources of these errors, their importance, and how they can be avoided/corrected for is discussed below.

Random Errors

Random errors occur in all measurement processes. They are the result of random, noisy phenomena, and they are different from one measurement to the next. The effects of noisy data can be reduced by taking more measurements, thereby leading to more accurate results.

There are many different factors which can contribute to random error within a given astrometric measurement. For ground-based optical measurements, air turbulence causes rays of light to bend slightly differently along their paths through the atmosphere, which can cause points of light to become more spread out or to shift slightly. Background noise in the images can be a result of photons leaking into the telescope from the atmosphere (due to light pollution, for example) or from thermal noise in the equipment. This type of noise can also cause some points of light to appear to be shifted to one side. Telescope tracking systems are not perfect, so the images of bodies may not be perfectly round, making the identification of their centres (also called centroids) difficult. Additionally, CCDs have pixels and thus provide a dicretised representation of the true continuous image being observed, meaning the identification of the position of the centroid is limited by the pixel size, especially if only $\sim$1 pixel is illuminated by the body. It can be even more difficult to identify the centroid of an asteroid if its motion causes a trail across the image (Owen 2000).

Some of these random errors can be reduced by changing how the measurements are taken. Errors from atmospheric disturbances can be reduced by increasing the exposure time (40% reduction in noise when doubling exposure time), though this is of course limited by the point at which stars would become overexposed or asteroids would leave trails in the image (Owen 2000). Taking multiple exposures of the same patch of sky can also be used to reduce noise: four observations can be combined to produce a result with half the random error that would result from a single observation (the final error is inversely proportional to the square root of the number of observations) (Owen 2000).

An important aspect of dealing with random errors is determining or estimating the extent of the observation noise and taking this into account when performing an orbit determination. This is often done on an observatory-by-observatory basis, since random errors can be dependent on the telescope quality and location, which are constant for a given observatory (excluding telescope upgrades or replacements). These assumed errors can be used to assign weights to the observations in the orbit-fitting process. See Subsection 2.4.4 for more details on observation weighting schemes.

Systematic Errors

Systematic errors are those which affect a set of measurements in the same way. While taking additional measurements can help with reducing the effects of random errors, the same is not true for systematic errors. However, if a systematic error is discovered and can be quantified, it can be corrected for in all measurements to which it applied.

There are several different potential sources of systematic errors that may apply to a given set of astrometric measurements. One major source of error is from star catalogues with biases (or certain biased zones of star catalogues): the calculation of the RA-DEC of a body observed in an image makes use of the relative positions of the background stars in the image. If there is a bias in the RA-DEC values of the reference stars in the catalogue used, that would introduce a bias into the RA-DEC values for the observed body too. Old star catalogues (pre-2000) could contain biases greater than 1 arcsecond (Owen 2000). Although more recent catalogues have improved upon these errors, even as recently as 2013, Desmars et al. (2013) found that "bias in stellar catalogues remains the main cause of uncertainty" when it comes to asteroid ephemerides. Random errors that are introduced in the creation of star catalogues will also eventually lead to systematic errors in measurements that use those star catalogues as reference.

Another source of systematic error comes from the fact that measured objects are not always viewed from the side illuminated by the Sun. The light recorded in an image may be from just the illuminated crescent of the body. This causes an offset between the centre of light and the centre of mass (Owen 2000). If the reported measurements are treated as the positions of the center of mass when it comes to orbit estimation, then a systematic error will be introduced.

Further systematic errors may stem from an inaccuracy in the reported position of the observatory, or from taking time measurements from a clock that is not perfectly synced with UTC (Langbroek 2024; Owen 2000). These have a parallax effect on the measurements taken, as they will be treated as though they were taken from a different time and/or location than they actually were, where the true measurements would have been different. This effect is most significant when observing objects that are particularly close to Earth.

As with the random errors, some methods can be applied to reduce some of the sources of systematic error. The effect of the random error in star catalogues can be reduced by taking into account more reference stars when calculating the RA-DEC of the asteroid(s). This may require using a wider field of view or taking a mosaic of images and then stitching them together (Owen 2000). Biases in old star catalogues can be very well accounted for by comparing them to more recent and more accurate star catalogues, such as those provided by the high quality astrometric data from the Gaia mission. These can then be used to debias old measurements, as long as the star catalogue used as reference was also recorded (Eggl et al. 2020). Biases in current-day star catalogues can be very difficult to avoid, but the most recent catalogues provided by Gaia data are extremely accurate, and all but eliminate star-catalogue-related errors in current and future astrometry (Eggl et al. 2020). Biases from incorrect position or time reporting can also be difficult to correct for, unless and until the bias can be noticed or measured, after which retroactive corrections can be applied to old measurements.

2.4.4. Initial Data Processing

It is clear from Subsection 2.4.3 that the physical aspects of how astrometric measurements are taken (telescope size, quality of parts, pixel resolution, and observation conditions) affect the eventual quality of astrometric measurements, as do other aspects such as which star catalogues are referenced and how accurately the time and location of the measurement are reported. When these measurements are used to construct estimations of asteroids' orbits, the quality of the data is a significant factor affecting how reliable the orbit estimations will be (Vereš et al. 2017). However, even with less-than-ideal input data quality, effective statistical treatment of the data can lead to improved results. This treatment often includes debiasing data that has been affected by systematic errors in star catalogues and applying weighting schemes so that the most reliable data is more highly valued in the orbit estimation process (Vereš et al. 2017).

Various debiasing and weighting schemes have been developed over the years, but two recent and commonly used schemes will be discussed here (both schemes are used by Farnocchia et al. (2021), Reddy et al. (2022), Hung et al. (2023), and Dziadura et al. (2023), to name a few examples). The weighting scheme is the one presented in Vereš et al. (2017) and the debiasing scheme is the one presented in Eggl et al. (2020).

Weighting Scheme

The format in which most MPC observations are reported and stored does not contain any information about the uncertainties in the measurements[8]. This is not ideal, as it is always true that some measurements are much more accurate than others, so not all data should be treated equally. Astrometric uncertainties therefore need to be assumed when using this data to perform orbit estimations. Therefore, the authors of Vereš et al. (2017) performed statistical analyses on the astrometric data from 13 major asteroid CCD surveys. The data was debiased prior to analysis. They investigated the residuals of these observations as a function of reported magnitude, epoch, and rate of motion. The magnitude of the target body can affect the residuals because low magnitudes correspond with low SNR and high magnitude objects may cause overexposure, both of which increase uncertainty. The observation epoch is relevant

[8]A newer format, the Astrometry Data Exchange Standard (ADES), was introduced which does allow observers to report uncertainties, SNR, et cetera (International Astronomical Union 2024). While this has been the MPC's preferred format for observations since 2018, it is not mandatory to use (Minor Planet Center 2018) and older observations are of course still in the older format.

because asteroid surveys can change over time through equipment upgrades, changes to measuring algorithms, and changes to the star catalogue used. Fast rate of motion across the sky can cause trails in the images, making the determination of position more difficult and prone to error. Based on how these factors affected residuals for the different surveys, a weighting scheme was developed. The details of how this weighting scheme is applied to the orbit determination process are given in Subsection 2.5.1.

The weights are defined as $1/\sigma^2$, where σ is the observation uncertainty, measured in arcseconds ($''$). The dependencies on brightness and rate of motion are not part of the final weighting scheme. The time/epoch dependence is relevant for three of the stations, so they are assigned one uncertainty value for observations before a particular epoch and another value for observations after that epoch. The weights for each survey are set according to the highest value of the RMS of the residuals as a function of target brightness (excluding some outliers). The most active stations and some selected follow-up observers are assigned σ values ranging from $0.1''$ to $1.0''$ (sometimes dependent on the star catalogue used), while other smaller stations get a blanket value of $\sigma = 1.0''$ (or $1.5''$ if the star catalogue used is not known). Uncertainty values are also given for non-CCD observations.

There appears to be a small error in the values reported in Table 3 of this paper. Based on the middle panels of Figure 6 and how the σ values are chosen, the reported σ values for stations 608 and J75 appear to be flipped, reporting values of $0.6''$ and $1.0''$, respectively, while they should be $1.0''$ and $0.6''$, respectively.

In order to avoid the introduction of unresolved systematic errors, a scaling factor is applied to the uncertainty values of observations which are part of a batch of more than four observations of an object in a single night from the same station. The scaling factor is $\sqrt{N/4}$, where N is the number of position observations, and it is only applied when $N \geq 5$.

The authors of the paper validated this weighting scheme on $\sim$200 multi-apparition (at least 5) NEOs. They did this by comparing orbital solutions that use observations from just 1 or 2 apparitions against orbital solutions using observations from all apparitions. They found that the scheme is somewhat conservative, but that it is better at handling outliers and faint detections and it reduces the need to manually set weights, as compared to previous weighting schemes. In terms of orbit predictions, the primary effect of the weighting scheme is on the uncertainty of the prediction, and not on the prediction itself.

Debiasing Scheme

As discussed in Subsection 2.4.3, systematic errors can be introduced when biased star catalogues are used as references when determining the RA-DEC measurements of observed bodies. These errors reduce the quality of the data and thereby the quality of the orbital fits that are calculated using that data, so it is desirable to have a method by which these errors can be reduced or removed. The authors of Eggl et al. (2020) used the second release of data from the Gaia mission (Gaia DR2) to identify biases in 26 star catalogues that have been used for small body astrometry. Based on this, they derived a scheme that can be used to remove these biases from old astrometric observations. This article is an update and extension of previous work (Farnocchia et al. 2015a) which used similar techniques but used a different star catalogue as its reference, as Gaia data was not available at that time. Eggl et al. (2020) updated the corrections provided by Farnocchia et al. (2015a), and included corrections for an additional 6 star catalogues.

The authors of Eggl et al. (2020) divided the celestial sphere into 49,152 tiles of equal area (around 0.8° by 0.8° each). In each tile, they compared the positions and proper motions (rate of change of position, not included in all star catalogues) of the stars given by the various star catalogues to the Gaia DR2 catalogue to identify systematic differences. The corrections for a given tile for position and proper motion are the median of the differences between the star catalogue's values and the Gaia DR2 values for the stars in that tile. These corrections can vary significantly between tiles, especially for tiles in different regions of the sky. The corrections identified are valid at the J2000 epoch, and are given as $\Delta\alpha_{J2000}$, $\Delta\delta_{J2000}$, $\Delta\mu_\alpha$, and $\Delta\mu_\delta$ for RA, DEC, and proper motion of RA and DEC, respectively. The position corrections are reported in milliarcseconds (mas) and the proper motion corrections are reported in milliarcseconds per year (mas/year). An astrometric observation that was taken at a time t and which used one of the 26 analysed star catalogues can be debiased by subtracting $\Delta\alpha$ and $\Delta\delta$, defined:

$$\begin{aligned} \Delta\alpha &= \Delta\alpha_{J2000} + \Delta\mu_\alpha \cdot \Delta t \\ \Delta\delta &= \Delta\delta_{J2000} + \Delta\mu_\delta \cdot \Delta t \end{aligned} \tag{2.26}$$

from the reported RA and DEC values, respectively. Δt is defined as the time between J2000 and the observation time t, in years. The correction values should be taken from the table corresponding to the star catalogue used for the observation and the row corresponding to the relevant area of the sky. These corrections are often on the order of 10^1 to 10^2 mas for position and 1 mas/year for proper motion, but can be as large as 1600 mas and 18 mas/year for certain star catalogues and certain regions of the sky.

In a similar process to that used by Farnocchia et al. (2015a) and Vereš et al. (2017), the proposed debiasing scheme was validated by performing orbit determination and prediction on a set of 700 multi-apparition NEOs. Although the quality of the star catalogue corrections is significantly better than previous work, applying it in the orbit determination process results in orbit fits that are only incrementally better than applying the debiasing scheme developed by Farnocchia et al. (2015a), in terms of predicting the error in the semi-major axis.

2.5. Orbit Estimation

Orbit estimation is the process in which observations of a celestial body are used to calculate the values of certain parameters relating to its orbit that allow the orbit to best match the observations. This draws on the observation data discussed in Section 2.4 and employs the dynamical modelling discussed in Sections 2.2 and 2.3.

The weighted least-squares process is outlined and described mathematically in Subsection 2.5.1. The steps needed to analyse the covariance and uncertainty properties of a weighted least-squares estimation are covered in Subsection 2.5.2. Finally, Subsection 2.5.3 contains an overview of methods that can be used to numerically integrate the equation of motion, including a discussions about errors, types of integrators, and integrator selection.

2.5.1. Weighted Least-Squares

The purpose of a weighted least-squares (WLS) estimation is to determine the values of a set of orbital parameters such that the orbit propagated using those parameters is the best possible fit to the observation data. Weights are applied to the observations because they may come from various sources or methods and so their reliability typically varies. It is an iterative process in which the estimated parameters are updated each iteration, ideally converging to the values which result in the best fit to the data. The 'best fit to the data' is the one for which the sum of the squares of the (weighted) residuals in minimised. A residual is the difference between the actual observation and the modelled observations. Residuals are also sometimes referred to as O-C, from Observed minus Calculated.

WLS orbit estimation is suited to situations where real-time solutions are not required and restrictions to memory and computation time are not too stringent, which is often the case in mission planning and scientific contexts. WLS estimation is a well-established and reliable method of orbit estimation for the purposes of determining the orbits of asteroids (see, for example Fenucci et al. 2024; Dziadura et al. 2023; Hung et al. 2023; Li et al. 2023; Reddy et al. 2022; Kretlow 2022). This method was developed by Carl Friedrich Gauss in the late 18th century. The explanations, mathematical process, and notation in this subsection and the following subsection are from Montenbruck and Gill (2001, Ch. 8) unless otherwise specified; a more complete mathematical description of the process can be found in that source.

The general process of WLS orbit estimation is outlined in the list below. For more detail on the implementation of points 1-5, see Subsection 3.1.3. The mathematical details of the iterative WLS estimation process (point 6.) are presented after the list.

1. Retrieve the observation data $\mathbf{z}$.
2. Apply any necessary debiasing settings to correct for known systematic errors in the observation data (see Subsection 2.4.4).
3. Define the weight matrix $\mathbf{W}$ by assigning weights to the observations according to their known or assumed errors (see Subsection 2.4.4).
4. Define the dynamical model, including the numerical propagation settings (see Subsection 2.5.3).
5. Define the parameters to be estimated $\mathbf{x}$ and their initial values $\mathbf{x}(t_0) = \mathbf{x}_0$.
6. Perform the iterative weighted least-squares estimation:
   - Numerically propagate the state over the relevant time span using the dynamic model and the initial parameter vector $\mathbf{x}_0$
   - Calculate the modelled observations $\mathbf{h}$
   - Calculate the residuals Δz
   - Calculate the Jacobian matrix $\mathbf{H}$
   - Solve the normal equations for $\Delta\mathbf{x}_{\text{lsq}}$
   - Update the initial state $\mathbf{x}_0$ by adding $\Delta\mathbf{x}_{\text{lsq}}$
   - Check for convergence (or another stopping condition)

The estimation of the trajectory and of various parameters relating to the force and measurement models of a target body can be performed together with the vector $\mathbf{x}(t)$ defined as:

$$\mathbf{x}(t) = \begin{pmatrix} \mathbf{r}(t) \\ \mathbf{v}(t) \\ \mathbf{p} \\ \mathbf{q} \end{pmatrix} \tag{2.27}$$

which is time-dependent and m-dimensional vector containing the body's position $\mathbf{r}$, its velocity $\mathbf{v}$, and the free parameters $\mathbf{p}$ and $\mathbf{q}$, which affect the force and measurement models respectively. This may be referred to as the state vector or the parameter vector. The ordinary differential equation:

$$\dot{\mathbf{x}} = f(t, \mathbf{x}) \tag{2.28}$$

describes the time evolution of $\mathbf{x}$, together with the value of $\mathbf{x}$ at the initial epoch t_0: $\mathbf{x}(t_0) = \mathbf{x}_0$. The observation data consists of n measurements taken at times $t_1, t_2, \ldots, t_n$. These are stored in the vector $\mathbf{z}$, defined as:

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix} \tag{2.29}$$

The ith observation is described by the equation:

$$z_i(t_i) = g_i(t_i, \mathbf{x}(t_i)) + \varepsilon_i = h_i(t_i, \mathbf{x}_0) + \varepsilon_i \tag{2.30}$$

where g_i is the value of the modelled observation (as a function of the time t_i and state $\mathbf{x}(t_i)$), and ε_i represents the difference between the actual observation z_i and the modelled observation g_i. ε_i is typically assumed to be randomly distributed with a mean of zero. h_i is similar to g_i but is a function of the time t_i and the initial state $\mathbf{x}_0$. Equation (2.30) can be written as a vector equation for all i as:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}_0) + \boldsymbol{\varepsilon} \tag{2.31}$$

The least-squares orbit determination problem is to find the initial state $\mathbf{x}_0 = \mathbf{x}_0^{\text{lsq}}$ that minimises the loss function $J(\mathbf{x}_0)$, defined as:

$$J(\mathbf{x}_0) = \boldsymbol{\rho}^T \boldsymbol{\rho} \tag{2.32}$$

where $\boldsymbol{\rho}$ is the residual vector, defined as the difference between the actual observation vector $\mathbf{z}$ and the modelled observation vector $\mathbf{h}(\mathbf{x}_0)$:

$$\boldsymbol{\rho} = \mathbf{z} - \mathbf{h}(\mathbf{x}_0) \tag{2.33}$$

Note that this is the case only if all observations are assumed to have equal weights. Weighting will be dealt with further ahead. Also note that the number of observations n must be greater than or equal to the number of parameters m in order for there to be a unique solution.

Finding $\mathbf{x}_0 = \mathbf{x}_0^{\text{lsq}}$ is not immediately straightforward because the function $\mathbf{h}$ is highly non-linear. However, using an approximate value of $\mathbf{x}_0^{\text{apr}}$, all quantities can be linearised around the reference state $\mathbf{x}_0^{\text{ref}}$ (which is initially given by $\mathbf{x}_0^{\text{apr}}$). The residual vector $\boldsymbol{\rho}$ can thus be approximated as:

$$\begin{aligned} \boldsymbol{\rho} &= \mathbf{z} - \mathbf{h}(\mathbf{x}_0) \\ &\approx \mathbf{z} - \mathbf{h}(\mathbf{x}_0^{\text{ref}}) - \frac{\partial \mathbf{h}}{\partial \mathbf{x}_0}\left(\mathbf{x}_0 - \mathbf{x}_0^{\text{ref}}\right) \\ &= \Delta z - \mathbf{H}\Delta\mathbf{x}_0 \end{aligned} \tag{2.34}$$

where $\Delta z = \mathbf{z} - \mathbf{h}(\mathbf{x}_0^{\text{ref}})$ is the difference between the actual observations and the observations modelled using $\mathbf{x}_0 = \mathbf{x}_0^{\text{ref}}$, and $\Delta\mathbf{x}_0 = \mathbf{x}_0 - \mathbf{x}_0^{\text{ref}}$. The Jacobian matrix $\mathbf{H}$ is defined as:

$$\mathbf{H} = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}_0}\right|_{\mathbf{x}_0 = \mathbf{x}_0^{\text{ref}}} \tag{2.35}$$

Equation (2.34) provides a way to predict the residuals $\boldsymbol{\rho}$ after applying a correction $\Delta\mathbf{x}_0$ to the reference state and using this updated reference state to calculated the modelled observations $\mathbf{h}$.

The orbit determination problem can now be re-formulated using the linear least-squares loss function, where the goal is to find $\Delta\mathbf{x}_0^{\text{lsq}}$ that minimises the loss function $J(\Delta\mathbf{x}_0)$, defined as:

$$J(\Delta\mathbf{x}_0) = \left(\Delta z - \mathbf{H}\Delta\mathbf{x}_0\right)^T \left(\Delta z - \mathbf{H}\Delta\mathbf{x}_0\right) \tag{2.36}$$

It can be shown that $\Delta\mathbf{x}_0^{\text{lsq}}$ can be found by solving the m-dimensional normal equations using standard techniques for positive definite linear systems of equations:

$$\mathbf{H}^T\mathbf{H}\Delta\mathbf{x}_0^{\text{lsq}} = \mathbf{H}^T\Delta z \tag{2.37}$$

Because $\mathbf{h}$ is highly non-linear, the solution obtained from minimising the linearised loss function (i.e. $\mathbf{x}_0^{\text{lsq}} = \mathbf{x}_0^{\text{ref}} + \Delta\mathbf{x}_0^{\text{lsq}}$) is not the actual solution of the orbit determination problem. However, this value of $\mathbf{x}_0^{\text{lsq}}$ may be substituted for the reference value $\mathbf{x}_0^{\text{ref}}$ and the procedure can be iterated (repeated) until a certain tolerance is met. Alternatively, the procedure may stop when a pre-defined maximum number of iterations have been performed. The Jacobian matrix $\mathbf{H}$ should be updated for each iteration for optimal convergence.

Thus far, observation weights have not been taken into account and all observation have been considered to be of equal quality. The treatment of these observation weights is now introduced. The weight of each observation i is defined at the inverse of the square of the measurement error σ_i. The values of σ_i may be known or assumed based on a pre-defined weighting scheme such as the one discussed in Subsection 2.4.4. The weight matrix $\mathbf{W}$ is a diagonal matrix with the weights in the diagonal entries:

$$\begin{aligned} \mathbf{W} &= \text{diag}\left(\sigma_1^{-2} \quad \sigma_2^{-2} \quad \dots \quad \sigma_n^{-2}\right) \\ &= \begin{pmatrix} \sigma_1^{-2} & & 0 \\ & \ddots & \\ 0 & & \sigma_n^{-2} \end{pmatrix} \end{aligned} \tag{2.38}$$

In this formulation, the measurements errors are assumed to be uncorrelated. The solution to the weighted least-squares problem may then be written as

$$\Delta\mathbf{x}_0^{\text{lsq}} = \left(\mathbf{H}^T\mathbf{W}\mathbf{H}\right)^{-1}\left(\mathbf{H}^T\mathbf{W}\Delta z\right) \tag{2.39}$$

2.5.2. Covariance & Uncertainty Analysis

If measurement and model errors are absent, the solution to the weighted least-squares method, $\mathbf{x}_0^{\text{lsq}}$, is equal to the true state. In this case, all residuals would also be zero. Of course, this is never the case in practice, as observations always contain some amount of measurement error. Let us investigate how these errors affect the solution to the WLS orbit estimation. For this purpose, the observation vector may be defined as:

$$\mathbf{z} = \mathbf{h}(\mathbf{x}_0) + \boldsymbol{\varepsilon} \tag{2.40}$$

where $\mathbf{x}_0$ is the true state at t_0 and $\boldsymbol{\varepsilon}$ are the measurement errors. The solution to the linearised least-squares problem is then:

$$\mathbf{x}_0^{\text{lsq}} = \mathbf{x}_0 + \left(\mathbf{H}^T\mathbf{W}\mathbf{H}\right)^{-1}\left(\mathbf{H}^T\mathbf{W}\boldsymbol{\varepsilon}\right) \tag{2.41}$$

Clearly, least-squares solution $\mathbf{x}_0^{\text{lsq}}$ is not equal to the true value of the state $\mathbf{x}_0$ when measurement errors $\boldsymbol{\varepsilon}$ are present, and this difference depends on the measurement errors.

If these measurement errors are assumed to contain no systematic errors (and thus a mean or expected value of zero, i.e. $E(\boldsymbol{\varepsilon}) = 0$), then the expected value of the least-squares solution $\mathbf{x}_0^{\text{lsq}}$ is equal to the true state $\mathbf{x}_0$:

$$E(\mathbf{x}_0^{\text{lsq}}) = \mathbf{x}_0 + \left(\mathbf{H}^T\mathbf{W}\mathbf{H}\right)^{-1}\left(\mathbf{H}^T\mathbf{W}E(\boldsymbol{\varepsilon})\right) = \mathbf{x}_0 \tag{2.42}$$

If the measurement errors are further assumed to be uncorrelated, then the covariance of the least-squares solution $\mathbf{x}_0^{\text{lsq}}$ is:

$$\text{Cov}\left(\mathbf{x}_0^{\text{lsq}}, \mathbf{x}_0^{\text{lsq}}\right) = E\left((\mathbf{x}_0^{\text{lsq}} - \mathbf{x}_0)(\mathbf{x}_0^{\text{lsq}} - \mathbf{x}_0)^T\right) \tag{2.43}$$

The variance of a random variable y is the expected value of the square of the difference between y and the mean value $\bar{y} = E(y)$, i.e. $\text{Var}(y) = E\left((y - \bar{y})^2\right) = \sigma^2$. The square root of the variance is σ, called the standard deviation of y. The covariance of two random variables y and z is $\text{Cov}(y, z) = E\left((y - \bar{y})(z - \bar{z})\right)$. The covariance of a vector $\mathbf{y}$ of random variables with itself, $\text{Cov}(\mathbf{y}, \mathbf{y})$, produces a covariance matrix where the diagonal components are the variances of the random variables and the off-diagonal components are the covariances of the random variables.

Substituting Equation (2.41) into Equation (2.43) and simplifying, the covariance matrix can be shown to be:

$$\mathbf{P}_0 = \text{Cov}\left(\mathbf{x}_0^{\text{lsq}}, \mathbf{x}_0^{\text{lsq}}\right) = \left(\mathbf{H}^T\mathbf{W}\mathbf{H}\right)^{-1} \tag{2.44}$$

which is the inverse of the normal equations matrix. This is true given that the weight matrix $\mathbf{W}$ is defined as in Equation (2.38). The diagonal components of the covariance matrix are the variances (σ^2) of the estimated parameters. Thus the square roots of these yield the standard deviations σ of the estimated parameters — this may be referred to as the formal error. The off-diagonal components are the covariances of the estimated parameters: positive covariances indicate positive relationships between parameters, negative covariances values indicate negative relationships, and zero indicates no relationship.

The correlation between two parameters $\mathrm{Cor}(\mathbf{x}_i, \mathbf{x}_j)$ is essentially a normalised version of the covariance, taking a value between -1 and 1, and indicating the direction and strength of the relationship between the parameters. The correlation matrix and can be obtained by calculating:

$$\mathrm{Cor}(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j)}{\sigma_i \sigma_j} \tag{2.45}$$

for each entry (i, j) in the covariance matrix. The diagonal entries of this matrix will all be equal to 1.

A number of assumptions have been made thus far: the measurement errors have a mean of zero (i.e. no systematic errors are present); measurement errors are uncorrelated; the measurement uncertainties chosen for the weight matrix are a true representation of the standard deviations of the observation errors; the dynamic model is complete and sufficient; and the distributions of the estimated parameters are Gaussian.

These assumptions may not always be entirely valid. Unmodelled or uncorrected systematic errors may in fact be present in the observation data, and certain observations may in fact be correlated with each other, for example consecutive measurements of an object from a single observatory in one night. The measurement uncertainties used to assign weights in the weight matrix are often based on relatively crude assumptions. The covariance matrix will also not reflect any model errors that may exist (i.e. errors due to deficiencies or inaccuracies in the acceleration models or the numerical integration method). Siltala and Granvik (2017) found that probability distributions of estimated parameters can differ significantly from Gaussian distributions (in this case, the relevant estimated parameter was the mass of the target body). For these reasons, it is often the case that the covariance matrix is overly optimistic about the errors of the orbit estimation.

The covariance matrix $\mathbf{P}_0$ in Equation (2.44) describes the covariances of the estimated parameters in $\mathbf{x}$ at time t_0. However, many of these parameters are time-varying (dynamical parameters), such as the body's position $\mathbf{r}$ and velocity $\mathbf{v}$ vectors and it is interesting to know their covariances at other times. The following equations are from Tapley et al. (2004).

The state transition matrix $\mathbf{\Phi}$ allows for the calculation of the approximate state $\overline{\mathbf{x}}$ at any time t_j based on the value of the state at time t_i:

$$\overline{\mathbf{x}}(t_j) = \mathbf{\Phi}(t_j, t_i)\mathbf{x}(t_i) \tag{2.46}$$

With this, it can be shown that the approximate covariance $\overline{\mathbf{P}}$ at time t_j is:

$$\overline{\mathbf{P}}_j = \mathbf{\Phi}(t_j, t_i)\mathbf{P}_i\mathbf{\Phi}(t_j, t_i)^T \tag{2.47}$$

This allows for the calculation of the approximate covariances of the dynamical parameters at any time t_j, using the initial covariance matrix $\mathbf{P}_0$ (Equation (2.44)) and the appropriate state transition matrix $\mathbf{\Phi}(t_j, t_0)$.

2.5.3. Numerical Integration

The propagations that are performed in each iteration of the WLS orbit estimation process (described in Subsection 2.5.1) require an acceleration model that defines the equation of motion of the target body, as well as a method that determines how the motion over time is numerically integrated. There are two parts to this: the propagator and the integrator. The selection of these is linked and is highly dependent on the motion involved and the time frame of the scenario being modelled.

Propagator

The propagation scheme is the way in which the state is defined and the formulation of the equations used to find the state derivative. All propagation schemes are mathematically equivalent to each other, but in practice some are more appropriate than others, depending on the situation. The following descriptions of propagation techniques are based on those in Wakker (2015, Sec. 20.6, Ch. 22) and Dirkx and Mooij (2019).

There are several methods for propagating the state, each with its own advantages and disadvantages. A general propagation scheme can be written as:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t; \mathbf{p}, \mathbf{u}) \tag{2.48}$$

where $\mathbf{p}$ represents the environmental parameters (e.g. positions of bodies not being propagated, gravity fields, et cetera) and $\mathbf{u}$ represents control parameters (mostly applicable to spacecraft which can produce thrust and thus not relevant to asteroidal dynamics). The vector $\mathbf{x}$ can take on many forms, as long as the elements therein fully describe the state of the bodies to be propagated. The simplest of these, the Cowell method, is explained below.

Using the Cowell method, the state $\mathbf{x}$ and its time derivative $\dot{\mathbf{x}}$ are defined as:

$$\mathbf{x} = \begin{pmatrix} \mathbf{r} \\ \dot{\mathbf{r}} \end{pmatrix}, \quad \dot{\mathbf{x}} = \begin{pmatrix} \dot{\mathbf{r}} \\ \ddot{\mathbf{r}} \end{pmatrix} \tag{2.49}$$

where $\mathbf{r}$, $\dot{\mathbf{r}}$, and $\ddot{\mathbf{r}}$ each contain three elements: one for each Cartesian direction. Equation (2.49) is representative of a situation where just one body's motion is propagated. If n bodies' motions are to be propagated together, then their states are simply concatenated together (such that $\mathbf{x}$ contains $6n$ elements). Not all bodies in a system need to be propagated if the positions of some are already known to a good degree of accuracy (for example, through planetary ephemerides), and as long as the motion of the propagated bodies would not have a significant effect on these bodies. As long as an initial state $\mathbf{x}_0$ is known, the state can be propagated over time by directly using the equation of motion as defined in Equation (2.17). Such an equation of motion can be used directly, as it is already expressed in terms of Cartesian vectors.

Advantages of the Cowell method include that it is straightforward and directly implements simple formulations of the equations of motion. It also makes no prior assumptions about the motion of the propagated body and thus is not susceptible to issues should those assumptions be broken. The formulation is also robust, in the sense that the occurrence of singularities are not a major concern. This method has a notable advantage over other schemes when the motion is highly perturbed, i.e. far from a Kepler orbit. However, this scheme typically results in very large values in the state and state derivative due to the direct use of Cartesian components, which can lead to the accumulation of rounding errors. The large and fast oscillation of these values also mean that the time step has to stay quite small (thus allowing more rounding error to accumulate) and can be difficult to adapt when using a variable-step integrator.

Other propagation schemes include Encke's method and the method of variation of orbital elements (Kepler elements). These will only be described briefly here. Encke's method makes the assumption that the propagated orbit is very close to a Kepler orbit and thus propagates only the difference between the Kepler orbit and the actual perturbed orbit. This overcomes the issues with the Cowell method relating to the large and fast-changing values in the state and state derivative, but the advantages diminish if the assumption of the orbit deviates too far from the idealised Kepler orbit. The state vector may also be formulated as a set of Keplerian elements: semi-major axis a, eccentricity e, inclination i, argument of periapsis ω, right ascension of the ascending node Ω, and true anomaly θ. An advantage of this method is that for Keplerian orbits (or near-Keplerian orbits), the Keplerian elements are constant (or near constant), aside from the true anomaly θ, making the solution more stable in general. However, this formulation can result in several singularities: for example, if $e \approx 0$, $e = 1$, $e = \infty$, $i \approx 0$, or angular momentum $H = 0$.

Integrator

A numerical integrator is a scheme for solving differential equations and is typically used when finding an analytical solution is not possible or feasible. In general, the problem will be of the form:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, t) \qquad \mathbf{x}(t_0) = \mathbf{x}_0 \tag{2.50}$$

That is, there is an equation for the state derivative that represents the motion and a known initial state vector $\mathbf{x}_0$, i.e. the state at the time t_0. In reality, the motion follows the continuous function $\mathbf{x}(t)$, but since we do not have an analytical solution, the best we can do is use a numerical integrator to find an approximation of the the state $\widetilde{\mathbf{x}}(t_i) \approx \mathbf{x}(t_i)$ at a set of discrete times t_i, with $i = 0, 1, \ldots N$. The difference between the true motion and the approximated motion is the error of the numerical solution, defined as $\varepsilon(t_i) = \widetilde{\mathbf{x}}(t_i) - \mathbf{x}(t_i)$. Since $\mathbf{x}(t_i)$ is not known, $\varepsilon(t_i)$ can only be approximated. The explanations of integration methods in this subsection are based on those in Montenbruck and Gill (2001, Ch. 4), Dirkx and Cowan (2019), and Dirkx and Mooij (2019).

There are two sources of error that arise from performing numerical integrations with a computer: truncation error and rounding error. **Truncation error** is an inherent mathematical property of the integration method being used. The magnitude and behaviour of this type of error is relatively predictable, and can be derived mathematically from the definition of the integrator. The local truncation error (LTE) is the error introduced during a single time step. The global truncation error (GTE) is the accumulation of the LTEs over all time steps. An integrator's truncation error is characterised by its 'order', which describes how the GTE scales with the size of the time step used (e.g. a '2nd-order' integrator's GTE approximately scales with the square of the time step, for a '3rd-order' integrator it scales with the cube of the time step, et cetera). An integrator's order does not, however, say anything about the magnitude of this error.

Rounding error comes from the finite precision of floating point numbers used by computers, which typically have 16 digits of resolution. Therefore, any digits after the 16th digit cannot be represented, so the value stored is rounded a small amount up or down. When completing many operations, the error from this rounding can accumulate to a significant magnitude. The behaviour of rounding error is unpredictable: even when using identical integration settings, the precise rounding error behaviour will differ between individual computers, operating systems, and compilers. While the magnitude of the rounding error is typically small (and not dominant compared to truncation error), it can become relevant when extreme precision is required. The magnitude increases (erratically) with the number of operations but generally scales with $\sqrt{N}$.

There are many kinds of integrators, each with its own advantages and disadvantages. The selection of an integrator is always a trade-off of their properties: truncation error, speed/computational complexity, step size flexibility, and stability.

Single-step integrators use only the current state $\mathbf{x}(t_i)$ to calculate the next state $\mathbf{x}(t_{i+1})$, while **multi-step** integrators also make use of previous states $\mathbf{x}(t_i)$, $\mathbf{x}(t_{i-1})$, $\mathbf{x}(t_{i-2})$, $\ldots$ to calculate $\mathbf{x}(t_{i+1})$. Single-step integrators are simpler and faster, but they can be less accurate. Multi-step methods can be more accurate and can make it easier to derive the state between time steps by interpolating, but they also require initialisation method for the first few time steps. Also, implementing a variable time step is not very straightforward.

As the name suggests, **fixed time step** integrators use the same sized time step $\Delta t = t_{i+1} - t_i$ for all i. This is a simple and straightforward approach, and may be faster for certain applications, but it can be insensitive to quick changes in the dynamics, and can result in wasted computational effort when the dynamics are changing very slowly. **Variable time step** integrators automatically update the size of Δt at each time step to match certain criteria (relating to the allowable LTE). This can be very useful for fully capturing fast-changing dynamics (i.e. when an acceleration vector's magnitude and/or direction changes significantly in a short time) while also not spending unnecessary computational effort on many small time steps when the dynamics are changing slowly. In scenarios where the dynamics are constantly relatively slow, however, these integrators introduce additional computational steps for little benefit.

The general methods used by different integrators vary significantly, and each has its own accuracy and computational complexity characteristics. The **Euler method** is the simplest, as it calculates the state at the next time step using $\mathbf{x}(t_{i+1}) = \mathbf{x}(t_i) + \dot{\mathbf{x}}(t_i)\Delta t$. This is a 1st-order method and is very simple and computationally light, but significant errors tend to accumulate very quickly.

Multi-stage methods, such as the Runge-Kutta (RK) methods, perform several function evaluations for each time step (the stages) and then combine them, resulting in a much more accurate $\mathbf{x}(t_{i+1})$ than the Euler method for an equal Δt, at the cost of a higher computational load. There are many different ways of calculating and combining the stages; each method is defined by a set of coefficients, which can be arranged in a Butcher Tableau (from which the order of the method can be calculated). The RK methods can be adapted to have variable time steps by performing two slightly different RK methods of different orders, using the difference in the results to estimate the error, and using this error together with pre-specified error tolerances to adjust the time step. Examples of this include the Runge-Kutta-Fehlberg (RKF) method and the Dormand-Prince (DP) method.

Multi-step methods, as discussed above, use several past state evaluations in the calculation of $\mathbf{x}(t_{i+1})$ (as opposed to the Euler, RK, RKF, DP, etc. methods, which are single-step methods). An example of this is the Adams-Bashforth (AB) method: this method defines $\mathbf{x}(t_{i+1})$ as a linear combination of the current state $\mathbf{x}(t_i)$ and s previous states. The set of coefficients of the previous states is different for different values of s. The order of this method is defined by the number of previous states that are used, making it very simple to adjust. An extended version of this method, Adams-Bashforth-Moulton (ABM), includes the implicit and explicit calculation of $\mathbf{x}(t_{i+1})$, and also allows for time step control (similar to the RKF method) and even order control. These methods can also be very accurate, but problems can arise when there are discontinuities in the dynamics (thrusters turning on/off, collisions), or when the dynamics are fast-changing (as the step size tends to decrease to tiny values).

Extrapolation methods, such as Bulirsch-Stoer (BS), take a large time step ΔT and divide it into a number of sub-steps, then evaluate the state at each of the sub-steps. It then increases the number of sub-steps (thereby decreasing the sub-step size), evaluates the states, and repeats several times. The sequence of numbers of sub-steps can vary depending on the specific method. With each repetition, the final state estimation (at $t_i + \Delta T$) will improve, assuming truncation error to be dominant. With this information, an extrapolation can be made to estimate what the state would be if we were to take infinitely short time steps. This method can be adjusted to have variable time step and variable order. Although this method involves many state evaluations, it allows for taking quite large time steps while maintaining high accuracy, so it is not necessarily a very computationally intensive method. These properties make this method suitable to long time-span integrations with high accuracy requirements.

When **selecting an integrator** to use for a particular problem, there are several factors that should be considered. The two main factors are speed and accuracy. Speed refers to the computational speed of running an integration using a certain integrator. This is largely determined by the number of times the state derivative function must be evaluated. Accuracy refers to the magnitude of error (truncation plus rounding error) which is acceptable to incur over the course of the integration. Note that truncation and rounding errors are not the only sources of model error: inaccuracies in the environment model can also lead to errors in the results of the integration, as compared to a 'true' solution. In general, it is good practice for the integration errors to be much smaller than the model errors. This is because it is generally relatively easy to achieve very low integration error by choosing more stringent settings, while model errors can depend on inputs that are harder to control, such as ephemerides and gravity fields. Generally, slower integrations will be more accurate and faster integrations will be less accurate, so the accuracy must be traded off against the time available to perform the integration. However, depending on the specific scenario, some integrators may be more favourable than others.

Another factor that may be relevant to integrator selection include whether dense output is required: it may be necessary to retrieve the state at times in between the discrete time steps. Some integrators are less suited to providing this data than others (e.g. BS, due to the large time steps it takes). Furthermore, one must consider which integrators are readily available for use: some more exotic or higher-order integrators may not already have an implementation in the software being used, in which case an implementation must be developed and tested, or a different (available) integrator must be chosen as a compromise.

The selection process can be carried out by comparing a selection of integrators and integrator settings under the same scenario(s) against a reference. This reference may be an analytical solution (if one exists) or it can be benchmark solution. A benchmark solution is one which is known to be very accurate. This may come from literature, where certain scenarios have been run with very high precision and accuracy, or it may be generated manually using very accurate methods and settings. Another way to estimate the integrator error is to propagate from the initial condition to the final condition, then make negative time steps to propagate backwards in time to the initial condition again. The difference with respect to the initial condition can be used to give an indication of the amount of integration error. The integrator and corresponding integrator settings that produce the lowest error and/or the most desirable error behaviour can then be chosen.

3
Methodology

This chapter presents and explains the processes and inputs that were used to generate the results of the research, which are estimations of the A_2 parameter of the Yarkovsky effect model. This primarily consisted of performing estimations for a set of asteroids in order to identify detections of the Yarkovsky effect. Section 3.1 first clarifies some terminology used in this chapter, then also describes the software and hardware used to perform the estimations, and provides an overview of the estimation setup and inputs (in Subsection 3.1.3). The details of some of these inputs are explained directly, while inputs that require further explanation are discussed in dedicated sections: numerical propagation settings in Section 3.3, acceleration settings in Section 3.4, and outlier rejection in Section 3.5. The selection of the set of asteroids on which these estimations were run is explained in Section 3.2.

3.1. Estimation Overview

Estimations are performed using a weighted least-squares procedure, as explained in Section 2.5. A brief recap of this and the distinction between propagations and estimations are given in Subsection 3.1.1. Estimations are implemented through Tudat: see Subsection 3.1.2 for some details about the software and hardware used for performing these estimations. The inputs to the estimation process are listed and explained in Subsection 3.1.3, with the help of a flow diagram.

3.1.1. Terminology

Before continuing further, some terminology used in this report should be defined. In this research, propagations and estimations are only ever carried out for one body at a time. This body may be referred to as the *target body* or just the *target*. A *propagation* is the process of numerically integrating a body's equation of motion, resulting in a history of states at discrete times which can be interpolated to define the body's approximate continuous orbital path. The inputs required for this are the initial state of the target body, the accelerations which affect its motion (see Section 2.2), and the numerical integration settings (see Subsection 2.5.3).

An *estimation* uses observations of the target body and involves an iterative process of multiple propagations to estimate a certain set of parameters (see Section 2.5 for more detail). At a minimum, these parameters typically include the state of the target body at a particular point in time (often the initial time t_0), but can also include parameters relating to the dynamic model. An estimation of n parameters may be referred to as an n-dimensional estimation. Each iteration, a propagation is performed, the computed trajectory is compared to the observations, and then the estimated parameters, formal uncertainties, and correlations are updated according to a weighted least-squares procedure. Ideally, the estimated parameters converge to the values which result in the best match between the observations and the target's propagated states. The inputs required for an estimation are the initial state of the target body and parameters, the accelerations which affect its motion (see Section 2.2), the numerical integration settings (see Subsection 2.5.3), the observation data to which the computed trajectories are to be compared (see Section 2.4), and a stop condition for the iterative process. The inputs used for the estimations performed in this research are outlined in more detail in Subsection 3.1.3.

The term 'epoch' refers to a specific point in time, usually one at which the equation of motion has been numerically integrated. The terms 'time step', 'step size' and Δt all refer to the same thing: the time gap between successive epochs.

3.1.2. Software & Hardware

The software used for setting up and performing orbit propagations and estimations in this research is Tudat, the TU Delft Astrodynamics Toolkit, developed at Delft University of Technology (Dirkx et al. 2022; Gisolfi et al. 2025). The core functionality, written in C++, is that of numerical state propagation and estimation. It has been used in published research for a wide range of topics including "orbital analysis of LEO spacecraft, trajectory optimization of interplanetary transfers, orbit determination and ephemeris generation for planetary missions, and high-accuracy natural satellite dynamics" (Tudat n.d.(a); Tudat n.d.(b)). The code is open-source and is available on Github[1]. A Python interface for Tudat, named Tudatpy has been available since 2020. The version of Tudatpy used is in this research is 1.0.0.dev52. All code for this research was written in Python 3.11. All project code is available on GitHub at: `https://github.com/ronanjmc/MScThesis_RonanMcIntyre_Public`.

The hardware used for running the estimations was a 2018 HP ZBook Studio x360 G5 laptop with an Intel Core i7-8750H processor and 16 GB of RAM. All code was run through Windows Subsystem for Linux (WSL) version 2 using Ubuntu 22.04.

3.1.3. Estimation Inputs

Figure 3.1 shows the required inputs to the estimation process in Tudat. A brief description of each input and the values/settings used in this research are given in the list below. For inputs that require minimal explanation, their details are given here. For any input that requires further elaboration, the (sub)section of this chapter in which it is discussed further is given.

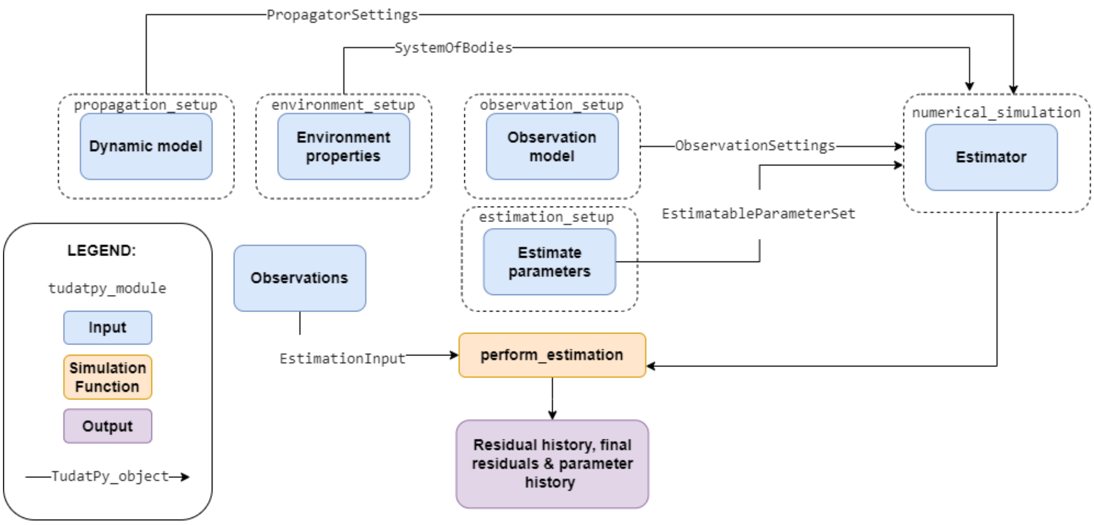


Figure 3.1: Flow diagram showing the inputs (in blue) and outputs of the estimation process in Tudat (Tudat n.d.(c))

- The **environment properties** include the ephemerides and gravity field settings of bodies that are not propagated.
  - The ephemerides of the Sun, the eight planets, the Moon, and Pluto are from the INPOP19a ephemerides (Fienga et al. 2019). For Mars, Jupiter, Saturn, and Pluto, the ephemeris of the planetary system barycentre was used.
  - The gravitational parameters of these bodies come from various sources. See Table 3.3 for the precise values.
    - ⋆ The gravitational parameters of the Sun, Uranus, and Neptune are from the INPOP19a ephemerides (Fienga et al. 2019), as are the gravitational parameters of the full planetary systems for Mars, Jupiter, and Saturn.
    - ⋆ The gravitational parameter of Mercury is derived from the spherical harmonic gravity field of Mercury determined by Konopliv et al. (2020), obtained from NASA's Planetary Data System (PDS)[2], `jgmess_160a`.
    - ⋆ The gravitational parameter of Venus is derived from the spherical harmonic gravity field of Venus determined by Konopliv et al. (1999), obtained from NASA's PDS[3], `shgj180u`.

[1]`https://github.com/tudat-team/`
[2]`https://pds-geosciences.wustl.edu/messenger/mess-h-rss_mla-5-sdp-v1/messrs_1001/data/shadr/`
[3]`https://pds-geosciences.wustl.edu/mgn/mgn-v-rss-5-gravity-l2-v1/mg_5201/gravity/`

    * The gravitational parameter of Earth is derived from the spherical harmonic gravity field of Earth determined by Fecher et al. (2017), obtained from the German Geosciences Centre (GFZ)[4].
    * The gravitational parameter of the Moon is derived from the spherical harmonic gravity field of the Moon determined by Lemoine et al. (2014) and Goossens et al. (2016), obtained from NASA's PDS[5], `gggrx_1200`.
    * The gravitational parameter of Pluto is from Brozović and Jacobson (2024) — the value used is that of the full Pluto system from Table 8.
  - Ephemerides of main-belt asteroids are taken from SPK files which were downloaded from JPL Horizons[6] (downloaded on 17 December 2025).
  - Masses of main-belt asteroids are taken from the SiMDA (Size, Mass and Density of Asteroids) catalogue (Kretlow 2020). See Table 3.4 for the precise values.
  - The gravitational constant is defined as $G = 6.672\,59 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ in Tudat.

- The **estimation parameters input** defines which parameters are to be estimated in the estimation. In this research, these are:
  - The Cartesian position and velocity at the initial time (6 parameters).
  - The Yarkovsky parameter A_2 (see Equation (3.9)).

- The **dynamic model** input contains the settings for which accelerations are to be included, the numerical integration settings, and related inputs such as the initial state, initial time, and final time.
  - The numerical integration settings and how these were selected are discussed in Section 3.3.
  - The included accelerations and how these were selected are discussed in Section 3.4.
  - The initial time is set to 00:00:00 TT of the day preceding the day of the earliest (used) observation (thus between 24 and 48 hours before the earliest observation).
  - The final time is set to 00:00:00 TT of the day which is two days after the last (used) observation (thus between 24 and 48 hours after the last observation).
  - The initial state (position and velocity) of the target body is taken from JPL Horizons.
  - The initial value of the Yarkovsky parameter A_2 is taken from the full version of Table B.1 of Fenucci et al. (2024).

- The **observations** input includes the observations themselves, their weights, and the number of estimation iterations to be performed.
  - The observations are retrieved from the MPC[7]. Only angular position (RA-DEC) observations taken between 1 January 1900 and 11 June 2026 from ground stations were used. Each observation has an associated UTC timestamp a 3-character MPC code indicating the observatory that made the measurement.
  - The filtering of observational outliers is discussed in Section 3.5.
  - The observation weights are defined using the weighting scheme of Vereš et al. (2017) (identical for all iterations), which was discussed in Subsection 2.4.4.
  - Star catalogue debiasing is applied to the observations according to the scheme of Eggl et al. (2020), which was discussed in Subsection 2.4.4.
  - The number of iterations was set to four. For a discussion on the convergence of the estimation results and how this was checked, see Subsection 4.2.1.

- The **observation model** input defines the type(s) of the observations that are used in the estimation and defines the "link ends" of the observation as well as any settings needed to correct for biases.
  - The observations are of the angular position type, specifically RA and DEC measurements (see Subsection 2.4.1 for how these are defined).
  - The link ends define the roles of the objects at each end of an observation: the passively reflecting target body is set as the "transmitter" and the ground observatory that made the measurement is set as the "receiver". The links also define the locations of the ground stations in an Earth-fixed reference system.
  - The straight-line light-time delay is taken into account, since observations do not represent the instantaneous position of the target due to the speed of light being finite.

[4] https://dataservices.gfz-potsdam.de/icgem/showshort.php?id=escidoc:1504398
[5] https://pds-geosciences.wustl.edu/grail/grail-l-lgrs-5-rdr-v1/grail_1001/shadr/
[6] https://ssd.jpl.nasa.gov/horizons/
[7] https://www.minorplanetcenter.net/

3.2. Asteroid Selection

The research of this thesis was significantly influenced by the methods and findings of Fenucci et al. (2024). This paper has been discussed in Subsections 2.2.4, 2.3.2 and 2.3.3. An overview of the methods used by Fenucci et al. (2024) to identify candidate asteroids for Yarkovsky detection and discard spurious detections is given in Subsection 3.2.1. The set of asteroids chosen for analysis in this thesis and the justifications for this choice are presented in Subsection 3.2.2.

3.2.1. Methodology of Fenucci et al. (2024)

The dynamic model used by Fenucci et al. (2024) is discussed in Subsection 2.2.4 and the Yarkovsky acceleration model specifically is discussed in Subsection 2.3.2. This subsection provides a general outline of their overall methodology and a summary of the results.

In order to define a range of plausible values for an asteroid's semi-major axis drift $\mathrm{d}a/\mathrm{d}t$ due to the Yarkovsky effect, the authors define a preliminary physical model. This model depends on physical properties of the asteroid: its diameter D, obliquity γ, rotation period P, density ρ, emissivity ε, absorption coefficient α, thermal inertia Γ, and heat capacity C. For γ, ε, α, and C, fixed values were chosen, which are assumed to be valid for all asteroids. For each of the other four parameters, a range of possible values was defined by a nominal value plus or minus one standard deviation (Gaussian distributions assumed). Γ was set to a fixed range for all asteroids. Where available, D, ρ, P, and their uncertainties were set per asteroid using values from databases. If these were not available, they were estimated using parametrised models. A Monte Carlo method was used to calculate a distribution of the expected Yarkovsky drift, by varying the values of the physical parameters within their defined ranges. The 95th percentile of this distribution is defined as Y_M, i.e. $P(\mathrm{d}a/\mathrm{d}t < Y_M) = 0.95$.

To identify candidates for Yarkovsky detection from the full population of NEAs, the authors first perform a 6-dimensional orbit determination for all NEAs and discard any for which the formal uncertainty in the semi-major axis σ_a is greater than 10^{-5} AU, as such an orbit is not sufficiently well determined for a Yarkovsky detection. For each of the remaining asteroids, if the following inequality is not fulfilled, that asteroid is discarded too:

$$\sigma_a < \Delta T \cdot Y_M \tag{3.1}$$

where ΔT is the length of the asteroid's observation arc (thus $\Delta T \cdot Y_M$ is the expected semi-major axis drift over the observation arc if $\mathrm{d}a/\mathrm{d}t = Y_M$).

The remaining asteroids are considered good candidates for Yarkovsky detection. For each one, a 7-dimensional orbit determination is performed in which the Yarkovsky parameter A_2 (see Equation (2.25)) is estimated. These estimations use the dynamic model outlined in Subsection 2.2.4 (with the Yarkovsky acceleration model as defined in Equation (2.25)) and observations from the MPC, including ground- and space-based optical observations and radar observations. The observations are weighed according to the scheme of Vereš et al. (2017) (see Subsection 2.4.4) and are debiased according to the scheme of Farnocchia et al. (2015a), which is very similar to the debiasing scheme of Eggl et al. (2020) (discussed in Subsection 2.4.4). Observation outliers are automatically removed using the iterative outlier rejection algorithm of Carpino et al. (2003).

If the resulting signal-to-noise ratio of the estimated A_2 parameter $\mathrm{SNR}_{A_2} = |A_2/\sigma_{A_2}|$ is less than 3, the detection is discarded on the grounds of low statistical significance. For each remaining detection, the authors check whether the estimated A_2 is compatible with the physical properties of the asteroid. To do this, they calculate the semi-major drift rate $(\mathrm{d}a/\mathrm{d}t)_{\mathrm{fit}}$ and uncertainty $\sigma_{(\mathrm{d}a/\mathrm{d}t)_{\mathrm{fit}}}$ corresponding to the fitted A_2 and σ_{A_2} using:

$$\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{2A_2}{na^2(1-e^2)} \tag{3.2}$$

which comes from the Gauss planetary equations, and where n is the mean motion. Then, if the following inequality is evaluated:

$$\left|\left(\frac{\mathrm{d}a}{\mathrm{d}t}\right)_{\mathrm{fit}}\right| - k\sigma_{(\mathrm{d}a/\mathrm{d}t)_{\mathrm{fit}}} < Y_M \tag{3.3}$$

where $k = 1.645$, representing the 5th percentile of the measured Yarkovsky drift. This value is chosen because the distribution resulting from the Monte Carlo method is not generally Gaussian. If the inequality is not fulfilled, it means that the value of $(\mathrm{d}a/\mathrm{d}t)_{\mathrm{fit}}$ is not considered compatible with the physical model and thus the detection is discarded. The physical model was defined using 'generous' values for the physical parameters (e.g. absorption coefficient $\alpha = 1$ and the obliquity γ was set to either 0° or 180°, values for which the transverse acceleration is maximised), so Y_M should be an upper bound of what is realistic. Therefore, if the 5th percentile of the measured distribution is greater than Y_M, it indicated that the measured semi-major axis drift rate is not realistically due to the Yarkovsky effect.

Finally, the remaining asteroids are checked for dependence on isolated tracklets, where an isolated tracklet is defined as a series of observations from the same observatory over a period of less than 15 days and where the time gap between these observations and the previous and following observations is at least 5 years. Another 7-dimensional orbit determination is performed using an observation data set where any isolated tracklets have been removed (with some exceptions for isolated tracklets than have been shown to be reliable). If the new SNR_{A_2} was less than 3, the detection was considered too reliant on isolated tracklets and the detection was discarded.

From the full set of 32 396 known NEAs as of 7 August 2023, 5308 fulfilled Equation (3.1) and were selected as candidates for Yarkovsky detection. Of these, there were 461 for which SNR_{A_2} was at least 3 after the 7-dimensional orbit determination. 88 of these were discarded on the basis of not fulfilling Equation (3.3), leaving 373. Of these, 25 were rejected due to high dependence on potentially unreliable isolated tracklets. In the end, the authors identified 348 asteroids with a reliable detection of the Yarkovsky effect.

3.2.2. Selected Asteroid Set

In this thesis, orbit estimations will be performed on the set of 348 asteroids for which Fenucci et al. (2024) detected the Yarkovsky effect. This set of 348 asteroids will henceforth be referred to as the F-Set (for ‘Fenucci et al.’ or ‘full’).

The asteroids in the F-set set have already been shown to have physical properties which are compatible with Yarkovsky effect detection. Furthermore, for each asteroid, the 7-dimensional orbit determination performed by Fenucci et al. (2024) detected a statistically significant ($\text{SNR}_{A_2} \geq 3$) Yarkovsky acceleration which was within the plausible range of possible values and this detection was shown to not to have high dependence on potentially unreliable isolated tracklets. The last point is relevant because there is a lot of overlap between the observation sets used in this research and used by Fenucci et al. (2024).

As outlined in Chapter 1, the purpose of this thesis is to investigate and determine the specific settings and methods that can be implemented using open-source software to produce comparable results to what has been produced in literature using proprietary software. Therefore, it is useful to use a set of asteroids for which the results can be directly compared. In order to answer the research questions, it is not necessary to reproduce the steps taken by Fenucci et al. (2024) to identify (new) candidate asteroids for Yarkovsky detection, but simply to compare results for asteroids for which detections have already been made. Using this F-set for this research also obviates the need to confirm that the detected orbital drift can be realistically attributed to the Yarkovsky effect by validating it against physical models of the asteroids, since this has already been done for these asteroids. Furthermore, the Yarkovsky acceleration model used by Fenucci et al. (2024) is exactly the same as the Yarkovsky acceleration model that is available to use in Tudat, making the estimated values of A_2 directly comparable. The full version of Table B.1 from Fenucci et al. (2024), containing extensive data on the results of their estimations, was downloaded and used for comparison purposes (see Section 4.3).

The reason for the set containing only NEAs and no other categories of asteroids is that the Yarkovsky effect seems to be most relevant to the motion of the NEAs. For example, when Dziadura et al. (2023) performed orbit estimations to search for Yarkovsky detections, they identified 49 NEAs with a Yarkovsky effect out of 446 NEAs tested, while identifying a Yarkovsky effect for zero inner main-belt asteroids out of 54 094 tested. It is notable that since the procedures of Fenucci et al. (2024) were carried out in 2023, a further 9382 NEAs have been discovered (as of the end of May 2026), which is an increase of around 29% (CNEOS 2024). However, it is likely that most of these objects have observational arcs that are too short for Yarkovsky detection anyway.

3.3. Integrator Selection

Selecting an appropriate numerical integration method (i.e. integrator) — and the accompanying integrator settings, such as tolerances and/or step size — is important in order to ensure that integrator error does not have a significant adverse effect on the orbit determination. Whilst some integrator error is unavoidable, it is desirable for the behaviour of that error to be predictable so that the level of error introduced by numerical integration can be estimated and demonstrated to be below a certain desired threshold.

3.3.1. Integrator Analysis Process

The integrator analysis for this research only considered fixed-step numerical integration methods. Specifically, the set of integrators that were considered comprises a range of Runge-Kutta (RK) methods of various orders. As explained in Subsection 2.5.3, variable time step integrators (such as the Runge-Kutta-Fehlberg (RKF) methods) can offer fast performance by predicting the local truncation error and adapting the time step to match the desired level of error. This means that time steps are short when necessary due to fast-changing dynamics, but can be relatively long when possible, potentially reducing the total number of function evaluations carried out as compared to a fixed-step integrator. However, there are several parameters which determine the behaviour of RKF integrators (minimum time step, maximum time step, initial time step, absolute tolerance, and relative tolerance). Tuning these parameters precisely is important to ensure that the integration process doesn't choose such small time steps that it gets "stuck", nor such large time steps that interpolation error begins to become significant (for example when interpolating the position in between time steps in order to calculate an observation residual). The benefit of such methods potentially taking a shorter time to compute (for a similar level of error as compared to fixed-step methods) was not considered to be a very important factor for this research, as the estimations don't need to be run a great number of times. For these reasons, variable time step integrators were not considered in this analysis. For similar reasons, multi-step methods were not considered either. And because of their suitability to longer-term integrations and potential interpolation issues, extrapolation methods were also not considered.

The process of selecting an integrator for this research was as follows. For a given body and fixed-step integrator, run a set of propagations, with each propagation using a different time step. All other propagation inputs and settings should remain the same, such as the initial position, initial and final times, dynamical model, and propagator. This is then repeated for a number of different integrators and for a number of different bodies.

Because the order N is a known property of an integrator and GTE is $O(\Delta t^N)$, the error of a propagation with time step Δt can be approximated by taking the difference in state with respect to a propagation that used a time step sufficiently smaller than Δt. A "sufficient" ratio of time steps will depend on the order of the integrator in question. For the integrators considered in this integrator analysis, $N \geq 4$, so a ratio of 2 was considered sufficient, as 2^4 is already more than an order of magnitude.

Therefore, for a set of propagations using a range of different time steps, the (position) error over time can be approximated for all but the propagation using the smallest time step. An analysis of how the error behaves with respect to the time step, the integrator, and the orbital properties of the bodies can then be used to inform the selection of an appropriate integrator and time step to use for the estimations. This process was carried out in two iterations: an initial quick integrator analysis — whose results were used to carry out some initial estimations and get some preliminary results — and then a more thorough and focused integrator analysis based on the results of the first iteration. Because the second iteration informed the integrator selection used for generating the final results, this iteration is discussed in more detail than the first.

3.3.2. First Iteration

First, an initial integrator analysis was conducted on a small number of bodies. The purpose of this first iteration was to define a maximum desired level of integrator error, in order to inform the final selection of integrator settings based on the next iteration. This first iteration's analysis included the following integrators: RK4, RK6, RK8, RK10, RK12, and RK14. Following the integrator analysis procedure discussed above, it was determined that RK6 provided comparable performance to higher-order integrators, at least for asteroids with relatively low eccentricity ($e < 0.4$). A time step of 1 day was chosen as it offered a balance of moderate computational speed and accuracy. Using these settings for three bodies with $e < 0.4$, the maximum position error over a propagation of 3.5 orbits was $\sim 10^{-4}$ km in all cases.

Estimations were performed for all asteroids in the F-set using the RK6 integrator and $\Delta t = 1$ day. The dynamic model used for these estimations was the same as the one outlined in Subsection 3.4.2 and the other inputs were the same as those outlined in Subsection 3.1.3, with the exception of outlier rejection, which had not yet been implemented. For each estimation, the true position error at each epoch was approximated by comparing to the position of that body at that epoch according to JPL Horizons. Figure 3.2 shows a histogram of the RMS of the total 'true' position error over time for these estimations.

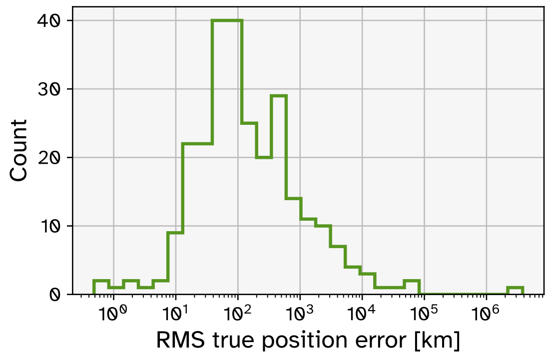


Figure 3.2: Histogram of the RMS of the 'true' error (assuming JPL Horizons' ephemeris represent a body's true position) for estimations performed using the RK6 integrator and $\Delta t = 1$ day. Only results for estimations with $\mathrm{SNR}_{A_2} > 3$ are included.

In order to be certain that integrator error does not significantly influence the results of the research, the maximum desired level of integrator error should be chosen to be significantly lower than the level of model error (e.g. two orders of magnitude lower). The lowest value of true error in Figure 3.2 was 0.483 km, for asteroid 437841 (1998 HD_{14}), which has $e = 0.3126$. Because the integrator settings were shown induce a very low level of integrator error for low eccentricity bodies, it can be assumed that this small amount of error can be main attributed to model error and observation error. Taking this value to be the best achievable level of model error using the defined dynamic model, the maximum desired level of integrator error was chosen to be 10^{-2} km (or 10 m). By not exceeding this level of integrator error, it can be ensured that integrator error will be significantly lower than model error in all estimations. In the second iteration of integrator selection, integrator settings

3.3.3. Second Iteration

The purpose of the second iteration is to define integrator settings which can be expected to induce less than the maximum desired level of integrator error (10^{-2} km, labelled as "Threshold" in the figures to follow) for all asteroids.

Body Selection

The first step was choosing a representative subset of bodies from the F-set. The integrator analysis involves many propagations, often using small time steps and thus it can be quite slow to run (up to ~3 hours per body, depending on the propagation duration). This makes it impractical to perform an integrator analysis for every single body. This is also not necessary, as many bodies with similar orbital properties and/or observation arc lengths will show similar error behaviour anyway. A reasonable approach, then, is to select a representative subset of bodies on which to run the integrator analysis. If the chosen set is sufficiently representative, it can be assumed that the results derived from the analyses are applicable to all bodies in the F-set. The following steps describe how that subset was selected.

The total integrator error accumulated over a propagation (GTE plus rounding error) can be increased by increasing the number of time steps n taken, since $GTE \sim n \cdot LTE$ (Colomés et al. 2023), and also because more time steps allows more opportunities for rounding error to accumulate. Fast-changing dynamics, i.e. when an acceleration vector's magnitude and/or direction changes significantly in a short time, can also cause LTE to increase. This may happen during close passes between bodies or at the periapsis passage of an orbit with high eccentricity. This is because LTE depends on the derivative(s) of the state vector, which are high at these moments. [Sec 6.1]

In order to choose a representative set of bodies for the integrator analysis, we can look at body properties which affect the factors mentioned above. As a proxy for determining whether a body has fast-changing dynamics, we can look at when the components of the state vector's derivative are high. Since the state vector comprises the position and velocity vectors $\mathbf{x} = \begin{pmatrix}\mathbf{r} & \mathbf{v}\end{pmatrix}^T$, the state derivate comprises the velocity and acceleration vectors $\dot{\mathbf{x}} = \begin{pmatrix}\dot{\mathbf{r}} & \dot{\mathbf{v}}\end{pmatrix}^T = \begin{pmatrix}\dot{\mathbf{r}} & \ddot{\mathbf{r}}\end{pmatrix}^T = \begin{pmatrix}\mathbf{v} & \mathbf{a}\end{pmatrix}^T$. From Equation (2.5), in a two-body system, acceleration is at a maximum when the distance between the bodies is at a minimum. The point when a small body is closest to its primary is called periapsis (denoted p) where $r = r_p$. This distance can be expressed as:

$$r_p = a(1-e) \tag{3.4}$$

The magnitude of the acceleration at this point is:

$$|\mathbf{a}_p| = \ddot{r}_p = \frac{\mu}{{r_p}^2} = \frac{\mu}{a^2(1-e)^2} \tag{3.5}$$

Acceleration magnitudes are written as $\ddot{r}$, in order to avoid confusion with the semi-major axis a. The vis-viva equation (Wakker 2015, Sec. 6.3) can be used to describe the magnitude of the velocity of a body in an elliptical orbit:

$$v = \dot{r} = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \tag{3.6}$$

Again, this shows that velocity is maximum at periapsis. Combining Equation (3.6) and Equation (3.4), it can be shown that the velocity at periapsis v_p is:

$$v_p = \dot{r}_p = \sqrt{\frac{\mu}{a}\left(\frac{1+e}{1-e}\right)} \tag{3.7}$$

For all of the bodies in the F-set, the Sun is the central body and thus $\mu \approx \mu_\odot = 1.327\,12 \times 10^{20}\ \mathrm{m^3\,s^{-2}}$ (Wakker 2015, App. B), and when the Sun is the primary body, periapsis may be called *perihelion*. Clearly, magnitude of the state derivative depends on the orbital eccentricity and semi-major axis of the body. Figure 3.3 shows the perihelion velocities and accelerations all of the asteroids in the F-set, according to their values of e and a, as retrieved from the JPL SBDB (JPL SBDB 2026). It shows that the state derivative reaches more extreme values for lower values of a and for higher values of e. Since the propagation interval of a body will be approximately the same length as its observation arc, bodies with long observations arcs are susceptible to having high GTE, due to the accumulation of LTE and rounding error over many steps. Figure 3.4 shows the distributions of eccentricity, semi-major axis, and observation arc for the bodies in the F-set. The observation arc was defined as the time between the earliest and most recent optical ground-based observations of the body available from the MPC.

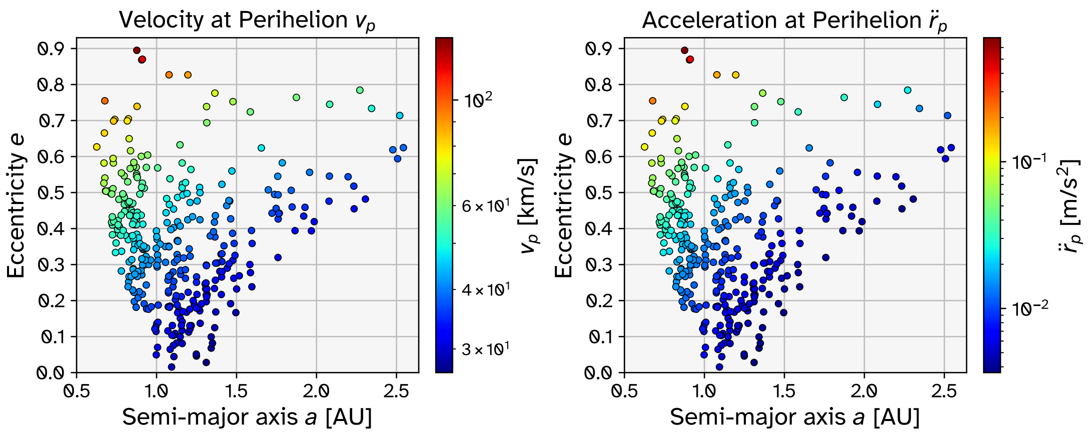


Figure 3.3: Distribution of the perihelion velocity magnitudes v_p and acceleration magnitudes $\ddot{r}_p$ for the 348 asteroids in the F-set, shown on scatter plots of e vs. a.

In selecting the set of bodies which are to be a representation of the full F-set for the purposes of integrator selection, an effort was made to choose a broad range of values of the observation arc, as well as a broad range of values of eccentricity and semi-major axis, in order to represent the full range of values that the state derivative may take on. In addition, care was taken to represent the extremes: all four bodies with $e > 0.8$ were included, which also includes the most extreme values of v_p and $\ddot{r}_p$. Figure 3.5 shows the selected subset of 22 bodies and distributions of parameters in a similar manner to Figure 3.4. This subset is henceforth referred to as the I-set (for ‘integrator’).

Integrators, Dynamic Model & Time Steps

A smaller set of integrators was used in this iteration of integrator analysis: RK6, RK8, RK10, and RK12. The first iteration indicated that the time steps required for RK4 and lower-order integrators would need

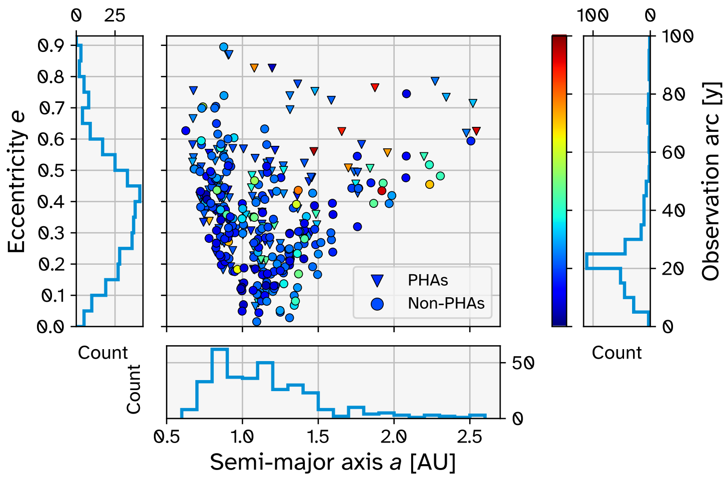


Figure 3.4: Overview of the eccentricity *e*, semi-major axis *a*, and observation arc distributions for the 348 asteroids in the F-set, shown with histograms of each parameter and a scatter plot of *e* vs. *a*, coloured by the observation arc length. There are 130 PHAs in the set (marked as triangles) and 218 non-PHAs (marked as circles).

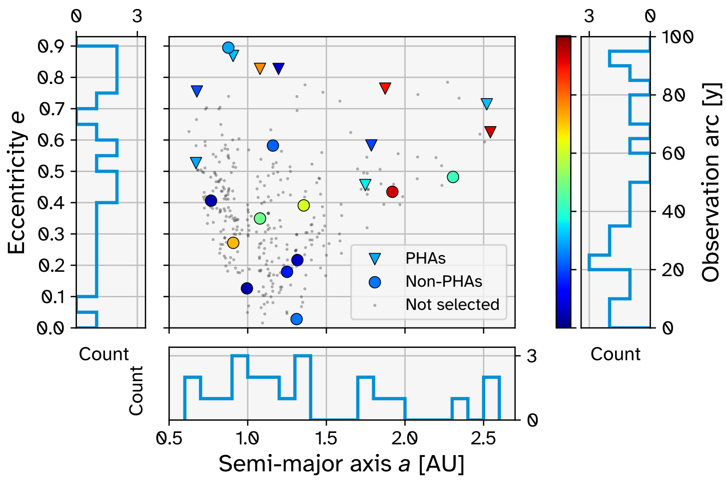


Figure 3.5: The same as Figure 3.4, but with only the asteroids selected for integrator analysis (I-set) highlighted. There are 10 PHAs in the I-set (marked as triangles) and 12 non-PHAs (marked as circles). The remaining 326 asteroids are marked as small grey dots in the scatter plot only and are not counted in the histograms.

to be very small to achieve reasonable levels of integrator error (thereby increasing run time and storage requirements for the output data). Very high order integrators also showed little overall benefit, so RK14 was left out. This selection still represents a reasonable range of orders. The dynamic model used for these propagations is the same as the one used for the full estimations (see Section 3.4). This was chosen in order to be representative of the actual scenario that is relevant for the eventual results-producing estimations. The range of Δt used was slightly different for different integrators (and bodies), but overall it spanned the range from 11.25 minutes to 64 days, increasing by a factor of 2 each time from the lowest value to the highest.

Integrator Analysis Results

The asteroid 1995 CR will be used as an example to demonstrate the figures that were generated from the integrator selection propagations. 1995 CR's orbit has a semi-major axis *a* of 0.9069 AU and an eccentricity *e* of 0.8684, and it is categorised as a PHA (JPL SBDB 2026).

Figure 3.6 is an example of the approximated position error over time for asteroid 1995 CR, using integrator RK8. Such a figure was generated for each integrator and for each body in the I-set. The most relevant information from Figure 3.6 can be condensed by taking the maximum (and/or RMS) of each curve, and then plotting this against the time step Δt. Figure 3.7 shows the curves of the maximum error against Δt for each of the four selected integrators for body 1995 CR. This figure demonstrates that a maximum error below the threshold value of 10^{-2} km is not achievable for the RK6 integrator, at least

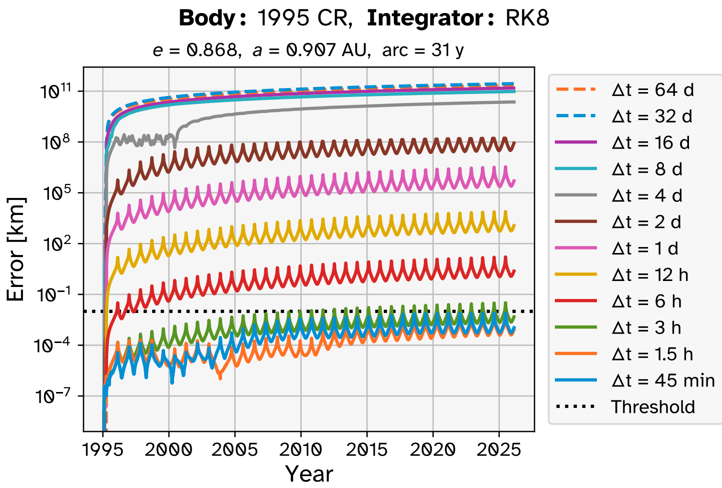


Figure 3.6: Approximated position error over time for body 1995 CR using integrator RK8 and a range of values for the time step Δt.

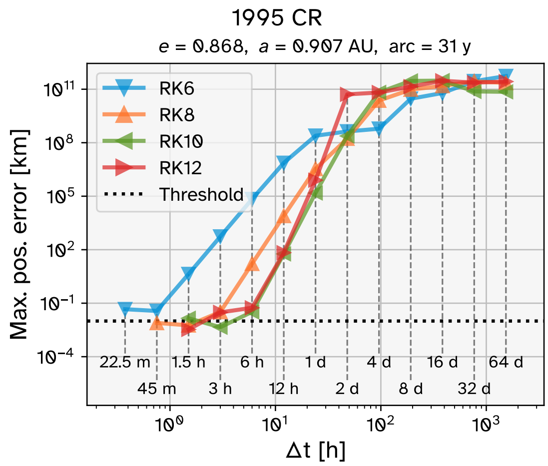


Figure 3.7: Maximum position error against the time step Δt for the four selected integrators for body 1995 CR.

for this high-eccentricity asteroid. Such a figure was generated for each body in the I-set.

Figures 3.8 to 3.10 show how the maximum position error varies with eccentricity, semi-major axis, and observation arc length. In each figure, there is a separate subplot for each integrator, and curves corresponding to various time steps. From Figure 3.8, it is clear that for any particular combination of integrator and time step, the position error generally trends upwards as eccentricity increases. Figure 3.9, however, shows that there is no such general trend for semi-major axis. The spikes in these curves come from the high eccentricity asteroids. In Figure 3.10, a slight trend can be seen by which position error increases alongside observation arc length. However, this is much more gradual than the eccentricity trend, and again the curves are dominated by spikes from the high-eccentricity asteroids.

Chosen Integrator Settings

After analysing the figures shown in this subsection (and comparable figures for the other bodies in the I-set), a scheme was selected for determining the combination of integrator and time step to be used for the estimations of the Yarkovsky effect, keeping in mind the desired level of error and trying to stay under that. This scheme, shown in Subsection 3.3.3, depends only on the target body's eccentricity e. For the higher eccentricities, there wasn't any combination of integrator and time step that had a maximum error below the threshold for all bodies over a reasonably wide range of e (see the highlighted points slightly above the threshold line in the RK10 subplots of Figures 3.8 to 3.10). Although the scheme could be made more specific to avoid having any points over the threshold line, this would likely be an instance of over fitting to these particular results.

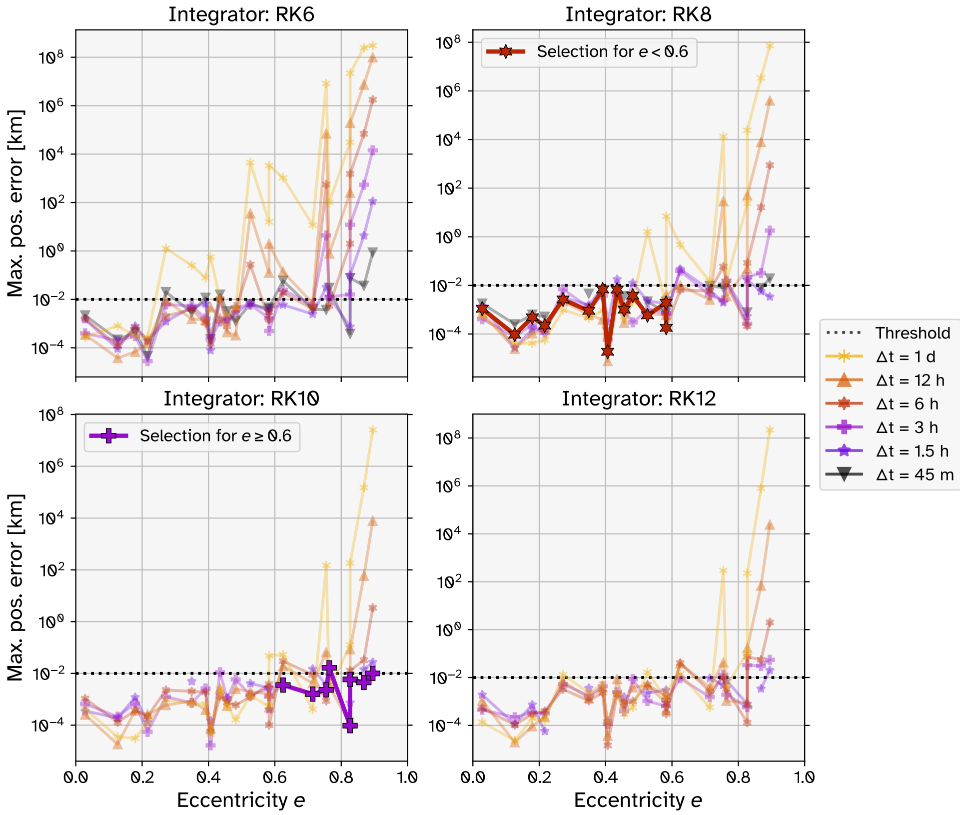


Figure 3.8: Maximum position error against eccentricity. Each subplot represents the results from using a particular integrator on the full I-set. Curves for high values of Δt are omitted for readability. The points corresponding to the chosen integrator scheme (see Subsection 3.3.3) are highlighted and labelled in the subplot legends.

Table 3.1: Chosen scheme for integrator and time step Δt to be used in an estimation based on the eccentricity e of the target body

Eccentricity e	**Integrator**	**Time step** Δt hour
$e < 0.6$	RK8	6.0
$e \geq 0.6$	RK10	3.0

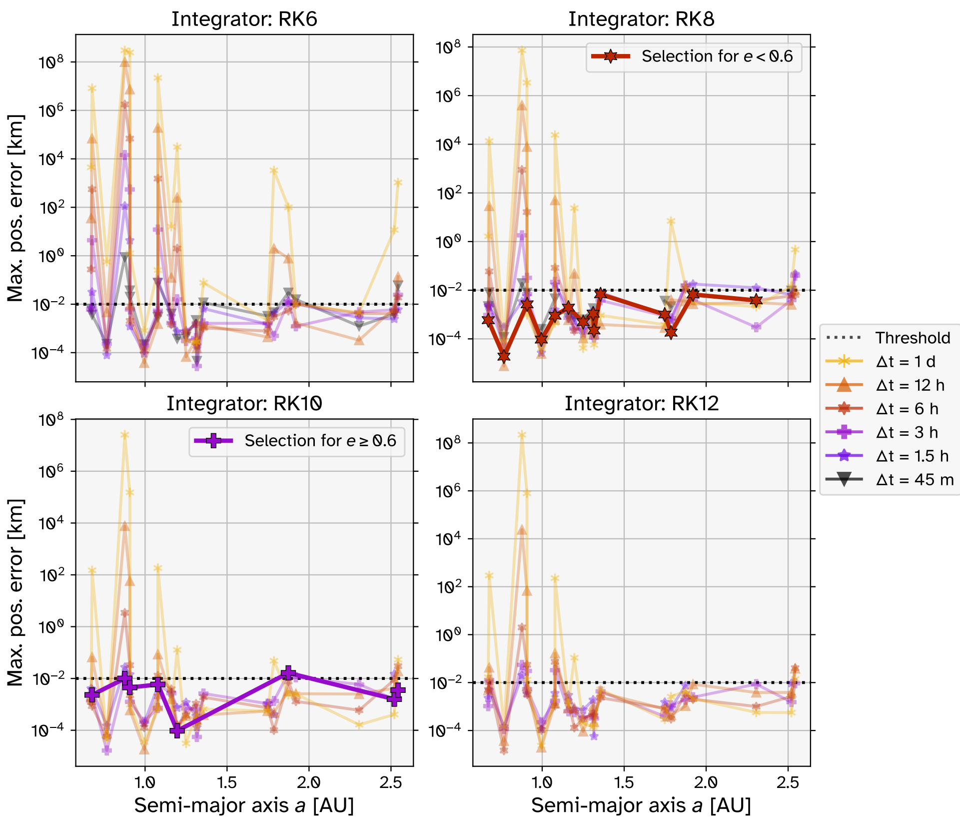


Figure 3.9: Maximum position error against semi-major axis. Each subplot represents the results from using a particular integrator on the full I-set. Curves for high values of Δt are omitted for readability. The points corresponding to the chosen integrator scheme (see Subsection 3.3.3) are highlighted and labelled in the subplot legends.

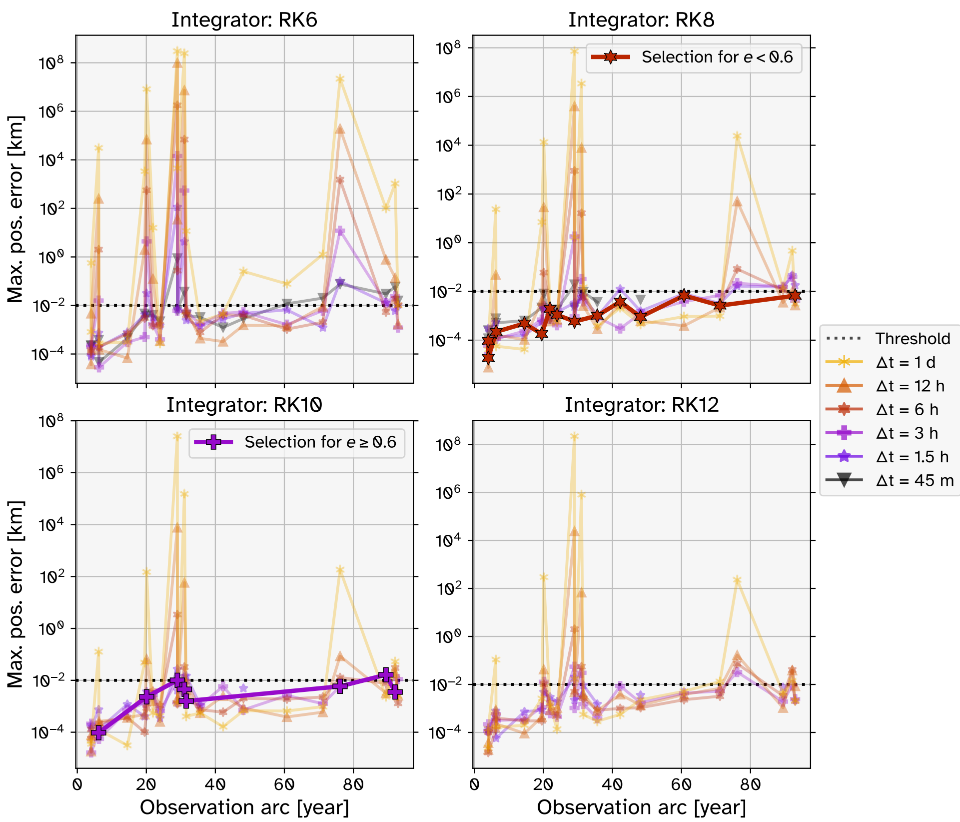


Figure 3.10: Maximum position error against observation arc length. Each subplot represents the results from using a particular integrator on the full I-set. Curves for high values of Δt are omitted for readability. The points corresponding to the chosen integrator scheme (see Subsection 3.3.3) are highlighted and labelled in the subplot legends.

3.4. Dynamic Model Definition

Selecting a suitable dynamic model is important in order to limit the effect of model error on the orbit estimation. This entails selecting the set of accelerations the affect the motion of the target body (see (Equation (2.17)). Too many unnecessary accelerations being included in the dynamic model will result in slow estimation run times, while any errors in these models could increase the overall model error. On the other hand, if significant accelerations are not included, the ability to model the motion of the orbit accurately and get a good fit to the observation data will be curtailed.

Subsection 3.4.1 outlines the process by which the dynamic model was chosen. Details about the selected model are given in Subsection 3.4.2.

3.4.1. Acceleration Selection

In order to examine which accelerations are necessary to include in the dynamic model, various sets of propagations and estimations were performed on a number of bodies, for a range of different acceleration scenarios. A number of abbreviations are used in this section. Before continuing further, they will be listed below:

- **BL:** Baseline. A set of accelerations, a propagation, or an estimation which is used as a reference for comparing other sets of accelerations, propagations, or estimations.
- **LB:** Large bodies. This refers to the set of the largest bodies in the Solar System: the Sun, the eight planets, and Earth's Moon. For Mars, Jupiter, and Saturn, the planetary systems (planet plus moons) are considered, rather than just the planets themselves. LB may also refer to the set of point-mass gravitational accelerations exerted by the LBs, plus the Schwarzschild relativistic correction for the Sun (see Equation (2.15)).
- **LMBA:** Large main-belt asteroid, also called 'largest MBA'. This refers to one of the largest objects in the asteroid belt. These are often considered as a set (LMBAs), defined by n_{LMBA}. The set contains the n_{LMBA} largest asteroids in the main-belt, according to the masses in the SiMDA catalogue (Kretlow 2020).
- **EIH:** Einstein-Infeld-Hoffmann. This refers to the use of Equation (2.16) for the formulation of the gravitational effects of the LBs, as opposed to using the point-mass gravitation. Specifically, LB+EIH means that the EIH formulation is used with all LBs included.
- **1b1:** One-by-one. For a set of accelerations, each one is turned on for one propagation and is off for all other propagations.
- **AB:** Additional bodies. This refers to Pluto (no other additional bodies were considered).
- **CP:** Close-passers. This refers to LMBAs which come within a certain threshold distance of the target body during the propagation period.

Regarding the propagations, first a propagation was done using a baseline (BL) acceleration model. Then a series of propagations were performed in which the only input that changed (compared to the BL) was the acceleration model. Specifically, one or more accelerations were added to the acceleration model as compared to the BL model. By comparing the position of the target body over time between the BL and the latter propagations, the effects of the added acceleration(s) can be analysed.

Regarding the estimations, a similar process was carried out in relation to the adding of accelerations and analysing the effects by comparing to a BL estimation. These estimations used 'simulated' observations: 3D positional information taken from JPL Horizons ephemerides. Although these aren't real observations, they can be treated in the same way as real observations in the least-squares orbit estimation process. The estimated parameters were the initial position and velocity. The inputs used for the propagations and estimations are generally the same as those listed in Subsection 3.1.3, with the exception of the dynamic model, the initial and final times, and the observations.

Under the assumption that the JPL Horizons ephemeris is a good representation of the true orbit of the target body, the results of these estimations can be analysed to give an indication of how far off the the estimated orbit is from the 'true' state of the target body i.e. an indication of the model error. Such an analysis can help to identify which accelerations should be included. Estimations can be more useful than propagations for the identification of important accelerations because the effects of very small perturbations can be 'absorbed' by the initial state estimation. On the other hand, this effect can also cause the impact of an acceleration to be underestimated due to over-fitting. Therefore, any differences observed or improvements made to the estimation fit due to the addition of acceleration are more likely to be a reflection that that acceleration has a significant effect. For this reason, only the figures resulting from the estimations will be discussed in the rest of this subsection.

The asteroid 2062 Aten will be used as an example to demonstrate the figures that were generated from the acceleration selection propagations and estimations. Aten's orbit has a semi-major axis a of 0.9670 AU and an eccentricity e of 0.1829.

Figure 3.11 is an example of the position difference over time, comparing the trajectory of the BL estimation (which used the LB accelerations, i.e. point-mass gravitation of the LBs plus the Schwarzschild relativistic correction for the Sun) to sixteen other estimations. In each of these estimations, the point-mass gravitation of one LMBA was included in addition to the LB accelerations. The LMBAs are sorted by mass in descending order in the figure's legend.

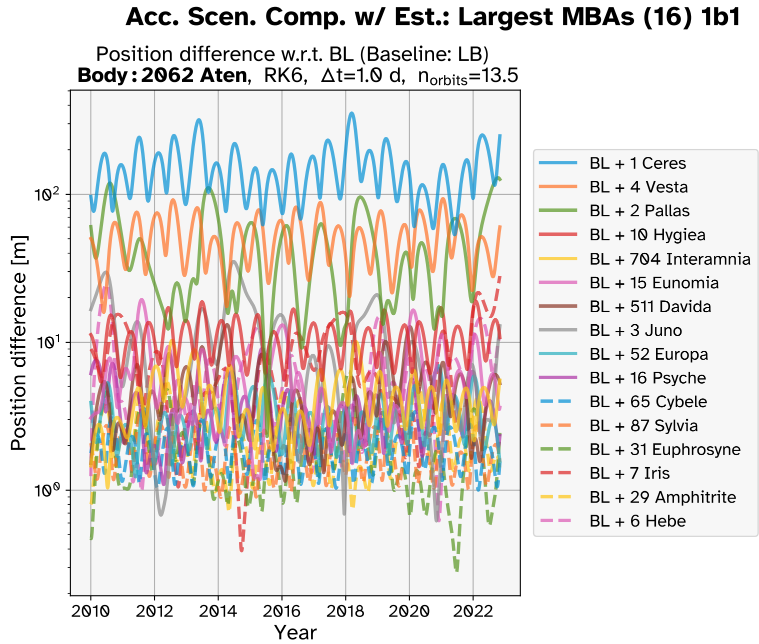


Figure 3.11: The position difference over time between the trajectory of the BL estimation and those of the estimations including one LMBA each.

For the same set of estimations, Figure 3.12 shows the difference in the magnitude of the position error of each estimation's trajectory as compared to the BL trajectory. The position error of each trajectory is calculated as the difference between the modelled position and the position taken from Aten's JPL Horizons ephemeris.

Figure 3.13 compares an estimation using the LB accelerations to one that uses EIH gravitation. The curve is mostly below the zero line, indicating that using the EIH formulation of gravitation provides a moderate benefit to the accuracy of the estimation.

Figure 3.14 shows the position difference over time, comparing the trajectory of the BL estimation to each possible combination of the following sets of accelerations: all sixteen LMBAs together; four close-passers (CP); and Pluto (AB). The figure demonstrates that the LMBAs have the greatest overall effect on the orbit, followed by Pluto, and then by the close-passers.

3.4.2. Chosen Accelerations

After analysing the figures shown in Subsection 3.4.1 (and comparable figures for the other bodies), the dynamic model to use for the estimation of the Yarkovsky effect was selected. The dynamic model, shown in Subsection 3.4.2 and Equation (3.8), will be the same for all estimations of the Yarkovsky effect.

The equation of motion of the target body under the chosen dynamic model is:

$$\mathbf{a}_i = \ddot{\mathbf{r}}_i = \mathbf{a}_\odot + \sum_{j=1}^{26} \mathbf{a}_{\text{3b pert. } j} + \mathbf{a}_{\text{rel.}} + \mathbf{a}_{\text{Yark.}} \tag{3.8}$$

where i represents the target body, $\mathbf{a}_\odot$ is the point mass gravitational acceleration due to the Sun (the central body), $\mathbf{a}_{\text{rel.}}$ is the Schwarzschild relativistic correction (see Equation (3.10)), $\mathbf{a}_{\text{Yark.}}$ is the Yarkovsky acceleration (see Equation (3.9)), and the 26 third-body perturbers j are the eight planets, Earth's Moon, Pluto, and the 16 LMBAs.

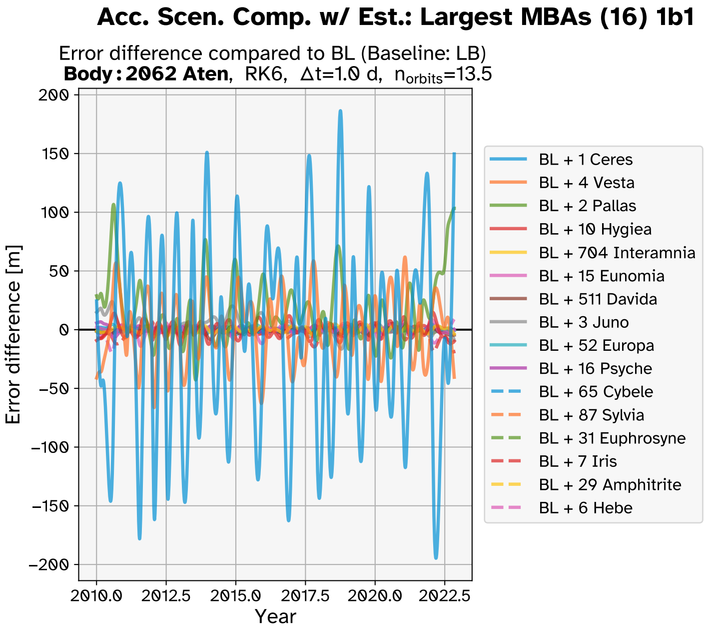


Figure 3.12: The error difference over time between the trajectory of the BL estimation and those of the estimations including one LMBA each. The position error of each trajectory is calculated as the difference between the modelled position and the position taken from Aten's JPL Horizons ephemeris. The figure shows the magnitude of the difference between the two position error vectors.

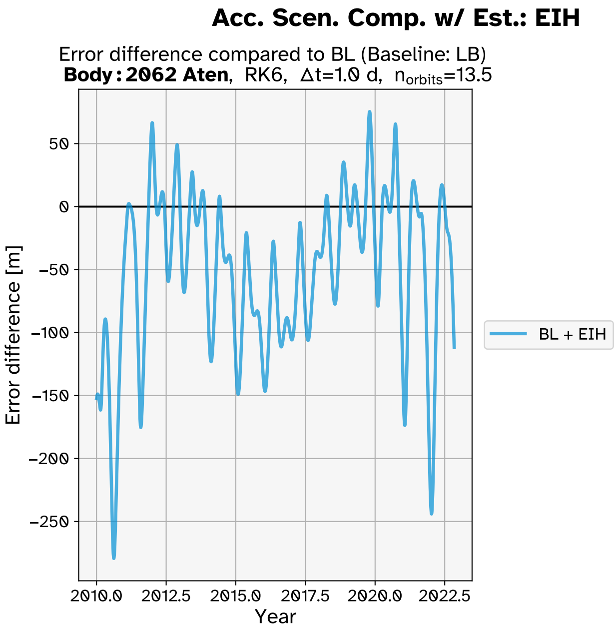


Figure 3.13: The error difference over time between the trajectory of the BL estimation the trajectory of an estimation that uses EIH gravitation. The position error is calculated as the difference between the modelled position and the position taken from Aten's JPL Horizons ephemeris. The figure shows the magnitude of the difference between the two position error vectors.

The model used for the Yarkovsky acceleration is (equivalent to Equation (2.25)):

$$\mathbf{a}_{\text{Yark.}} = A_2 \left(\frac{1\text{AU}}{r}\right)^2 \hat{\mathbf{t}} \tag{3.9}$$

where $\hat{\mathbf{t}}$ is the transverse direction, r is the heliocentric distance in AU, and A_2 is the parameter that is estimated in a weighted least-squares orbit estimation.

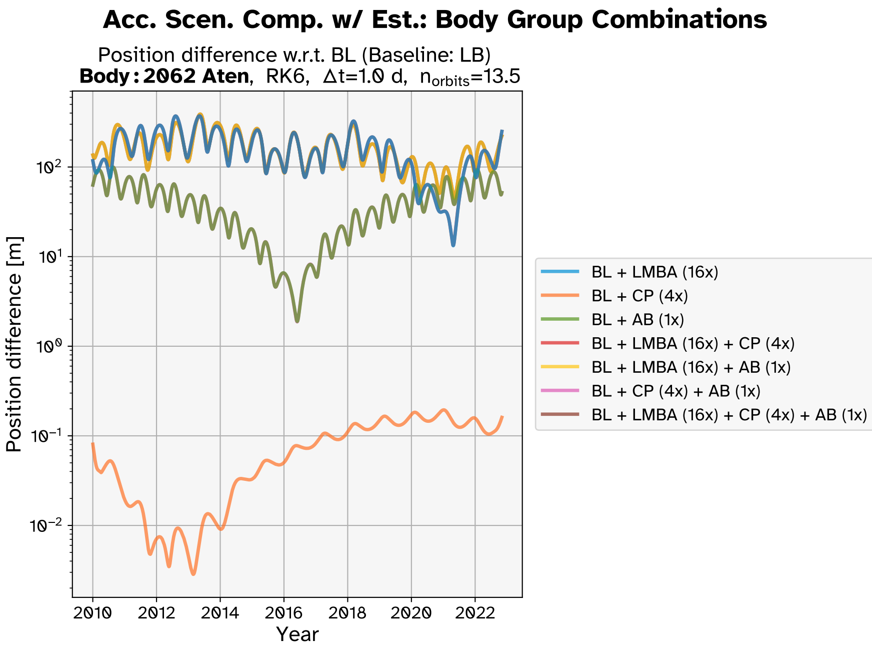


Figure 3.14: The position difference over time between the trajectory of the BL estimation and the estimations in which different combinations of sets of accelerations are included.

Table 3.2: Accelerations chosen to be used in the dynamic model. For explanations of the abbreviations, please refer to the summary in Subsection 3.4.1. See Equation (3.8) for the full equation of motion for this dynamic model.

Source(s)	Acceleration	Note
LB	Point-mass gravity	
LMBA	Point-mass gravity	$n_{\text{LMBA}} = 16$
Pluto	Point-mass gravity	
Sun	Schwarzschild relativistic correction	See Equation (3.10)
Sun	Yarkovsky	See Equation (3.9)

The model used for the relativistic correction acceleration is (equivalent to Equation (2.15)):

$$\mathbf{a}_{\text{rel.}} = \frac{\mu_\odot}{c^2 {r_{\odot i}}^3}\left[\left(\frac{4\mu_\odot}{r_{\odot i}} - {v_{\odot i}}^2\right)\mathbf{r}_{\odot i} + 4(\mathbf{r}_{\odot i}\cdot\mathbf{v}_{\odot i})\mathbf{v}_{\odot i}\right] \tag{3.10}$$

where $\mu_\odot$ is the gravitational parameter of the Sun $\odot$, c is the speed of light, and $\mathbf{r}_{\odot i}$ and $\mathbf{v}_{\odot i}$ are the position and velocity vectors of the target body with respect to the Sun.

Table 3.3 gives the masses and gravitational parameters of the eight planets, Earth's Moon, and Pluto that are used in the dynamic model. Table 3.4 gives the masses, mass uncertainties, and gravitational parameters of the 16 LMBAs that are used in the dynamic model. The ephemerides used for these bodies' positions were given and cited in Subsection 3.1.3.

Table 3.3: Masses M and gravitational parameters GM of the Sun, eight planets, the Moon, and Pluto included in the dynamical model, given to 13 significant digits.

Body	M kg	GM $\mathrm{m^3\,s^{-2}}$	Source
Sun	$1.988\,919\,445\,703 \times 10^{30}$	$1.327\,124\,400\,420 \times 10^{20}$	1
Mercury	$3.301\,846\,612\,948 \times 10^{23}$	$2.203\,186\,869\,109 \times 10^{13}$	2
Venus	$4.868\,553\,171\,692 \times 10^{24}$	$3.248\,585\,920\,790 \times 10^{14}$	3
Earth	$5.973\,698\,990\,947 \times 10^{24}$	$3.986\,004\,415\,000 \times 10^{14}$	4
Moon	$7.347\,671\,776\,397 \times 10^{22}$	$4.902\,800\,121\,847 \times 10^{12}$	5
Mars*	$6.418\,553\,397\,406 \times 10^{23}$	$4.282\,837\,521\,400 \times 10^{13}$	1
Jupiter*	$1.899\,004\,206\,762 \times 10^{27}$	$1.267\,127\,648\,000 \times 10^{17}$	1
Saturn*	$5.686\,035\,737\,247 \times 10^{26}$	$3.794\,058\,520\,000 \times 10^{16}$	1
Uranus	$8.684\,107\,070\,868 \times 10^{25}$	$5.794\,548\,600\,000 \times 10^{15}$	1
Neptune	$1.024\,568\,735\,765 \times 10^{26}$	$6.836\,527\,100\,580 \times 10^{15}$	1
Pluto*	$1.461\,801\,190\,000 \times 10^{22}$	$9.754\,000\,000\,000 \times 10^{11}$	6

Notes:
- Planets marked with (*) indicate that the mass and ephemeris of the planetary *system* (planet plus moons) was used.
- The value of the gravitational constant used is $G = 6.672\,59 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$.
- Source 1: INPOP19a (Fienga et al. 2019).
- Source 2: Konopliv et al. 2020.
- Source 3: Konopliv et al. 1999.
- Source 4: Fecher et al. 2017.
- Source 5: Lemoine et al. 2014; Goossens et al. 2016.
- Source 6: Brozović and Jacobson (2024, Table 8).

Table 3.4: Masses M, mass uncertainties σ_M, and gravitational parameters GM of the 16 largest main-belt asteroids (LMBAs), sorted by mass.

Body	M kg	σ_M kg	GM $\mathrm{m^3\,s^{-2}}$
1 Ceres	9.38×10^{20}	2.11×10^{16}	$6.258\,889\,420 \times 10^{10}$
4 Vesta	2.59×10^{20}	1.76×10^{17}	$1.728\,200\,810 \times 10^{10}$
2 Pallas	2.05×10^{20}	4.48×10^{18}	$1.367\,880\,950 \times 10^{10}$
10 Hygiea	8.42×10^{19}	4.91×10^{18}	$5.618\,320\,780 \times 10^{9}$
704 Interamnia	3.54×10^{19}	5.23×10^{18}	$2.362\,096\,860 \times 10^{9}$
15 Eunomia	3.16×10^{19}	1.25×10^{18}	$2.108\,538\,440 \times 10^{9}$
511 Davida	3.08×10^{19}	1.02×10^{19}	$2.055\,157\,720 \times 10^{9}$
3 Juno	2.63×10^{19}	2.37×10^{18}	$1.754\,891\,170 \times 10^{9}$
52 Europa	2.34×10^{19}	3.77×10^{18}	$1.561\,386\,060 \times 10^{9}$
16 Psyche	2.29×10^{19}	1.60×10^{18}	$1.528\,023\,110 \times 10^{9}$
65 Cybele	1.63×10^{19}	2.03×10^{18}	$1.087\,632\,170 \times 10^{9}$
87 Sylvia	1.51×10^{19}	1.30×10^{18}	$1.007\,561\,090 \times 10^{9}$
31 Euphrosyne	1.49×10^{19}	5.47×10^{18}	$9.942\,159\,100 \times 10^{8}$
7 Iris	1.41×10^{19}	2.41×10^{18}	$9.408\,351\,900 \times 10^{8}$
29 Amphitrite	1.33×10^{19}	1.65×10^{18}	$8.874\,544\,700 \times 10^{8}$
6 Hebe	1.28×10^{19}	2.51×10^{18}	$8.540\,915\,200 \times 10^{8}$

Notes:
- The value of the gravitational constant used is $G = 6.672\,59 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$.
- Source for masses and uncertainties: SiMDA catalogue Kretlow (2020).

3.5. Outliers

As explained in Subsection 2.4.3, the observations used in the estimation contain random noise and systematic error. They can also contain human errors such as misidentifications or typographical mistakes (Carpino et al. 2003). Such large errors can have a significant negative effect on the quality of an orbit estimation. As a result, observation outliers must be identified and removed. Carpino et al. (2003) introduced an automatic outlier rejection algorithm, which is commonly used in literature for removing problematic observations for the purposes of asteroid orbit determination (see, for example, Fuentes-Muñoz et al. 2025; Fenucci et al. 2025; Fuentes-Muñoz et al. 2024; Farnocchia et al. 2024; Fenucci et al. 2024; Dziadura et al. 2023). However, this algorithm is not (yet) implemented in Tudat and cannot be straightforwardly applied manually, as it is an iterative process that is meant to be performed alongside the iterations of the estimation. As an alternative, this research uses a more blunt procedure for outlier rejection; this procedure is outlined in Subsection 3.5.1. The selection of the value to use for the residual cutoff is explained in Subsection 3.5.2.

3.5.1. Outlier Rejection Process

Observation outliers can be identified by comparing them to an existing orbital solution which is known to be accurate. One such source of existing orbital solutions for asteroids is JPL Horizons, which is already used to define the initial state of the target body and the ephemerides of large main-belt asteroids. The process of outlier removal used in this research is as follows. With the observations retrieved from the MPC and the observation models defined as usual, the ephemeris of the target body over the full observation period is loaded from JPL Horizons. Simulated residuals are then calculated by taking the difference between the observations and the RA-DEC positions of the target's ephemeris.

A filter is be applied that removes any observation (RA-DEC pair) if either the RA or DEC simulated residual is larger than a certain pre-defined cutoff value. A check is performed to see if the first and/or last observations have been removed due to the filter. If so, the initial state, initial time, and final time of the estimation are updated accordingly. These updated values and the filtered observation set are passed into the estimation process.

3.5.2. Residual Cutoff Selection

In order to choose an appropriate value to use for the pre-defined residual cutoff input, an analysis was performed. The process described above of calculating and filtering residuals was carried out for all bodies in the F-set, using a range of different cutoff values. These values were: 1, 2, 3, 4, 5, 6, 8, 10, 15, 20, 30, and 50 arcseconds. In each case, the number of observations available and the number of observations filtered out was recorded. Figure 3.15 shows the percentage of observations removed per body as a histogram for several cutoff values as well as the rate of optical observation rejection reported by Fenucci et al. (2024).

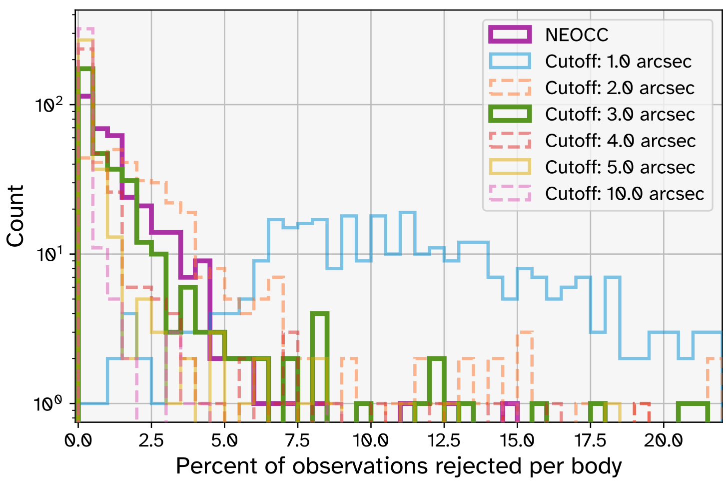


Figure 3.15: Histograms of the percentage of optical observations removed by the outlier rejection process per body, with separate curves for different values of the cutoff. Note that for legibility, a selection of six relevant cutoff values are represented here out of the full set of twelve. The rate of optical observation rejection (i.e. not including radar observations) reported by Fenucci et al. (2024) is labelled as NEOCC and is highlighted in purple. The selected residual cutoff value of 3 arcsec is highlighted in green.

Using the automatic outlier rejection algorithm of Carpino et al. (2003), Fenucci et al. (2024) rejected an average of 1.37% of the available optical observations, and rejected zero optical observations for 20.2% of the asteroids in the F-set. In Figure 3.15, the curves for the cutoff values of 2, 3, and 4 arcsec are the closest to the NEOCC curve. For these cutoff values, the mean rejection rate is 3.22%, 1.39%, and 0.81% respectively. The rate of rejecting zero observations is 10.4%, 34.3%, and 49.6% respectively.

The cutoff value of 3 arcsec was selected for use in the estimations because this curve is the closest match to the NEOCC curve in Figure 3.15, and it is the value that has the closest average rate of rejection compared to NEOCC. A cutoff value which results in similar behaviour to the NEOCC observation filtering was chosen because Fenucci et al. (2024) uses a more sophisticated automatic outlier rejection algorithm, so it is reasonable to assume that they generally correctly remove the right (amount of) true outliers. It should be noted that choosing a cutoff value that results in similar rejection rates to Fenucci et al. (2024) does not necessarily mean that the same observations that are being removed by the two methods. Outliers from very accurate sources may be left in because their residuals still fall below the cutoff, while some observations may be removed despite their residuals fitting inside the expected noise profile of certain lower accuracy observation sources (the noise of these observations would be accounted for by the weighting scheme applied in the estimation).

One consequence of removing outliers can be that the observation arc is decreased. With a cutoff value of 3 arcsec, there are 11 bodies for which the reduction in observation arc is greater than 1 week. The most extreme of these is 69230 Hermes: for this asteroid, removing a handful of observations in the 1930s results in the observation arc being reduced by 62.9 years because no (optical ground-based) observations are available between then and 2000. Another example is asteroid 66400 (1999 LT7), whose observation arc is reduced by 12.1 years due to the filtering out of the earliest available observations. The simulated residuals for asteroid 66400 are shown in Figure 3.16, where it can be seen that two observations before 1990 with high RA simulated residuals are filtered out, meaning the first used observations are in 1999.

While the automatic outlier rejection procedure used by Fenucci et al. (2024) rejects more than 10% of observations for only 3 bodies, the procedure used here rejects more than 10% of observations for for 8 bodies (with a cutoff value of 3 arcsec). The most extreme of these is the asteroid 2010 VK_{139}, for which 21.1% of observations are rejected. Its simulated residuals are shown in Figure 3.17, where it can be seen that a large number of observations from this asteroid's first observed opposition are rejected. This is typical for all 8 of the bodies with observation rejection rates over 10%.

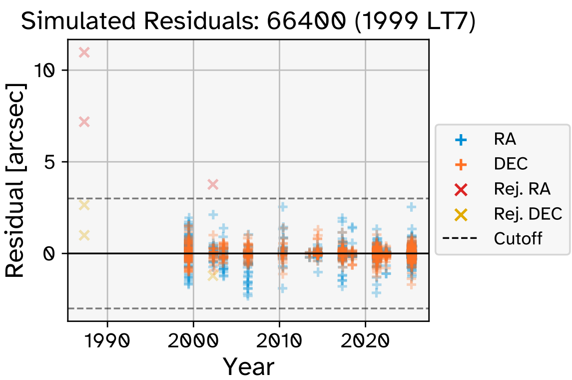


Figure 3.16: Simulated RA-DEC residuals for asteroid 66400 (1999 LT7), whose observation arc is reduced due to observation filtering. Accepted observations are marked with '+'. Rejected observations are marked with 'X'.

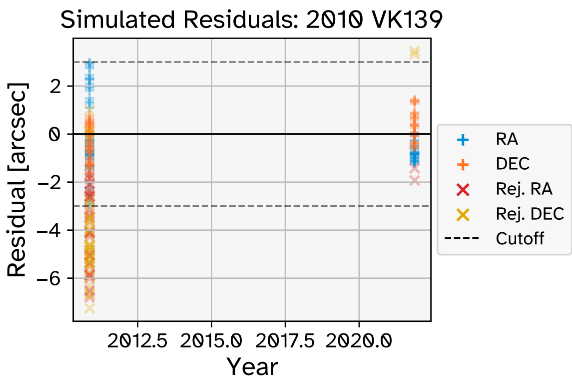


Figure 3.17: Simulated RA-DEC residuals for asteroid 2010 VK_{139}, for which a particularly high proportion of observations were rejected. Accepted observations are marked with '+'. Rejected observations are marked with 'X'.

4
Results

This chapter presents and discusses the results that were generated in this research. Section 4.1 gives details about the criteria that were applied to decide whether an estimation could be accepted as a valid detection of the Yarkovsky effect. Section 4.2 concerns the verification and validation of the estimations: by applying convergence tests, analysing observation residuals, and comparing the fitted orbits to accurate ephemerides, it is shown that the estimations perform well, providing context to the subsequent results. Section 4.3 presents a wide range of results pertaining to the performance of the estimations overall (as opposed to analysing individual asteroid's estimations), and compares these results against two different literature sources: Fenucci et al. (2024) (also referred to as NEOCC) and JPL SBDB.

Note that figures in this chapter follow a general colour code for consistency and readability:

- **Blue** represents the entire set of 347 completed estimations (E-set)
- **Green** represents the set of 281 accepted/successful estimations (S-set)
- **Red** represents the set of 66 estimations that were rejected (R-set)
- **Purple** represents results from Fenucci et al. (2024), labelled as NEOCC in figures and variables
- **Grey** represents data from JPL SBDB, labelled as JPL in figures and variables

Any deviations from the scheme defined above will be noted in the figure captions. The results of the E-, S-, R-sets may be referred to with the label 'Tudat', as opposed to results from literature sources (NEOCC and JPL SBDB). Note that this colour scheme applies to figures representing many asteroids/estimations. For figures of individual asteroids (e.g. residuals over time or position error over time), this colour scheme does not apply.

A distribution of the time it took to run each estimation can be seen in Figure 4.1. The median runtime was just over 5 minutes. 15% of the estimations took longer than 10 minutes. The longest single estimation took 61 minutes, for asteroid 2101 Adonis, which has an observation arc length of (and thus approximately a propagation duration of) 89.4 years and an eccentricity of 0.7641, meaning the time step used was 3 hours. The total runtime for all estimations was approximately 39 hours.

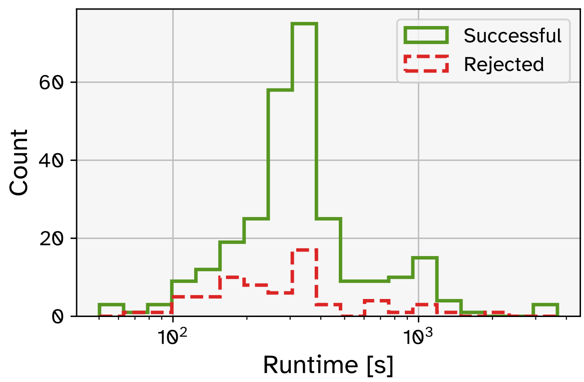


Figure 4.1: Distribution of run times of the estimations (E-set), divided by successful and rejected estimations.

4.1. Accepted & Rejected Detections

Of the 348 asteroids in the F-set, estimations were completed for 347 of them. The one missing asteroid is 101955 Bennu[1]. This set of 347 asteroids and their estimations is henceforth referred to as the E-set (for (completed) 'estimation').

Two rejection criteria were applied to determine whether the detected Yarkovsky effect was valid: one relating to the SNR of A_2 ($\mathrm{SNR}_{A_2} = |A_2|/\sigma_{A_2}$) and the other relating to the quality of the fit of the normalised residuals. 66 estimations (19%) were rejected due to failing the SNR_{A_2} criterion. None of the estimations failed on the normalised residuals criterion. The two different rejection criteria are outlined in more detail in Subsections 4.1.1 and 4.1.2. It is valuable to analyse these in order to understand what, if anything, may have gone wrong with these estimations. Note that a rejection on the basis of low SNR_{A_2} does not mean that the orbit determination was poor, just that it cannot be said definitively that a Yarkovsky effect is necessary to describe the motion of the body.

The remaining 281 estimations (81%) were accepted, having passed both rejection criteria. These estimations are therefore considered to be valid detections of the Yarkovsky effect. This set of 281 asteroids and their estimations is referred to in figures with the label 'Successful' and is referred to in the text as the S-set (for 'successful estimation'). A discussion of the

4.1.1. Rejection Criterion 1: Low SNR

The first criterion that must be passed for an estimation to be accepted is that SNR_{A_2} must be greater than 3. This criterion for the statistical significance of the estimated A_2 is the same as the one applied in Fenucci et al. (2024). For 66 (19%) of the 347 completed estimations, $\mathrm{SNR}_{A_2} < 3$. The SNR_{A_2} distribution for these estimations can be seen in the bottom plot of Figure 4.2 (red histogram). This set of 66 asteroids and their estimations is referred to in figures with the label 'Rejected' and is referred to in the text as the R-set (for 'rejected').

Note that these 66 estimations are rejected only in terms of their Yarkovsky detection. Thus, they will be considered separately or not at all in further discussions relating to the A_2 parameter. They are not necessarily poor orbital fits, however (as will be shown in Subsection 4.1.2), so they will be included in discussions about the residuals and errors in position and velocity.

4.1.2. Rejection Criterion 2: High Residuals

Not all observations are of equal quality, hence why the least-squares orbit estimation was weighted (see Subsection 2.5.1). This fact should also be taken into account when analysing the fit of an orbit estimation to the observation data. Thus, rather than looking at the angular residuals directly, they are normalised by their measurement error σ_i, which, in this research, is defined according to the weighting scheme of Vereš et al. (2017) (discussed in Subsection 2.4.4).

Assuming that (a) the observation uncertainties assumed by the applied weighting scheme (Vereš et al. 2017) are accurate reflections of the standard deviations of their true error distributions, (b) model error is negligible, and (c) the observations do not contain systematic errors, then the probability distribution of the normalised residuals (NR) is Gaussian with $\mu = 0$ and $\sigma = 1$. What is more important however are the magnitudes of the normalised residuals, or absolute normalised residuals (ANR). The probability distribution of the magnitude of a normally-distributed random variable with zero mean is a half-normal distribution. If a variable Y is normally distributed with $\mu = 0$ and $\sigma = 1$, then $|Y|$ follows a half-normal distribution with expected value $E(|Y|) = \sqrt{2/\pi} = 0.797\,885$ and variance $\mathrm{Var}(|Y|) = 1-(2/\pi) = 0.363\,380$ (Olive 2005, Sec 3.11). Following such a distribution, 68.3% of ANRs would be expected to be between 0 and 1, and 27.2% between 1 and 2. Only 4.28% of observations are be expected to have ANRs between 2 and 3, and only 0.27% can be expected to have ANRs larger than 3.

For a given estimation with n observations, consider the distribution of ANR. For each observation, there is a corresponding ANR, so there are n samples of the ANR distribution. Let the mean of these samples be called MANR (mean of the absolute normalised residuals). According to the central limit theorem (CLT), with sufficiently large n, the distribution of MANR can be approximated as a normal distribution with $\mu = E(\mathrm{ANR})$ and $\sigma^2 = \mathrm{Var}(\mathrm{ANR})/n$ (and thus $\sigma = \sqrt{\mathrm{Var}(\mathrm{ANR})/n}$). If the measured MANR is significantly larger than $E(\mathrm{ANR})$, then it is unlikely that this is just due to random chance.

Thus, the second criterion was applied by evaluating the following inequality for each estimation:

$$\mathrm{MANR} < E(\mathrm{ANR}) + 3\sqrt{\frac{\mathrm{Var}(\mathrm{ANR})}{n}} \tag{4.1}$$

[1]Bennu's JPL Horizons ephemeris contains discontinuities in its position in (in ∼2017 and ∼2031). This causes issues with how observation outliers are processed (see Section 3.5) and also causes an error when retrieving the ephemeris file via the Horizons API in the automated way that was employed for the rest of the estimations. This is a notable omission because the SNR_{A_2} achieved for Bennu in literature tends to be among the highest of any NEAs (Farnocchia et al. 2013; Dziadura et al. 2023; Fenucci et al. 2024).

If the inequality was fulfilled, the estimation passed the second criterion. If not, it failed. If MANR is more than 3σ larger than the expected value, then there is only a 0.135% probability that this is due to random chance. If this is the case, it is an indication that there is likely a problem with that estimation — such as uncorrected systematic errors in the observations, insufficient integrator settings, insufficient dynamic model, or insufficient outlier removal — and it should therefore not be considered a valid estimation. It may also be an indication that the observation uncertainties assumed in the weighting scheme underestimate the true noisiness of the data.

All 347 estimations passed the second criterion, and thus none were rejected on the basis of being a poor fit to the observation data. The MANR distribution for all estimations can be seen in the left plot of Figure 4.2.

4.1.3. Detection Discussion

The results from applying the rejection criteria described above indicate that the estimations provide reasonably good fits to the observation data in all cases. The extent to which this is true will be discussed further in Section 4.2. Furthermore, in a large majority of cases (81%), a statistically significant Yarkovsky effect could be detected. Section 4.3 presents comparisons between these Yarkovsky results and the results reported in literature.

Figures 4.2 and 4.3 show how the accepted and rejected estimations are distributed with respect to MANR, SNR_{A_2}, eccentricity, and semi-major axis. The highest MANR of any estimation was just 0.7178, which comfortably well below the expected value of $E(\text{ANR}) = 0.7980$. This is likely due to the fact that the uncertainties assumed in the weighting scheme of Vereš et al. (2017) do not perfectly represent the noisiness of the observation data, and that the actual random error is generally lower than assumed. This aligns with the authors' finding that the weighting scheme is somewhat conservative.

In the estimations performed by Fenucci et al. (2024), $\text{SNR}_{A_2} > 3$ for all 348 asteroids in the E-set. There may be several different reasons as to why the SNR_{A_2} achieved in the Tudat estimations is lower than that of the NEOCC estimations in these cases. These include the different treatment of outliers and the inclusion of radar and space-based optical observations in the NEOCC estimations. These reasons are discussed further in Section 4.3 (see Figures 4.28a and 4.28b). The shape of the SNR_{A_2} curve is also different from the NEOCC results: in Fig. 3 of Fenucci et al. (2024), the peak of the SNR_{A_2} curve is at 3, and the number of estimations with higher SNR_{A_2} drops off quite quickly. In contrast, the SNR_{A_2} curve at the bottom of Figure 4.2 has its peak at around 5, with values dropping off relatively steadily to either side of the peak. Again, this different behaviour may be due to differences in outlier rejection and observation datasets.

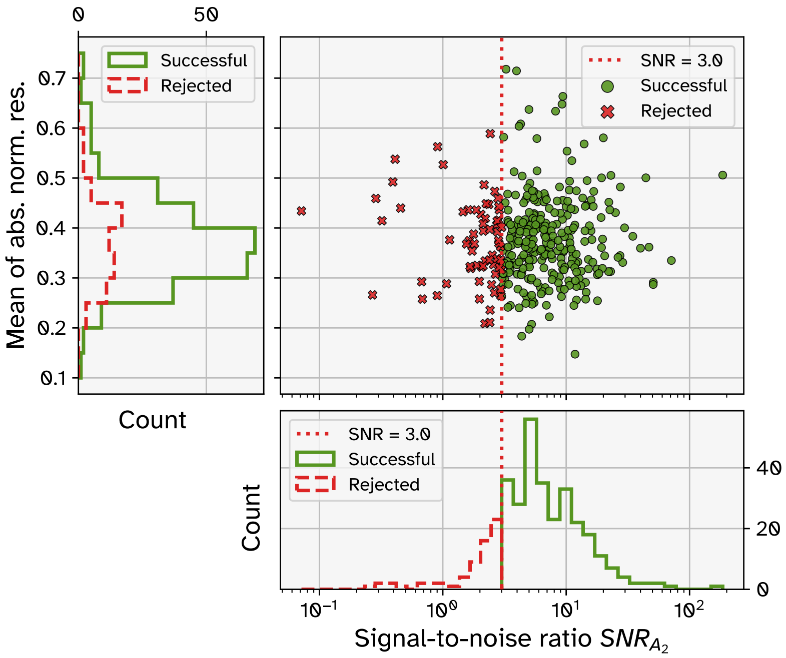


Figure 4.2: Distribution of the mean of the absolute normalised residuals (MANR) and the signal-to-noise ratio of A_2 (SNR_{A_2}) for all estimations (E-set), divided by acceptance and rejection.

Figure 4.3 shows that the asteroids in the S-set and R-set are fairly well distributed in both eccentricity and semi-major axis. There are no major differences in the rate of rejection for estimations within any particular range of e or a. This is an indication that these orbital characteristics (nor the numerical integration settings, which depend on e) are not critical factors affecting whether an estimation has low SNR_{A_2}.

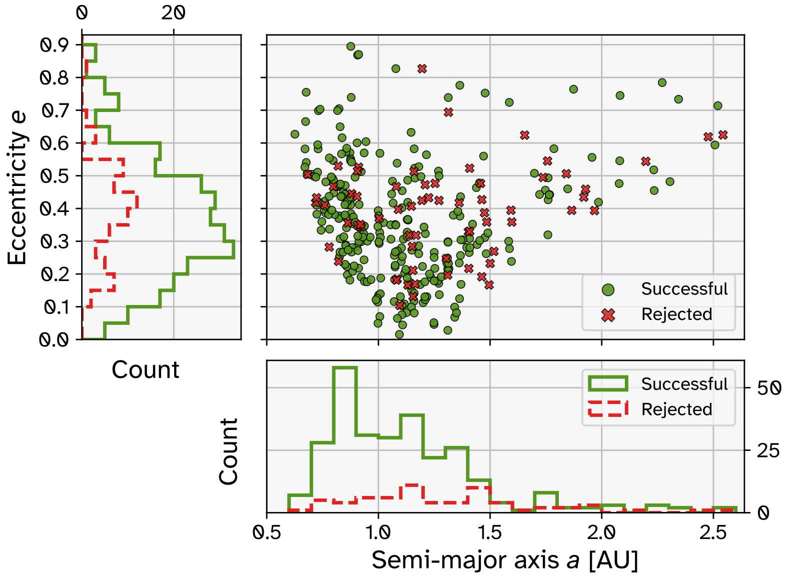


Figure 4.3: Distribution of the eccentricity e and semi-major axis a of the asteroids in the E-set, marked by acceptance and rejection.

4.2. Verification & Validation

This section presents evidence and discussions of the extent to which the estimations performed have correctly processed the observation data and produced accurate orbit estimations. It is important to demonstrate this in order to put the other results in context and to understand any shortcomings. Subsection 4.2.1 describes the criteria that were applied to check whether the estimations converged, as well as the results of this convergence testing. Subsection 4.2.2 presents an analysis of the observation residuals by using the results of a few asteroids as representative examples. Subsection 4.2.3 presents comparisons between the orbits of the estimations from this research and the asteroid ephemerides available via JPL Horizons.

4.2.1. Convergence Testing

Three criteria were applied to check whether the estimation process had properly converged. By default, the number of iterations was set to four. Convergence was checked by analysing the set of estimated parameters — three position components, three velocity components, and the Yarkovsky parameter A_2 — for each of the four iterations. As mentioned in Subsection 3.1.3, the initial state (position and velocity) of the target body is taken from JPL Horizons and the initial value of A_2 is taken from Table B.1 of Fenucci et al. (2024). Updates to these parameters are calculated after each iteration according to the weighted least-squares process (see Subsection 2.5.1), and the next iteration uses these updated values for its propagation. Updates to the parameters are also calculated after the last iteration, defining the parameters that would have been used for the next iteration, if the set number of iterations was higher. The best iteration is defined as the one for which the RMS of the normalised residuals (RMSNR) is the lowest.

The following are the criteria which were used to check the convergence of the estimations:

- **Criterion 1:** This criterion is applied to check whether the estimation has converged with respect to the observation residuals. Calculate the difference in the RMS of the residuals (arcsec) between the last iteration and the second-to-last iteration. This criterion is passed only if this difference is less than 0.001 arcsec.
- **Criterion 2:** This criterion is applied to check whether the individual estimated parameters have converged, using their formal uncertainty for scale. For each parameter in the parameter vector, calculate the difference between the value in the best iteration and the value in the iteration after the best iteration. Divide this by the formal error of that parameter from the best iteration. This criterion is passed only if this value is less than 0.001 for all parameters. This represents a change of less than 0.1% of the formal error of the parameter.
- **Criteria 3a/b/c:** These criteria are only applied if the last iteration was the best iteration. If the absolute change in the magnitude of the (a) initial position, (b) initial velocity, or (c) A_2 parameter between the best iteration and the 'next' iteration is less than (a) 1 m, (b) 10^{-7} m s^{-1} (100 nm s^{-1}), and (c) 10^{-17} m s^{-2} (10 am s^{-3}), then this criterion is passed.

If the last iteration was the best iteration, criterion 3 is applied because a significantly large parameter change was determined for the next iteration, but this iteration was not performed. It may be that this would be an improvement upon the best iteration, so failures of this criterion are marked as not converged.

For criteria 2 and 3, the parameters of the best iteration are compared with those of the iteration immediately after the best iteration. This was not possible for criterion 1 because in the cases where the last iteration is the best, the RMS of the residuals for the next iteration is not available. Instead, the results of the last two iterations are compared, regardless of which iteration was the best.

Using four iterations, all estimations passed criteria 1, 2, and 3a. The estimation for asteroid 85953 (1999 FK_{21}) failed criteria 3a and 3b and 152563 (1992 BF) failed criterion 3b. These estimations were re-run with 6 iterations. In both cases, the best iteration remained the 4th one. Therefore, all estimations can be confirmed to be converged.

Figure 4.4 shows that a considerable number of estimations converge to their best solution in less than the maximum number of iterations. This is likely because the initial state estimates tend to be quite good. As can be seen in Figure 4.5, the initial position difference between first iteration and the best iteration is less than 100 km for most estimations, and that the distribution is similar for the sets of accepted and rejected estimations.

4.2.2. Observation Residuals

Observation residuals are the differences between the reported observations of a body' position and the modelled position as calculated during the orbit estimation. Analysing these allows us to assess the

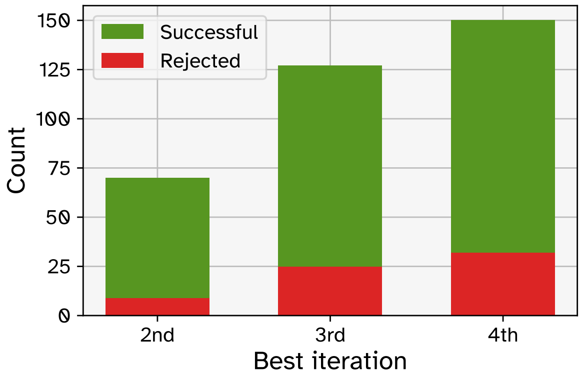


Figure 4.4: Bar chart showing which of the iterations of the estimation process resulted in the lowest RMS of the normalised residuals.

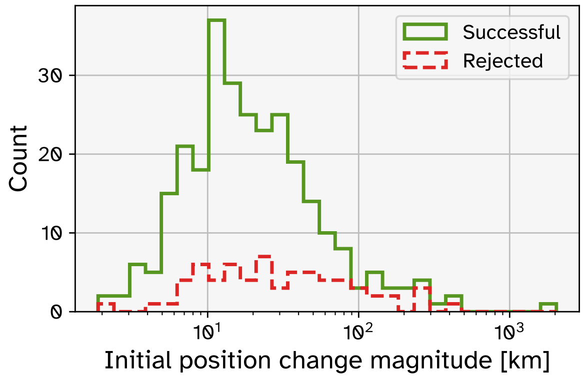
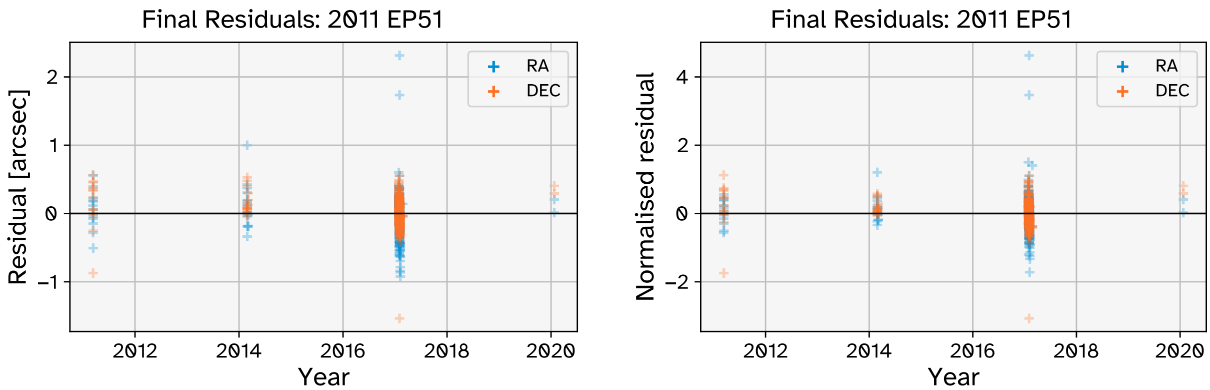


Figure 4.5: Distribution of the magnitude of the change in initial position between the first iteration and the best iteration, for all estimations (E-set), divided by successful and rejected estimations. The initial position of the first iteration is taken from the JPL Horizons ephemeris of the target body.

quality of the orbital fit to the real-life observations of the asteroid's position. A good orbit determination will produce a close match to the observations, ideally with the only remaining residuals being attributable to observation error. This section demonstrates the residuals achieved by the estimations performed in this research.

In this report, the term 'O-C residual', for 'Observed minus Calculated', refers to the raw, un-normalised residuals. Since all of the observations used in this research were angular RA-DEC observations, O-C residuals have angular units (arcseconds are the chosen unit of angle used here). As mentioned in Subsection 4.1.2, normalised residuals (NR) are calculated as the O-C residuals divided by their measurement error σ_i (in this research, these errors are defined according to the weighting scheme of Vereš et al. (2017), discussed in Subsection 2.4.4). Note that the terms 'final residuals' and 'post=fit' residuals both refer to the residuals of the best iteration of the estimation. 'Initial residuals' or 'pre-fit' residuals may be used to refer to the residuals of the first iteration.

Observation outliers were filtered out according to the procedure defined in Subsection 3.5.1. In short, the JPL Horizons ephemeris (see Subsection 4.2.3 for more details on these ephemerides) were used to calculate 'simulated' residuals, i.e. the differences between the observed position and the position according to the ephemeris. Any observations whose RA or DEC simulate residual was greater than 3 arcsec was rejected and removed from the observation dataset. Figures 3.16 and 3.17 demonstrates these simulated residuals and how observations with residuals larger than the cutoff are removed.



(a) O-C Residuals

(b) Normalised residuals

Figure 4.6: Residuals for the best iteration of the estimation of asteroid 2011 EP_{51}.

Figure 4.6 shows the observation residuals (O-C and normalised) for the asteroid 2011 EP_{51}. This is a case with a particularly good fit to the observation data: the RMS of the O-C residuals is 0.2616 arcsec, and the RMS of the normalised residuals (RMSNR) is 0.4663. It is noteworthy that the majority of the observations have absolute O-C residuals of less than 1 arcsec. For the small number of observations which have larger O-C residuals, the corresponding normalised residuals are quite large, over 4 in one case. These are outliers which nevertheless passed the rejection procedure, indicating that this procedure is not perfect and that the results of the estimations may be somewhat contaminated by the presence of outliers.

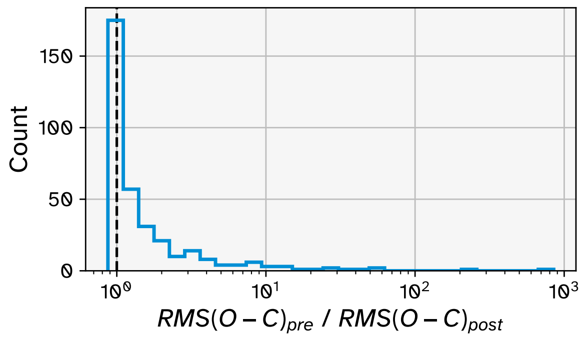


Figure 4.7: Distribution of the ratio of the RMS of the O-C residuals in the first and best iterations.

For 2011 EP_{51}, the quality of the fit to the observations was very similar in the first iteration as in the best iteration: from an RMS of O-C residuals of 0.2695 arcsec in the first iteration to an RMS of O-C residuals of 0.2616 arcsec in the best iteration (a ratio of 1.030). Although this ratio was close to 1 for many estimations, this was not the case for a significant number of them, as Figure 4.7 shows.

To understand this, asteroid 2020 PP_1 will be used as an example. For this asteroid $\text{RMS(O-C)}_{\text{pre}} = 4.039$ arcsec and $\text{RMS(O-C)}_{\text{post}} = 0.3329$ arcsec, meaning the ratio is is 12.13. Figures 4.8 and 4.9 show how the O-C residuals change between the first and best iterations. In the pre-fit residuals, there is a clear negative drift in RA of about 10 arcsec and some large deviations from zero in DEC, up to about 12 arcsec. This kind of behaviour may be an indication that something is missing from the dynamic model, causing the propagated state to gradually differ from the true state by more and more over time. It may also be due to errors in the initial state, especially if the initial epoch of the propagation is one where the state derivative is large. However, depending on the extent of these effects, they may be corrected for in the estimation anyway by adjusting the estimable parameters such that the effect is ‘absorbed’. In the post-fit residuals (Figure 4.9), the large trends and deviations from zero have been eliminated, making the orbit determination a very good fit to the observation data.

Figures 4.10a and 4.10b summarise the RMS of the residuals (both O-C and normalised) for all of the estimations in the E-set. These figures show that the orbit estimations performed in this research provide very good fits to the observation data. The highest RMS of O-C residuals is 0.9843 arcsec and the highest RMSNR is 1.0990, and only 4 estimations have $RMSNR > 1$. In these 4 cases, the O-C residuals tend to be quite small but a certain proportion of observations appear to be outliers. These outliers are either quite large, as indicated by high |NR|, or there are relatively few observations in total and the outlier observations make up a significant proportion of the total observations. One example of this is for asteroid 2011 CE_{50}, whose residuals are shown in Figure 4.11. The number of observations for this asteroid is only 73. None of these were filtered out by the outlier rejection process; all but one of the (absolute) simulated residuals were <1 arcsec, with the remaining one still being <2 arcsec. Similarly, the absolute O-C residuals resulting from the estimation are all <1 arcsec. However, of the 73 observations, 54 come from the Pan-STARRS 1 survey (Hawaii, USA) and 14 come from the Mt. Lemmon survey (Arizona, USA). These observation sources are known to be quite accurate and are accordingly assigned given low observation uncertainties (0.2 arcsec and 0.5 arcsec, respectively) by the weighting scheme of Vereš et al. (2017). The result is that, despite the O-C residuals being small, several observations actually have quite high absolute normalised residuals, thereby resulting in a large RMSNR. Thus, the presence of relatively high RMSNR for these estimations can be attributed to insufficient outlier rejection, rather than to a poor overall orbital fit.

Figures 4.12 to 4.14 present the distributions of the RMSNR against eccentricity, semi-major axis, and observation arc length. None show strong trends that indicate that RMSNR is correlated with any of these parameters, though it is noteworthy that all of the estimations with $RMSNR > 0.8$ occur for eccentricities less than 0.6. The numerical integration settings are more relaxed for $e < 0.6$, indicating that they may not be sufficient in these cases. Although there may appear to trends of smaller RMSNR with larger semi-major axis or longer observation arc, there are far fewer points in these areas of the figures, indicating that such a trend may not continue to be seen if more estimations were to be carried out on asteroids with large semi-major axes or long observation arcs.

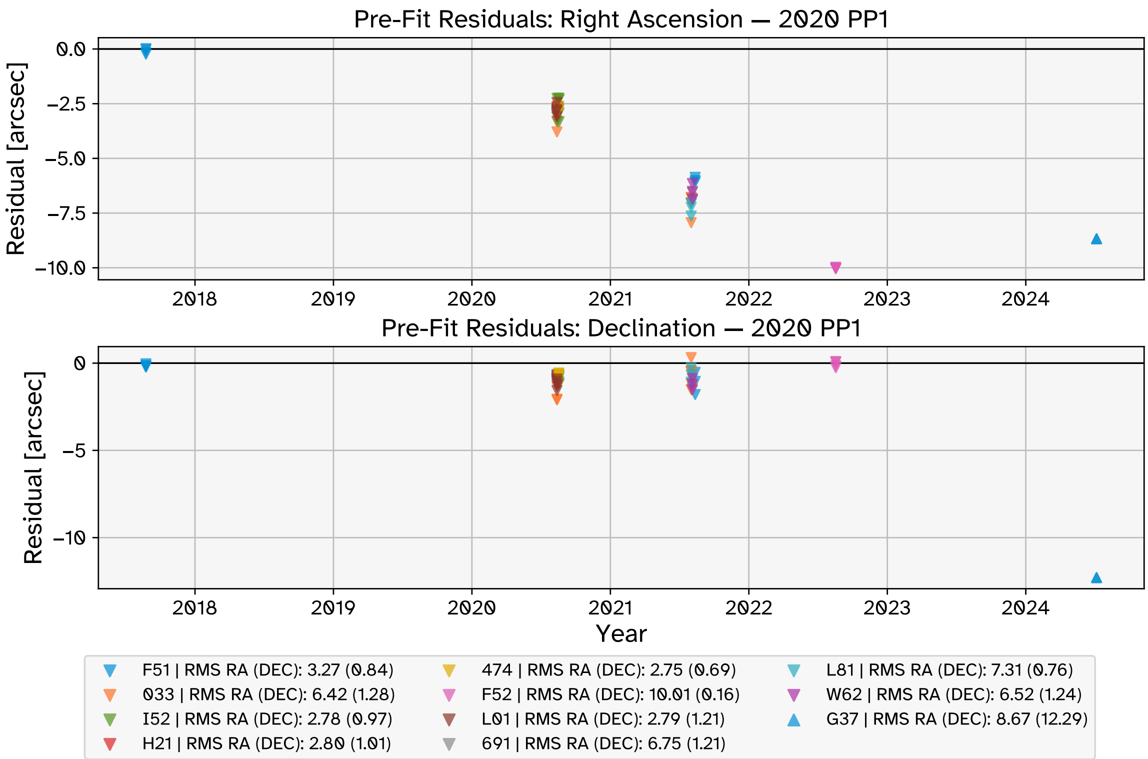


Figure 4.8: O-C residuals for the first iteration of the estimation of asteroid 2020 PP_1. The observations are marked by the observatory that made them, referred to using their 3-character MPC observatory codes. For the full names and coordinates of these observatories, please refer to the MPC website: `https://www.minorplanetcenter.net/iau/lists/ObsCodesF.html`.

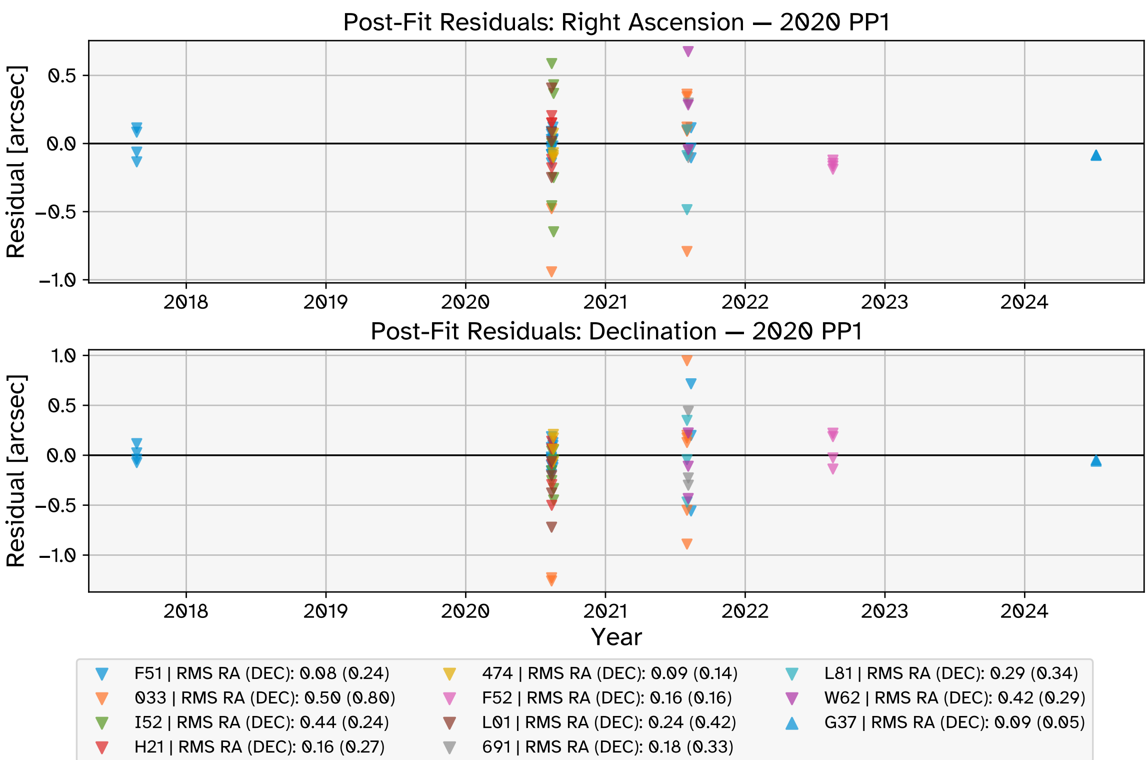
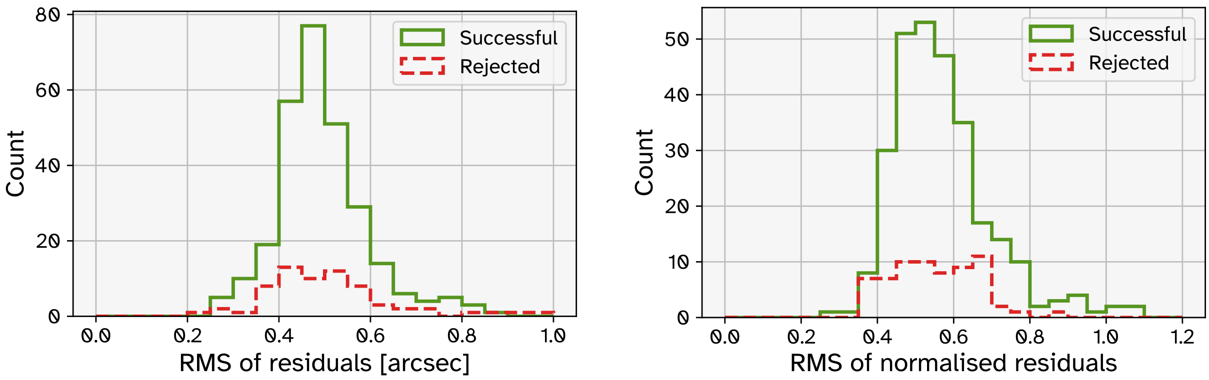


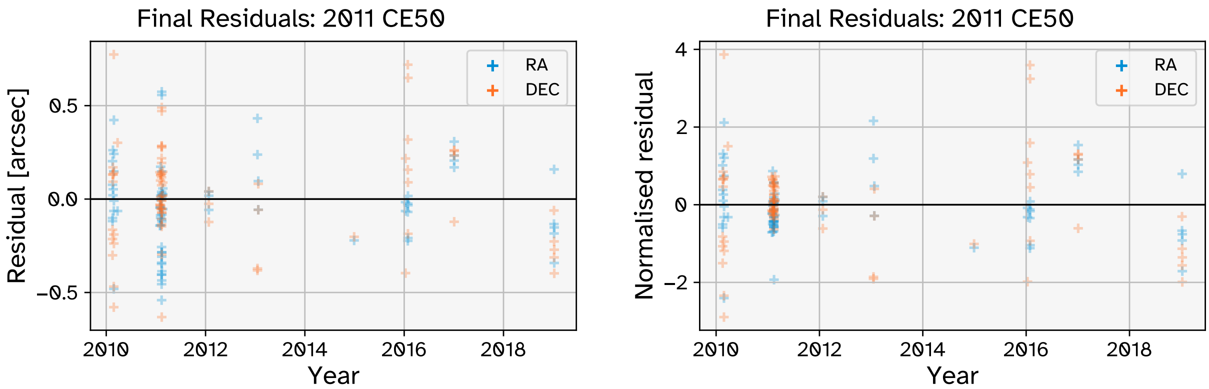
Figure 4.9: O-C residuals for the best iteration of the estimation of asteroid 2020 PP_1. The observations are marked by the observatory that made them, referred to using their 3-character MPC observatory codes.



(a) O-C Residuals

(b) Normalised residuals

Figure 4.10: RMS of residuals for all estimations (E-set), divided by successful and rejected estimations



(a) O-C Residuals

(b) Normalised residuals

Figure 4.11: Residuals for the best iteration of the estimation of asteroid 2011 CE_{50}.

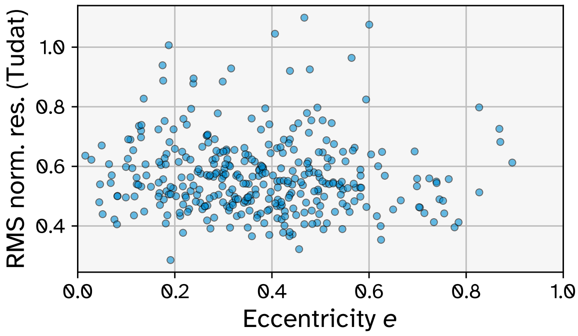


Figure 4.12: Distribution of RMS of the normalised residuals against eccentricity *e*.

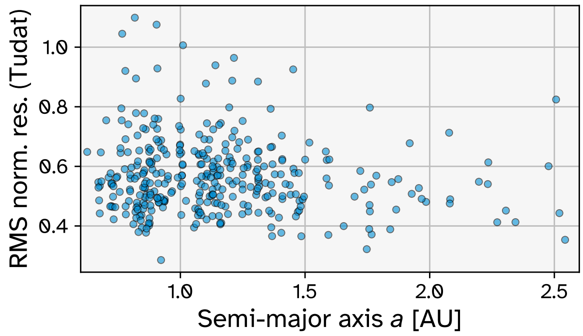


Figure 4.13: Distribution of RMS of the normalised residuals against semi-major axis *a*.

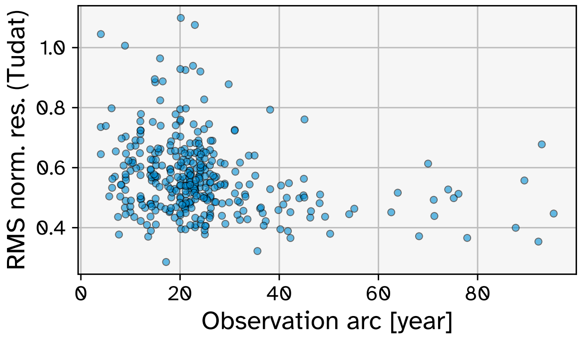


Figure 4.14: Distribution of RMS of the normalised residuals against observation arc length.

4.2.3. Comparison with JPL Horizons

JPL Horizons[2] (JPLH) is an online database that contains, amongst other things, accurate ephemerides for small Solar System bodies, generated by the Solar System Dynamics Group of NASA's Jet Propulsion Laboratory (JPL). JPLH is a useful resource because research publications that perform orbit estimations (e.g. Fenucci et al. 2024) generally only publish high-level results and not the entire state histories of the orbits, so directly comparing the state at any point in time to analyse differences is not typically possible. Using JPLH ephemerides, such comparisons are possible. This section compares the fitted orbits of this research to the JPLH ephemerides to demonstrate similarities and differences between the two orbit sources. The ephemerides of the asteroids in the E-set, including their uncertainties, were accessed via the Horizons API and downloaded as SPK files on 20 April 2026.

While the exact inputs and methods used to generate these ephemerides are not publicly available, it is known that JPLH makes use of a number of high-accuracy observation sources which are not available through the MPC for their orbit determinations, including occultation observations (Dunham et al. 2021), astrometry provided by the Gaia mission (Fenucci et al. 2024), and positional data from the space missions (e.g. Cheng et al. 2018). JPLH also publishes the formal errors of the ephemerides, which act as an indication of their accuracy, which is generally very good. In this section, the JPLH ephemerides are assumed to be a good representation of the true orbits of the bodies in the E-set. Of course, this is not entirely true, as the JPLH orbit determinations do contain some model error and observation error, as indicated by the published formal uncertainties. Furthermore, there is significant overlap between the observation datasets used in the Tudat and JPLH orbit determinations, meaning errors may be correlated. The interpretations of the results must therefore take into account the limited validity of the assumption that the JPLH ephemerides represent the true orbits.

The state difference between the Tudat estimations and the JPL Horizons ephemeris of a given body can be compared at any time within the propagation period. However, in order to condense the overall level of the difference between the two orbits, the following terms are defined:

$$\mathrm{TE}_{\mathrm{pos.}}(t) = \sqrt{(x - x_{\mathrm{ref}})^2 + (y - y_{\mathrm{ref}})^2 + (z - z_{\mathrm{ref}})^2} \tag{4.2}$$

$$\mathrm{TE}_{\mathrm{vel.}}(t) = \sqrt{(v_x - v_{x\mathrm{ref}})^2 + (v_y - v_{y_{\mathrm{ref}}})^2 + (v_z - v_{z\mathrm{ref}})^2} \tag{4.3}$$

where x, y, z (v_x, v_y, v_z) are the position (velocity) components of the estimated trajectory at time t and $x_{\mathrm{ref}}, y_{\mathrm{ref}}, z_{\mathrm{ref}}$ $(v_{x\mathrm{ref}}, v_{y_{\mathrm{ref}}}, v_{z\mathrm{ref}})$ are the position (velocity) components taken from the ephemeris at time t. Under the assumption that the ephemeris is an accurate representation of the true orbit of the target body, TE is a measure of the true error. $(\mathrm{TE}_{\mathrm{RMS}})_{\mathrm{pos.}}$ is defined as the RMS of $\mathrm{TE}_{\mathrm{pos.}}$ over all epochs of the estimation and $(\mathrm{TE}_{\mathrm{RMS}})_{\mathrm{vel.}}$ is defined as the RMS of $\mathrm{TE}_{\mathrm{vel.}}$ over all epochs of the estimation. Each of these gives an indication of the overall level of true error for this estimation's propagated trajectory.

It can be useful to analyse the true error measures directly to understand how much the estimated orbits deviate from the JPL Horizons ephemerides. Arguably more important, however, is analysing how well the calculated (formal) error approximates the true error. Let S_r and S_v be defined as:

$$S_r(t) = \sqrt{{\sigma_x}^2 + {\sigma_y}^2 + {\sigma_z}^2} \tag{4.4}$$

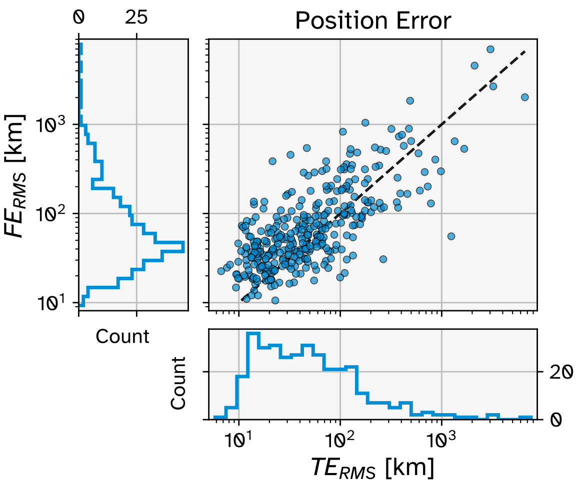
$$S_v(t) = \sqrt{{\sigma_{v_x}}^2 + {\sigma_{v_x}}^2 + {\sigma_{v_x}}^2} \tag{4.5}$$

where $\sigma_x, \sigma_y, \sigma_z$ $(\sigma_{v_x}, \sigma_{v_y}, \sigma_{v_z})$ are the formal errors of the position (velocity) components. These formal errors are calculated for any time t using the process discussed in Subsection 2.5.2 (see Equation (2.47)). Thus S_r and S_v give an indication of the total formal error in position and velocity. Note that S_r and S_v are not quantities that describe the probability distribution of the position and velocity vectors; they are merely used as indications to distil multiple uncertainty terms into single values.

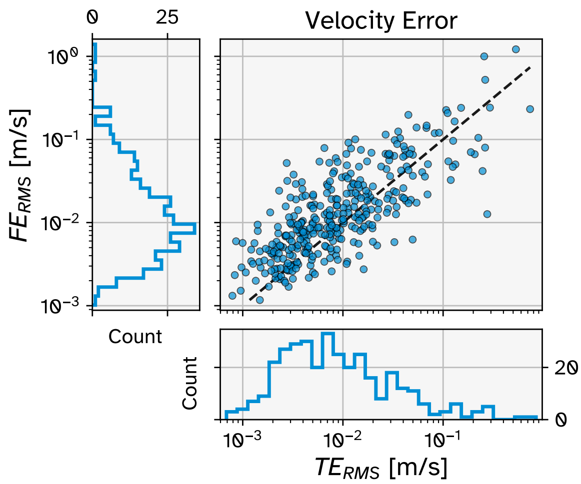
$(\mathrm{FE}_{\mathrm{RMS}})_{\mathrm{pos.}}$ is defined as the RMS of S_r over all epochs of the estimation and $(\mathrm{FE}_{\mathrm{RMS}})_{\mathrm{vel.}}$ is defined as the RMS of S_v over all epochs of the estimation. Each of these gives an indication of the overall level of the formal error for this estimation's propagated trajectory.

By comparing $\mathrm{FE}_{\mathrm{RMS}}$ and $\mathrm{TE}_{\mathrm{RMS}}$, we can get a sense of how well the formal errors represent the true errors. This can be seen in Figures 4.15 and 4.16, for the errors in position and velocity. In both Figures 4.15a and 4.15b, the distributions are generally near the diagonal but with a wide spread, and with somewhat more points lying above the diagonal than below. This indicates that, on average, the formal error is pessimistic and overestimates the true error. The majority of the formal and true position errors are in the range 10-100 km, with a significant number in the range 100-1000 km, and a small number

[2] `https://ssd.jpl.nasa.gov/horizons/`



(a) Position error



(b) Velocity error

Figure 4.15: Distributions of a measure for the overall formal error (FE_{RMS}) against a measure for the overall true error (TE_{RMS}) of each estimation's output trajectory. The black dashed lines are the diagonals.

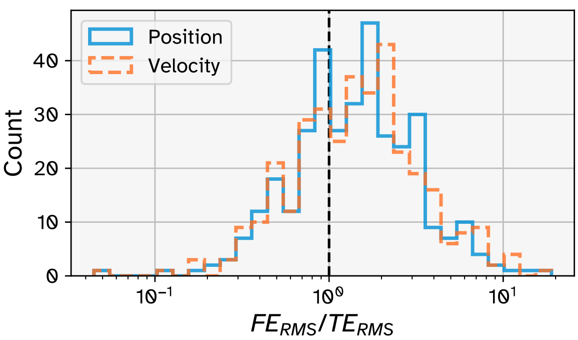


Figure 4.16: Distribution of the ratio FE_{RMS}/TE_{RMS} for position and velocity.

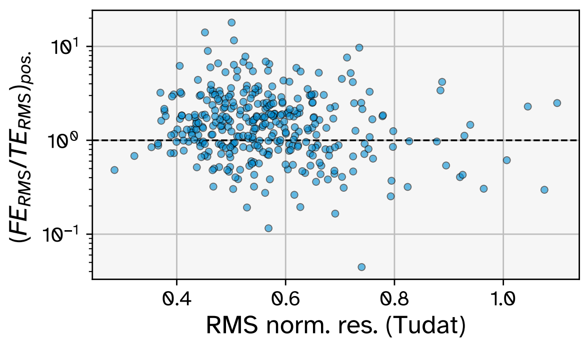


Figure 4.17: Distribution of the ratio $(FE_{RMS}/TE_{RMS})_{pos.}$ against the RMS of the normalised residuals.

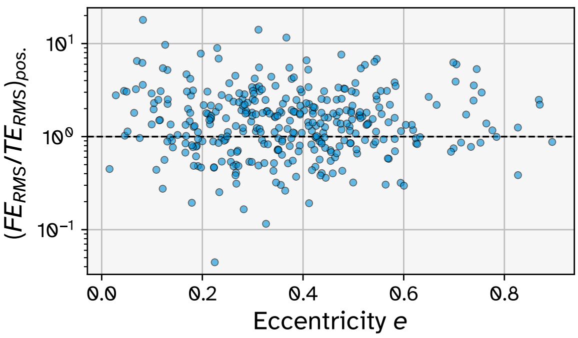


Figure 4.18: Distribution of the ratio $(FE_{RMS}/TE_{RMS})_{pos.}$ against eccentricity.

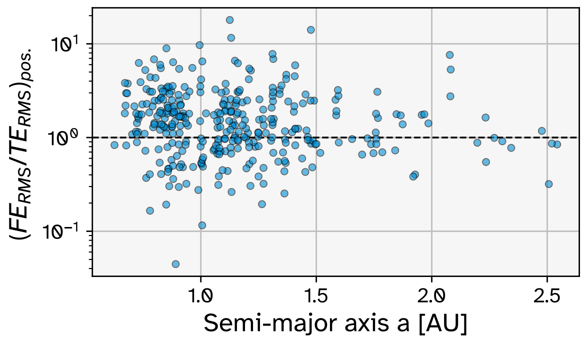


Figure 4.19: Distribution of the ratio $(FE_{RMS}/TE_{RMS})_{pos.}$ against semi-major axis.

larger than 1000 km. Figure 4.16 shows the ratio of the formal-to-true error (FE_{RMS}/TE_{RMS}) for both position and velocity. In almost all cases, this ratio is between 0.1 and 10, and again there is a visible skew towards values larger than 1.

The FE_{RMS}/TE_{RMS} distributions for position and velocity are very similar. Therefore, only the position error ratio was plotted for Figures 4.17 to 4.19. These figures were made to investigate whether a correlation could be found between the formal-to-true error ratio and any other orbital parameters or other estimation results. As can be seen, there is no strong correlation between the formal-to-true error ratio and the eccentricity, semi-major axis, nor the RMSNR. The same is true for the observation arc length (not plotted).

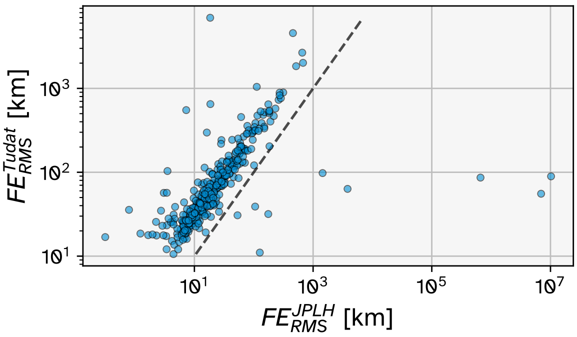


Figure 4.20: Distribution of the $(FE^{\mathrm{Tudat}}_{\mathrm{RMS}})_{\mathrm{pos.}}$ against $(FE^{\mathrm{JPLH}}_{\mathrm{RMS}})_{\mathrm{pos.}}$, each measures of the overall formal error of the orbit determinations produced by Tudat and JPLH. The dashed black line is the diagonal.

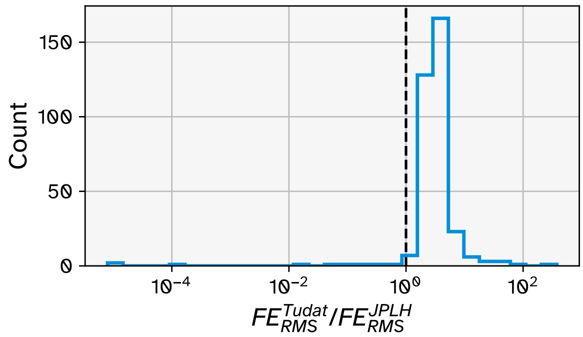


Figure 4.21: Distribution of the ratio $(FE^{\mathrm{Tudat}}_{\mathrm{RMS}}/FE^{\mathrm{JPLH}}_{\mathrm{RMS}})_{\mathrm{pos.}}$.

Figures 4.20 and 4.21 show how the formal error measure $(FE_{\mathrm{RMS}})_{\mathrm{pos.}}$ compares between the Tudat and JPLH orbit determinations. Clearly, the formal error of the JPLH orbit determinations are generally considerably lower than those of the Tudat estimations. In Figure 4.20, most of the points cluster around a line which is shifted upwards from the diagonal line, but there is a significant spread for both orbit determination sources, with an especially large spread in $(FE^{\mathrm{JPLH}}_{\mathrm{RMS}})_{\mathrm{pos.}}$, seemingly due to anomalies in the JPLH ephemeris uncertainty tables. 338 points lie above the diagonal and 9 lie below it. Since the precise way in which observation data is handled (i.e. outlier rejection and potentially iterative observation weighing) in the generation of the JPLH ephemerides is not publicly available, it is not straightforward to clearly interpret the meaning of these figures, but they indicate that the JPLH ephemerides generally have lower uncertainty and can be reasonably used as a reference for the 'true' orbit in most cases.

Figure 4.22 is a demonstration of the position error of with respect to the JPLH ephemeris (true error) for asteroid 612050 (1997 GL_3), as well as the formal errors of the Tudat and JPLH orbit determinations in the RSW directions and for the position vector. The zero line of each subplot represents the JPLH asteroid ephemeris, and the coloured regions above and below represent the $\pm 3\sigma$ uncertainty range of the JPLH ephemeris. For the most part, the true error curves are within these coloured regions, meaning the Tudat orbit estimation is quite accurate. Asteroid 612050 has a period of 1249.9 days (3.422 years), and it can be seen that many of the true and formal errors oscillate with this period. This is typical and is likely attributable to differences in the dynamic model used. This asteroid was chosen as an example for which the ratio $(FE_{\mathrm{RMS}}/TE_{\mathrm{RMS}})_{\mathrm{pos.}}$ is close to 1 — in this case it is 0.9922 — meaning the calculated formal errors are a very good approximation of the true error. For this asteroid, the ratio $(FE^{\mathrm{Tudat}}_{\mathrm{RMS}}/FE^{\mathrm{JPLH}}_{\mathrm{RMS}})_{\mathrm{pos.}}$ is 2.679, which is also quite typical amongst all estimations.

Figure 4.23 demonstrates the true position error and formal errors for asteroid 2008 LG_2. This asteroid is an example of an estimation where the ratio $(FE_{\mathrm{RMS}}/TE_{\mathrm{RMS}})_{\mathrm{pos.}}$ is high — in this case it is 11.41. It is clear that the formal errors of the Tudat estimations are large in comparison to the true errors, which are close to the zero line in all subplots. For this asteroid, the ratio $(FE^{\mathrm{Tudat}}_{\mathrm{RMS}}/FE^{\mathrm{JPLH}}_{\mathrm{RMS}})_{\mathrm{pos.}}$ is 2.177.

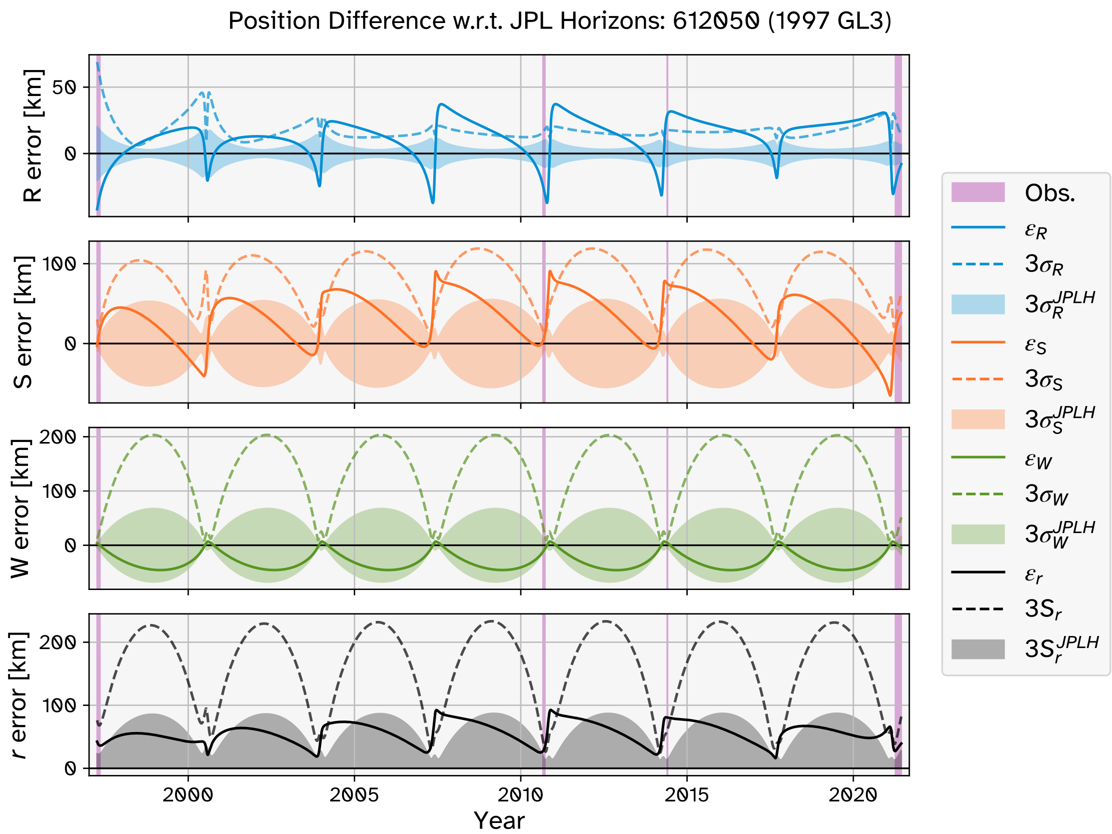


Figure 4.22: Position error ε with respect to JPL Horizons ephemeris of asteroid 612050 (1997 GL_3). The dashed coloured curves represent the formal errors (3σ) of the Tudat estimation in the RSW directions. The coloured regions represent the formal error in the JPLH ephemeris ($\pm 3\sigma$). In the bottom plot, the dashed black curve and grey region are the $3S_r$ measure of the overall formal error of the Tudat and JPLH orbit estimations, respectively. Pink regions indicate periods in which observations are available.

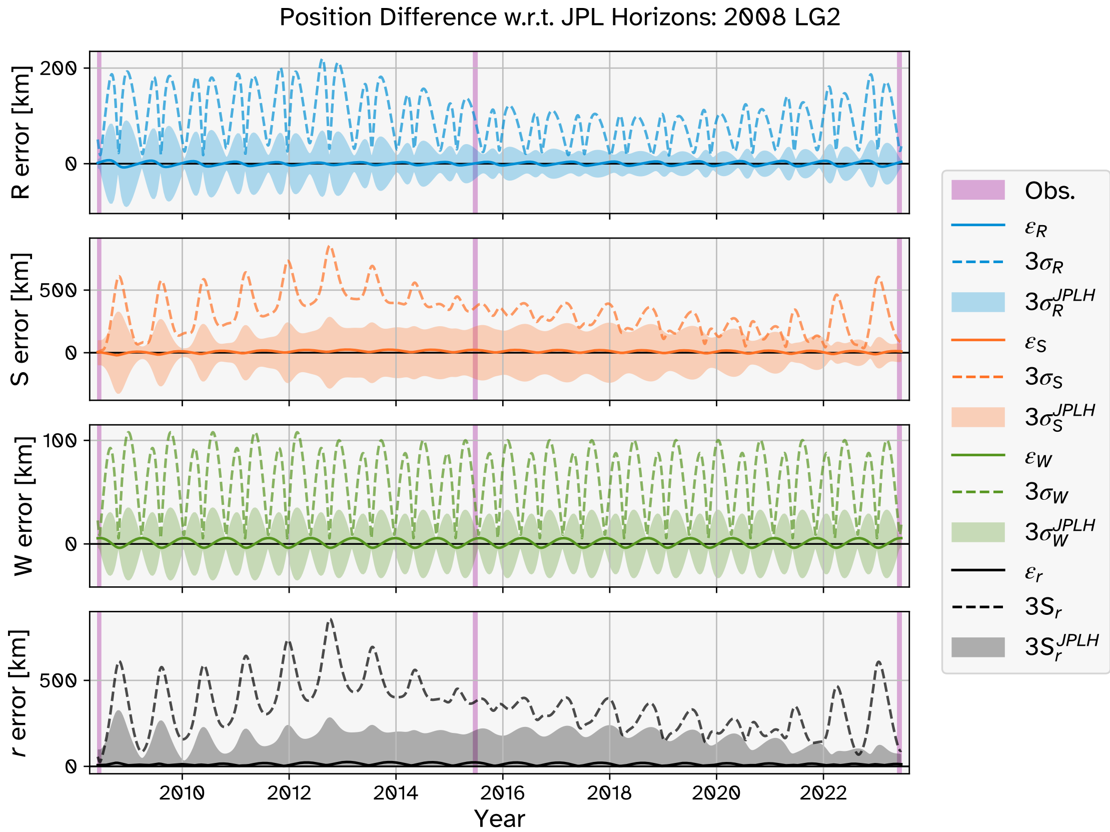


Figure 4.23: Position error ε with respect to JPL Horizons ephemeris of asteroid 2008 LG_2. The dashed coloured curves represent the formal errors (3σ) of the Tudat estimation in the RSW directions. The coloured regions represent the formal error in the JPLH ephemeris ($\pm 3\sigma$). In the bottom plot, the dashed black curve and grey region are the $3S_r$ measure of the overall formal error of the Tudat and JPLH orbit estimations, respectively. Pink regions indicate periods in which observations are available.

4.3. Comparisons With Literature

This section presents comparisons between the results of this research (labelled 'Tudat') and the results of comparable work in literature. Specifically, results concerning the full set of estimations are compared against those of Fenucci et al. (2024), as well as against data published in the JPL Small-Body Database (SBDB).

Seeing as the set of asteroids chosen for running the estimations in this research was based on the set of asteroids for which Fenucci et al. (2024) detected a Yarkovsky effect (see Section 3.2), it is natural to compare the two sets of results. This is possible thanks to the full version of Table B.1 in Fenucci et al. (2024) that has been published online[3]. Note that results from this research may be labelled as 'NEOCC'.

The JPL SBDB[4] contains extensive information about asteroids and other small Solar System bodies, including naming information, orbital data, and physical data where available. The orbital data includes the A_2 parameter that was used in the creation of the JPL Horizons ephemeris of the body. Specific information about the inputs and methods used by JPL to perform their orbit determinations is not readily available, so direct comparisons between Tudat results and the JPL SBDB data are not as straightforward as when comparing against Fenucci et al. (2024).

Similarities and differences between three sources of results (Tudat, NEOCC, and JPL) regarding the methodology and treatment of observations are discussed in Subsection 4.3.1, to provide context to the results later in this section. A comparison of the observation residuals achieved by the different sources is given in Subsection 4.3.2, followed by comparisons of the signal-to-noise ratio (SNR) and its dependence on the use of radar observations in Subsection 4.3.3. The Yarkovsky detections are compared in Subsection 4.3.4 by looking at the determined A_2 parameter and its uncertainty. Finally, a discussion of the observed differences in the results is presented in Subsection 4.3.5.

4.3.1. Context

There is a lot of overlap in the observation datasets used between the different orbit determination sources, primarily consisting of optical astrometry from the MPC. One big difference is that Fenucci et al. (2024) and JPL SBDB make use of radar observations (from the JPL Solar System Dynamics website[5]) and space-based optical observations in their estimations where available, whilst the Tudat estimations use only ground-based optical astrometric observations. As mentioned in Subsection 4.2.3, JPL also makes use of observation sources such as the Gaia mission, occultation observations, and positional data from the space missions.

Another notable difference is that the estimations performed by Fenucci et al. (2024) were completed $\sim$3 years before the estimations performed for this research. Because of observations made in the mean time, there were more optical observations available to be used in the Tudat estimations than in the Fenucci et al. (2024) estimations. The JPL SBDB is kept up-to-date, with orbits of small bodies being re-computed within hours of new observations being published. The SBDB data used for comparisons in this section were downloaded on 26 May 2026.

Furthermore, there are differences in how outliers are treated. Fenucci et al. (2024) reports that they use the outlier rejection algorithm of Carpino et al. (2003), whilst the less sophisticated outlier rejection procedure described in Subsection 3.5.1 is used in this research. However, Fenucci et al. (2024) use the observation weighting scheme of Vereš et al. (2017), which is the same as the one used in this research, and the debiasing scheme of Farnocchia et al. (2015a), which is very similar to the one used in this research (Eggl et al. 2020). Details about the treatment of observations by JPL are not publicly available.

There are also similarities in the methodology between this research and that of Fenucci et al. (2024). In both cases, the same Yarkovsky acceleration model is used, and estimations are performed by means of a 7-dimensional weighted least-squares orbit determination. JPL also uses the same Yarkovsky acceleration model, though precise details about their orbit determination process are not publicly available.

4.3.2. Observation Residuals

Comparing the overall residuals of the estimations from the three different sources allows us to compare how well each source manages to fit the orbit to the real-life observation data. This section compares the RMS of the normalised residuals (RMSNR) resulting from the estimations of each source.

Figure 4.24 shows the distributions of RMSNR for each of the three sources of orbit determinations. The mean of the Tudat distribution (0.5467) is marginally lower than that of the NEOCC distribution (0.5502), whilst both are significantly higher than the median of the JPL distribution (0.4427). The maximum RMSNR for estimations in the E-set for Tudat, NEOCC, JPL are 1.099, 1.114, and 0.7186, respectively,

[3] https://cdsarc.cds.unistra.fr/viz-bin/cat/J/A+A/682/A29#/description
[4] https://ssd.jpl.nasa.gov/tools/sbdb_lookup.html#/
[5] https://ssd.jpl.nasa.gov/sb/radar.html

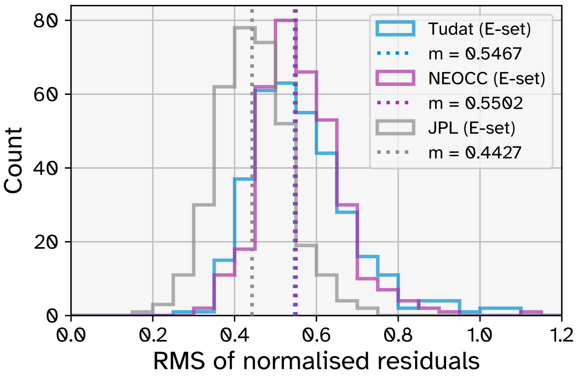


Figure 4.24: RMS of normalised residuals for all estimations (E-set), resulting from the orbit determinations of this research (Tudat), Fenucci et al. (2024) (NEOCC), and JPL. The dotted vertical lines represent the median values of the distributions, whose values are given in the legend.

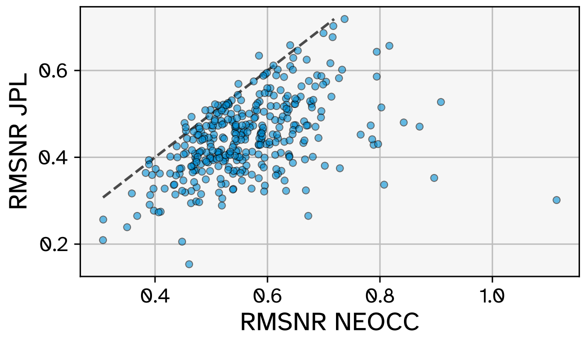


Figure 4.25: Comparison of the RMSNR resulting from the JPL and NEOCC orbit determinations, for all estimations (E-set). The black dashed line is the diagonal.

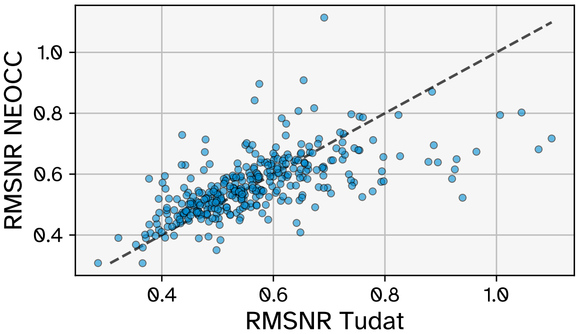


Figure 4.26: Comparison of the RMSNR resulting from the NEOCC and Tudat orbit determinations, for all estimations (E-set). The black dashed line is the diagonal.

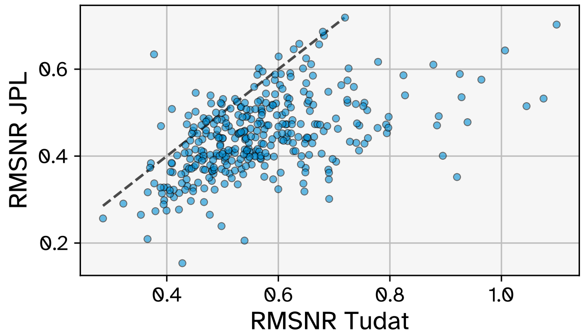


Figure 4.27: Comparison of the RMSNR resulting from the JPL and Tudat orbit determinations, for all estimations (E-set). The black dashed line is the diagonal.

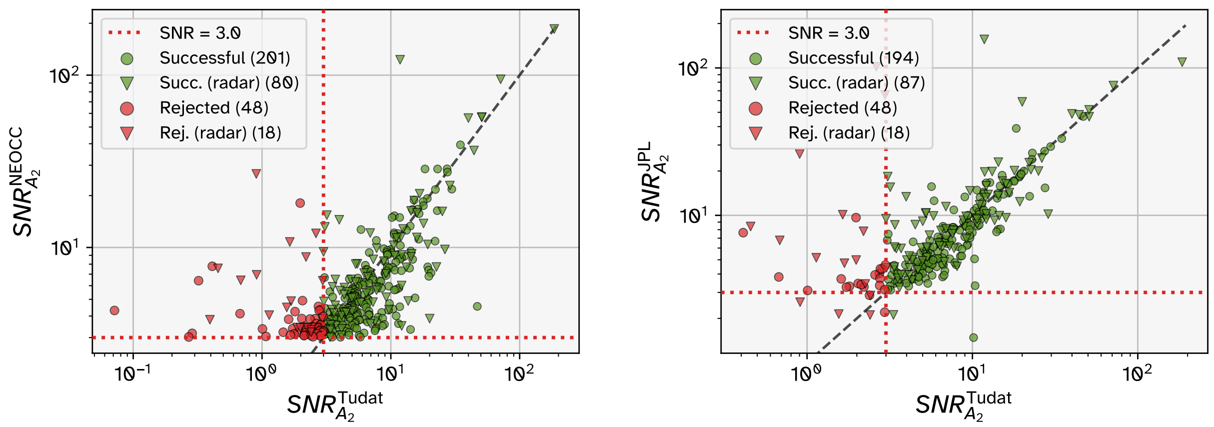
and the minimum RMSNR are 0.2856, 0.3075, and 0.1536, respectively.

Comparing the RMSNR distributions for Tudat and NEOCC (see Figures 4.24 and 4.26), they are generally very similar, though the NEOCC distribution is somewhat more concentrated near the median, while the Tudat distribution is more spread out, with more instances of low and high RMSNR. The similarities can be attributed to the fact that the observation datasets, debiasing, weighing, and overall methodology is very similar between the two sources. As discussed in Subsection 4.2.2, the instances where Tudat has high RMSNR can be attributed to insufficient filtering of outlier observations, which is less likely to occur in the NEOCC estimations due to their use of an automatic statistical outlier rejection algorithm.

Comparing the RMSNR distributions for Tudat and JPL (see Figures 4.24 and 4.27), they are quite different. The JPL distribution is generally much lower, with the vast majority of values being below the median of the Tudat distribution. In Figure 4.27, there are 33 points above the diagonal, compared to 314 points below the diagonal. The exact process of observation weighing and outlier rejection used by JPL is not publicly available, and it may be that the observation weights are adjusted iteratively based on the previous iteration's residuals. This would by definition result in lower residuals, making the direct comparison between the residuals of these sources not particularly meaningful, apart from being able to attribute a large part of the differences to this difference in observation treatment. The same reasoning applies when comparing the RMSNR distributions for JPL and NEOCC (see Figure 4.25), since the observation treatment by the NEOCC is overall more similar to that of the Tudat estimations than to the JPL estimations.

4.3.3. Signal-to-Noise Ratio of A_2

The SNR of the A_2 parameter, defined $\mathrm{SNR}_{A_2} = |A_2|/\sigma_{A_2}$, is an important aspect for the orbit determination of the Yarkovsky effect. In this research and in Fenucci et al. (2024), any estimations for which $\mathrm{SNR}_{A_2} < 3$ were rejected as Yarkovsky detections: with a low SNR_{A_2}, it cannot be confidently asserted that the fitted value of A_2 is a reflection of a true physical effect.



(a) NEOCC vs. Tudat

(b) JPL SBDB vs. Tudat

Figure 4.28: Comparison of SNR between literature and Tudat estimations for the E-set. Markers are coloured by whether the Tudat estimations were accepted (green) or rejected (red). Triangular markers indicate that the literature orbit determination made use of radar observations. Circular markers indicate that no radar observations were used in the literature orbit determination. The numbers in the plot legends indicate the corresponding number of points in the figure. The dashed black lines are the diagonals. The dotted red lines are where SNR=3 (used as the cutoff rejection criterion 1 in this research, and also used similarly in Fenucci et al. (2024)).

Figure 4.28 shows how SNR_{A_2} compares between the Tudat estimations and those found in literature. Amongst the successful estimations (green), the points in both figures are generally close to the diagonal. For those that were rejected (red), the distribution is erratic. The points are marked by whether the literature estimation used radar observations or not. For both NEOCC and JPL, there is a wide range of SNR_{A_2} for both estimations with and without radar, though the estimations which achieve the highest SNR_{A_2} are one that make use of radar data — this is also evident in Figure 4.29. It is clear, however, that the mere inclusion of radar data is not enough to achieve good SNR_{A_2}, as there are several such estimations which have $SNR_{A_2} < 3$ for JPL. 27.2% of the rejected estimations are ones for which NEOCC used at least one radar observation in their estimation. For the accepted/successful estimations, this proportion is 28.5%. For JPL, these numbers are 27.2% and 31.0% respectively. This demonstrates a point made by Fenucci et al. (2024): “the presence of radar observations generally improves the SNR, even though they are not necessarily needed to obtain good detections”. NEOCC and JPL also include space-based optical observations in their datasets, which are not considered in this research. However, data relating to the number of these space-based observations is not provided by Table B.1 of Fenucci et al. (2024), so it is difficult to analyse the effect of this.

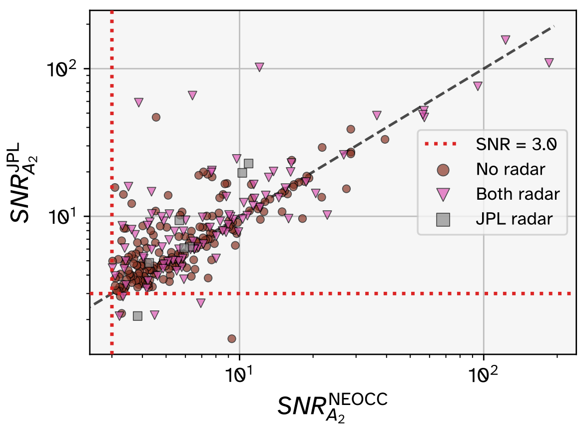


Figure 4.29: Comparison of SNR between the two literature sources, NEOCC and JPL SBDB, for the E-set. Brown circular markers indicate that neither source used radar observations in their orbit determination (242 points). Pink triangular markers indicate that radar observations were used by both sources (98 points). Grey square markers indicate that radar observations were used by the JPL orbit determination, but not by NEOCC (7 points). There were no cases where radar observations were used by NEOCC but not by JPL. The dashed black line is the diagonal.

The Tudat estimations with the highest SNR_{A_2} (>50) are for the asteroids: 1998 SD_9 ($SNR_{A_2} = 185.5$),

Table 4.1: The 8 asteroids for which $\mathrm{SNR}_{A_2}^{\mathrm{NEOCC}} > 30$, with the number of observations used and the SNR_{A_2} resulting from the NEOCC and Tudat estimations.

Body	NEOCC			Tudat	
	SNR_{A_2}	*n* optical	*n* radar	SNR_{A_2}	*n* optical
1998 SD_9	185.3	145	1	185.5	149
99942 Apophis	122.9	9505	50	11.80	8167
480883 (2001 YE_4)	94.61	429	7	71.21	442
2340 Hathor	57.11	1334	7	50.82	708
524522 Zoozve	56.93	598	10	50.74	599
2002 BF_{25}	56.35	143	4	40.00	143
2000 EB_{14}	39.41	81	0	34.67	83
483656 (2005 ES_{70})	36.52	202	1	44.22	228

480883 (2001 YE_4) ($\mathrm{SNR}_{A_2} = 71.2$), 2340 Hathor ($\mathrm{SNR}_{A_2} = 50.8$), and 524522 Zoozve ($\mathrm{SNR}_{A_2} = 50.7$). These are all amongst the bodies with the highest SNR_{A_2} in the NEOCC estimations too. Table 4.1 shows some data for the 8 bodies for which the $\mathrm{SNR}_{A_2}^{\mathrm{NEOCC}} > 30$. 7 out of these 8 use radar observations. 99942 Apophis is an example of an asteroid for which a relatively good SNR_{A_2} can be achieved without radar (11.80 for the Tudat estimation), but the inclusion of some radar observations (far fewer than the number of optical observations) greatly improves the SNR_{A_2}, with NEOCC achieving 122.9, an improvement of more than 10 times. For the asteroids 480883, Hathor, Zoozve, 2002 BF_{25}, the inclusion of radar observations appears to provide a modest to significant boost to SNR_{A_2}. 1998 SD_9 is an example of an asteroid for which very high SNR_{A_2} can be achieved regardless of the inclusion of the single available radar observation. For 2000 EB_{14}, no radar observations are used by NEOCC. Although there are slightly more optical observations used in the Tudat estimation of this asteroid, $\mathrm{SNR}_{A_2}^{\mathrm{Tudat}} < \mathrm{SNR}_{A_2}^{\mathrm{NEOCC}}$. This demonstrates that the differences in SNR_{A_2} between the two sources should not be solely attributed to the presence or absence of radar observations. Other factors, such as the different outlier rejection process, are likely responsible for at least part of these differences.

4.3.4. Yarkovsky Detections

This subsection present figures and analysis relating to the determination of the A_2 parameter and comparisons with determinations of this parameter from literature, namely from Fenucci et al. (2024) and from the JPL SBDB.

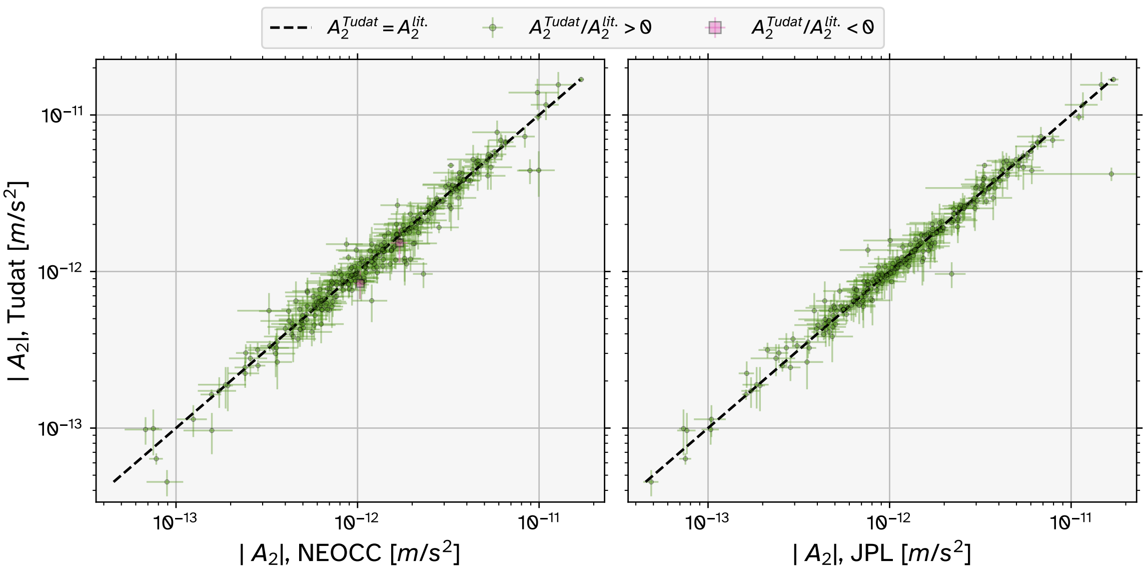
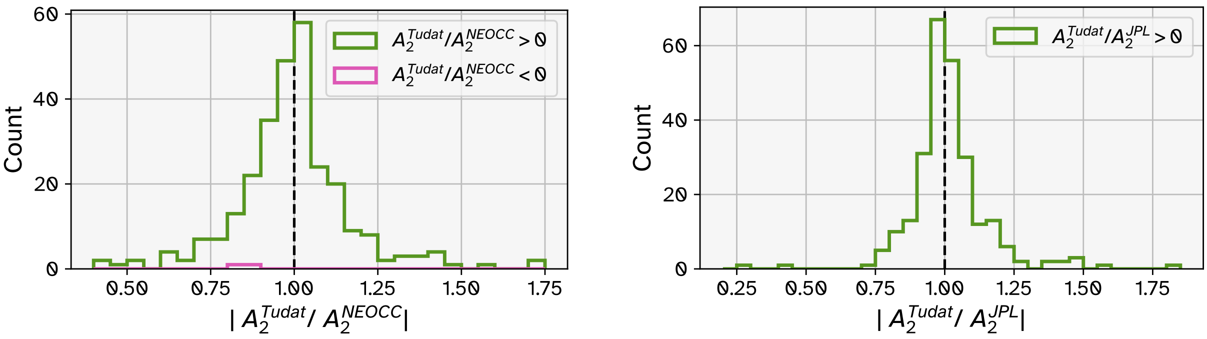


Figure 4.30: Comparisons between the absolute values of A_2 resulting from the Tudat estimations and from literature (NEOCC in left subplot and JPL in right subplot). The green and pink markers represent results from the S-set. A green marker indicates that the A_2 estimations share the same sign. A pink marker indicates that the A_2 estimations have opposite signs (there are two such points in the left subplot and zero in the right subplot). Note that there are 281 points (S-set) in the left subplot, but only 257 points in the right subplot. The error bar size is 1σ. The black dashed lines are the diagonals.



(a) Tudat vs. NEOCC **(b)** Tudat vs. JPL

Figure 4.31: Distributions of the ratio $|A_2^{\mathrm{Tudat}}/A_2^{\mathrm{lit.}}|$ for the literature sources (lit.) NEOCC and JPL SBDB. Again, the 281 points (S-set) are represented in (a), but only 257 points are represented in (b).

Figures 4.30 and 4.31 show that generally, the A_2 values estimated in this research are reasonably close to those that are reported in literature, as most of the points in each subplot of Figure 4.30 are close to the diagonal, and the distributions in Figure 4.31 are generally close to 1. There are several estimations which show a significant difference as compared to the values from literature, although many of these are accompanied by appropriately larger error bars. There is a wider spread from the diagonal in the left subplot (NEOCC) that in the right subplot (JPL), but it should be noted that there are more points in the NEOCC plot (281) than there are in the JPL subplot (257). This is because the JPL SBDB does not define an A_2 value for all asteroids in the S-set (nor for all asteroids in the R-set, for which JPL SBDB defines an A_2 value for only 35 out of the 66 asteroids in the set).

Figure 4.30 shows the absolute value of the Yarkovsky parameter for readability, but this hides the distribution of positive versus negative A_2 values. For the successful Tudat estimations (S-set), 74 of the estimated A_2 values are positive (corresponding to prograde (P) rotation), and 207 of them are negative (corresponding to retrograde (R) rotation), for a ratio of $R/P = 2.797$, which is in line with the findings of La Spina et al. (2004), whose model predicts $R/P = 2^{+1}_{-0.7}$.

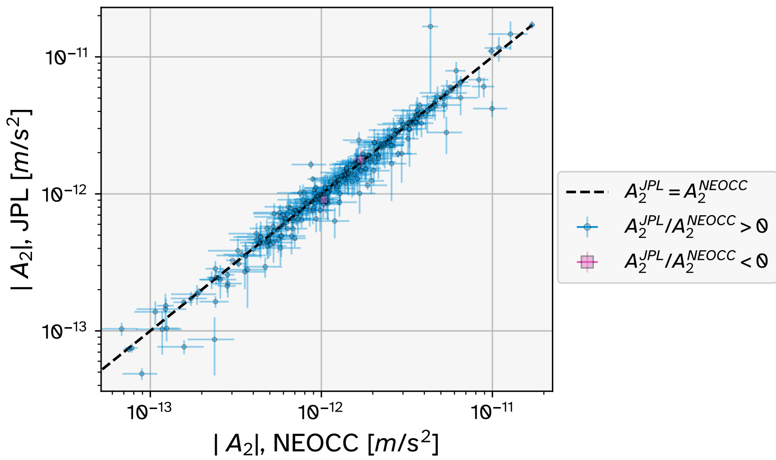


Figure 4.32: Comparisons between the absolute values of A_2 as reported by JPL SBDB and Fenucci et al. (2024) (NEOCC). A blue marker indicates that the A_2 estimations share the same sign. A pink marker indicates that the A_2 estimations have opposite signs (there are two such points). Note that there are 292 points in this figure rather than the full 347 of the E-set, because the JPL SBDB does not define an A_2 value for all asteroids in the E-set. The error bar size is 1σ. The black dashed line is the diagonal.

Figures 4.32 and 4.33 show a comparison of the A_2 values from JPL and NEOCC. Although a similar figure is reported in Fenucci et al. (2024), it is interesting to note that there are some differences. The spread of points around the diagonal appears slightly wider here, and there are more points which could be considered outliers, with the error bars not touching the diagonal. Furthermore, just like in Figure 4.30, there are two points for which the A_2 is of opposite sign. No mention was made of results with opposite signs in Fenucci et al. (2024) (similar to the figures in this subsection, the Figure 6 of Fenucci et al. (2024) shows the absolute values of the A_2 parameter, but does not make any distinction between points for which the two different values share a sign and points for which the values have opposite signs). In

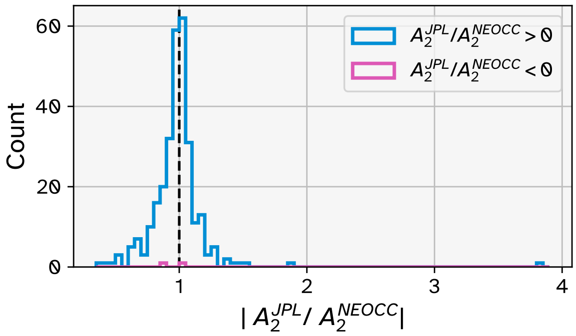
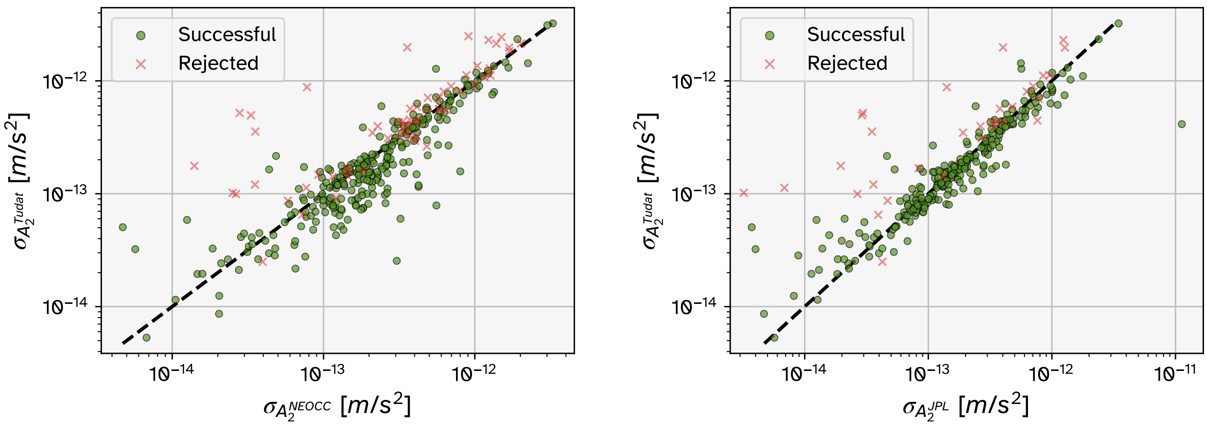


Figure 4.33: Distribution of the ratio $|A_2^{\mathrm{JPL}}/A_2^{\mathrm{NEOCC}}|$. Again, there are only 292 points represented in this figure out of the full 347 of the E-set.

both Figures 4.30 and 4.32, these points correspond to the asteroids 452807 (2006 KV_{89}) and 488803 (2005 GB_{120}). In these cases, the ratio $|A_2^{\mathrm{Tudat}}/A_2^{\mathrm{NEOCC}}|$ is 0.8951 and 0.8127, respectively, and the ratio $|A_2^{\mathrm{JPL}}/A_2^{\mathrm{NEOCC}}|$ is 1.043 and 0.8778, respectively. While it is difficult to say exactly why this may have happened, the fact that these ratios are relatively close to 1 may suggest that there was simply a typographical error in Table B.1 of Fenucci et al. (2024), and a minus sign was omitted in two places. Other typographical errors were noticed in this table, such as the insertion of a space in the middle of two asteroids' permanent designation numbers. There may also have been updates to these values due to more recent observational data. Fenucci et al. (2024) used data downloaded from JPL SBDB in August 2023. The JPL SBDB data used in this thesis was downloaded in May 2026. There may have been updates to the orbit fits since then, perhaps based on more recent observations, which caused the sign of one or more of the Yarkovsky parameters to change. This may also explain the presence of more outliers.



(a) Tudat vs. NEOCC. Full S-set and R-set.

(b) Tudat vs. JPL. 257 points from the S-set (out of 281) and 35 points from the R-set (out of 66) are represented.

Figure 4.34: Comparisons of the estimated (formal) uncertainties σ_{A_2} of the Yarkovsky parameter resulting from the Tudat estimations and from literature (NEOCC and JPL SBDB). The green circular markers represent results from the S-set and red 'X' markers indicate results from the R-set. The black dashed line is the diagonal.

Figure 4.34 shows how the formal uncertainties in the A_2 parameter compare between the Tudat estimations and the formal uncertainties reported in literature. The green points (corresponding to the successful Tudat estimations) in Figure 4.34b are generally quite close to the diagonal line, With 102 green points above the diagonal and 155 below, this indicates a good match between Tudat's and JPL's A_2 uncertainties, with the Tudat uncertainties being somewhat more optimistic. There are . However, there are a significant number of points above the diagonal line, for which $\sigma_{A_2}^{\mathrm{Tudat}}$ is considerably larger than $\sigma_{A_2}^{\mathrm{JPL}}$. As discussed in Subsection 4.3.1, this may be largely attributable to the differences in the observation weighting and filtering processes used by JPL. Figure 4.34a shows a substantially messier distribution of green points, with more points spreading relatively far from the diagonal. There are 97 green points above the diagonal and 184 below, indicating the Tudat formal error in A_2 may be too optimistic.

As introduced in Subsection 4.2.3, the ratio $\mathrm{FE}_{\mathrm{RMS}}/\mathrm{TE}_{\mathrm{RMS}}$ quantifies in a single value how well the

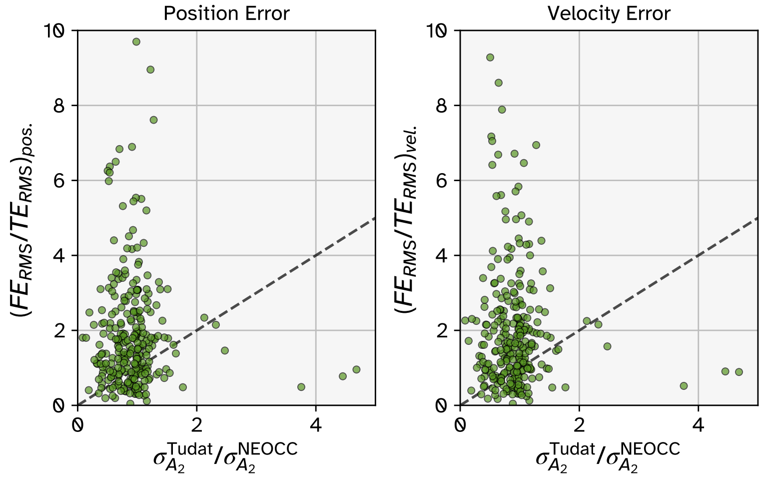


Figure 4.35: Distribution of the ratio $\mathrm{FE_{RMS}}/\mathrm{TE_{RMS}}$ (a measure of the match between the formal and true errors in position/velocity) against $\sigma_{A_2}^{\mathrm{Tudat}}/\sigma_{A_2}^{\mathrm{NEOCC}}$ (a measure of the match between the formal and true errors in A_2). There are 5 points missing from each subplot due to being outside of the plot range. The black dashed lines are the diagonals.

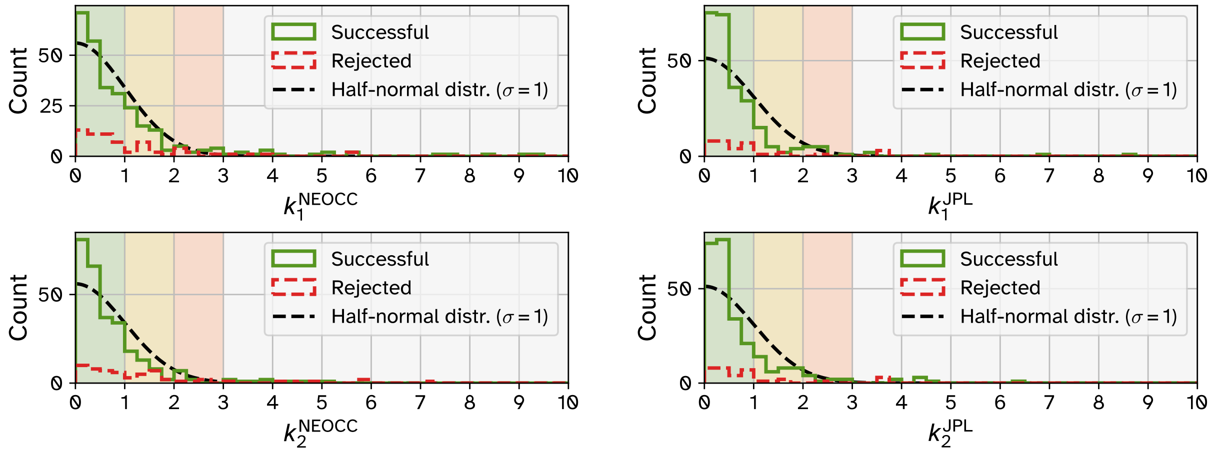
formal errors approximate the true errors, for either the position or velocity vectors. Similarly, the ratio $\sigma_{A_2^{\mathrm{Tudat}}}/\sigma_{A_2^{\mathrm{NEOCC}}}$ quantifies how well the formal error in A_2 approximates the true error in A_2, under the assumption that the A_2 estimations and their uncertainties reported by Fenucci et al. (2024) are a good representation of the true values. Figure 4.35 shows how these two quantities compare against each other. Interestingly, there does not seem to be any correlation between the position (velocity) error ratio and the A_2 error ratio, and in fact many of the points cluster around the line $\sigma_{A_2^{\mathrm{Tudat}}}/\sigma_{A_2^{\mathrm{NEOCC}}} = 1$. This implies that the A_2 uncertainty estimate can be reasonably good even when the formal error in position (velocity) is a relatively poor reflection of the true error. However, the assumptions which underlie this conclusion may not be valid. The Tudat and NEOCC formal uncertainty in the A_2 parameter are highly correlated with each other, as seen in Figure 4.34a, presumably due to the large overlap in the observation datasets and methodological similarities. Thus, $\sigma_{A_2^{\mathrm{NEOCC}}}$ may not be a particularly good measure of the 'true' error. The same can be said about $\mathrm{TE_{RMS}}$: again, the Tudat orbit estimations and the JPLH ephemerides are highly correlated and use similar datasets, so $\mathrm{TE_{RMS}}$ may also not be a great measure of the true error. Finally, The lack of a correlation may be related to the comparison not being a 'fair' one: the horizontal axis is a ratio of two formal errors taken directly from the estimation results, while the vertical axis is the ratio of measures which take into account several error components which are each time-varying over the full propagation period.

The relative errors k_1 and k_2, are defined here as:

$$\begin{aligned} k_1 &= \frac{|A_2^{\mathrm{lit.}} - A_2^{\mathrm{Tudat}}|}{\sigma_{A_2^{\mathrm{Tudat}}}} \\ k_2 &= \frac{|A_2^{\mathrm{lit.}} - A_2^{\mathrm{Tudat}}|}{\sigma_{A_2^{\mathrm{lit.}}}} \end{aligned} \tag{4.6}$$

The superscript 'lit.' will be replaced with 'NEOCC' and 'JPL' accordingly. These values can be used to quantify the differences between the Tudat estimations of the A_2 parameter and those found in literature. The same terms were used in Fenucci et al. (2024) (Equation (15)) to compare their estimations with those of JPL. k_1 is essentially the difference between the two estimations, normalised using the Tudat estimation's uncertainty in A_2. It describes how many 'steps' of $\sigma_{A_2^{\mathrm{Tudat}}}$ are between A_2^{Tudat} and $A_2^{\mathrm{lit.}}$. A small value of k_1 indicates that the literature estimation of A_2 is compatible with the Tudat estimation of A_2 and its uncertainty, while a large value indicate the opposite. k_2 can be described in the same way, with the roles of Tudat and literature estimations reversed.

Under the assumption that A_2^{NEOCC} and A_2^{Tudat} are random variables that follow Gaussian distributions, k_1 and k_2 are expected to follow a half-normal distribution with $\sigma = 1$. As mentioned in Subsection 4.1.2, if a variable Y is normally distributed with $\mu = 0$ and $\sigma = 1$, then $|Y|$ follows a half-normal distribution with expected value $E(|Y|) = \sqrt{2/\pi} = 0.797\,885$ and variance $\mathrm{Var}(|Y|) = 1 - (2/\pi) = 0.363\,380$ (Olive 2005, Sec 3.11). Following such a distribution, 68.3% of k values would be expected to be between 0 and 1, 27.2% between 1 and 2, 4.28% between 2 and 3, and 0.27% larger than 3.



(a) k_1^{NEOCC} and k_2^{NEOCC}. Full S-set and R-set represented in each subplot, except for a few points which lie outside of the plot range of the figure: $k_1^{\text{NEOCC}} > 10$ for 1 estimation in the S-set, and $k_2^{\text{NEOCC}} > 10$ for 3 estimations in the S-set and for 7 estimations in the R-set.

(b) k_1^{JPL} and k_2^{JPL}. 257 points from the S-set (out of 281) and 35 points from the R-set (out of 66) are represented. Additionally, a few points lie outside of the plot range of the figure: $k_1^{\text{JPL}} > 10$ for 1 estimation in the S-set, and $k_2^{\text{NEOCC}} > 10$ for 8 estimations in the R-set.

Figure 4.36: Actual and expected distributions of the two relative error measures $k_1^{\text{lit.}}$ and $k_2^{\text{lit.}}$ (see Equation (4.6)) for the two literature sources (NEOCC and JPL SBDB). In each case, the expected distribution of k for estimations in the S-set is a half-normal distribution with $\sigma = 1$.

As seen in Figure 4.36a, the distributions of k_1^{NEOCC} and k_2^{NEOCC} are quite close to the expected half-normal distribution, but with more data points than expected in the 0-0.5 range and slightly fewer than expected in the 0.5-2 range. Both distributions also have larger tails than would be expected, indicating large differences between the estimations: 19 successful estimations have $k_1^{\text{NEOCC}} > 3$ and 13 successful estimations have $k_2^{\text{NEOCC}} > 3$. Considering there are 281 successful estimations, the expected number of estimations with $k > 3$ is just 0.7587. Taking the k_1^{NEOCC} values, the A_2 determinations from this research are compatible with the NEOCC values in 93.2% of cases.

Figure 4.36b shows that the distributions of k_1^{JPL} and k_2^{JPL} are somewhat different to those of NEOCC. Now there are even more data points in the 0-0.5 range than expected, and even fewer in the 0.5-2 range. The tails are also smaller: 6 successful estimations have $k_1^{\text{JPL}} > 3$ and 7 successful estimations have $k_2^{\text{JPL}} > 3$. Taking the k_1^{JPL} values, the A_2 determinations from this research are compatible with the JPL values in 97.7% of cases. Again, it should be noted that there are fewer asteroid represented in Figure 4.36b as in Figure 4.36a: just 257 compared to 281. The expected half-normal distribution is adjusted accordingly.

The high presence of low k values (< 0.5) is likely attributable to some of the point discussed in Subsection 4.3.1: large overlaps in observation datasets and methodological similarities which result in very similar A_2 determinations between different sources.

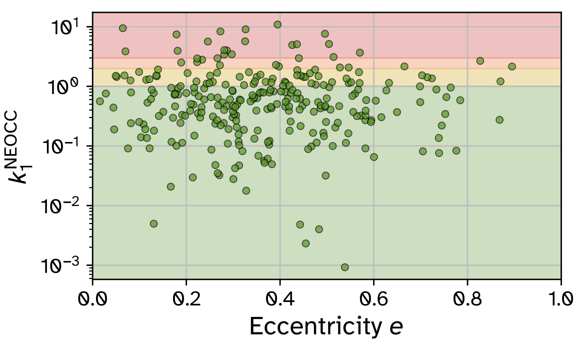


Figure 4.37: Distribution of k_1^{NEOCC} against the range of eccentricity e for the S-set.

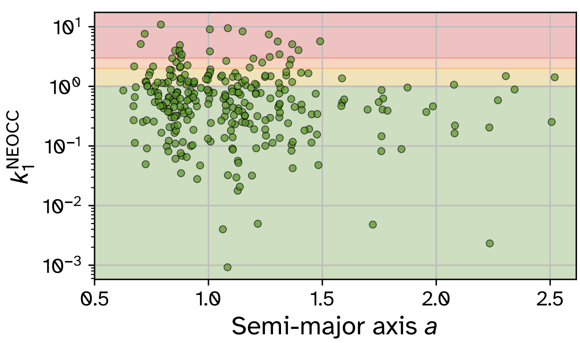


Figure 4.38: Distribution of k_1^{NEOCC} against the range of semi-major axis a for the S-set.

Figures 4.37 to 4.40 show a variety of plots comparing the k_1^{NEOCC} values against eccentricity e, semi-major axis a, observation arc length, and RMSNR. In each of these figures, the coloured regions represent ranges of values of k: green for $0 \leq k < 1$; yellow for $1 \leq k < 2$; orange for $2 \leq k < 3$; red for $k > 3$. Figure 4.37 shows that most of the large values of k_2^{NEOCC} occur for asteroids with eccentricity less than 0.6. This may be a coincidence, since there are relatively few asteroids with eccentricity greater than 0.6

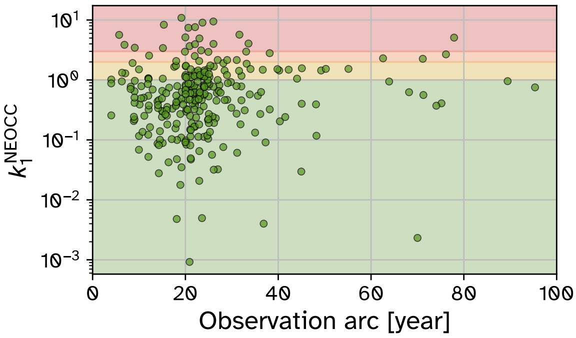


Figure 4.39: Distribution of k_1^{NEOCC} against the range of the observation arc length for the S-set.

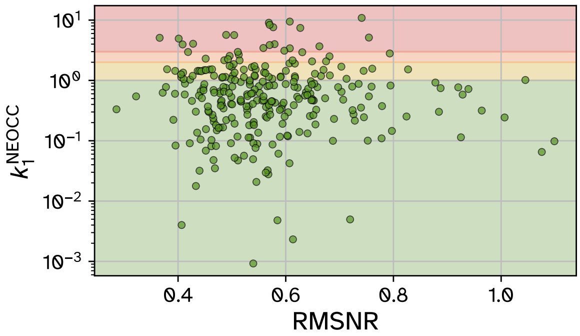


Figure 4.40: Distribution of k_1^{NEOCC} against RMSNR for the S-set.

(26 asteroids out of the 281 in the S-set), but it may also be an indication that the integrator and time step settings are not always sufficient, since these settings are more relaxed for asteroids with $e < 0.4$. There is no strong correlation visible between k_1^{NEOCC} and any of the parameters plotted in Figures 4.38 to 4.40.

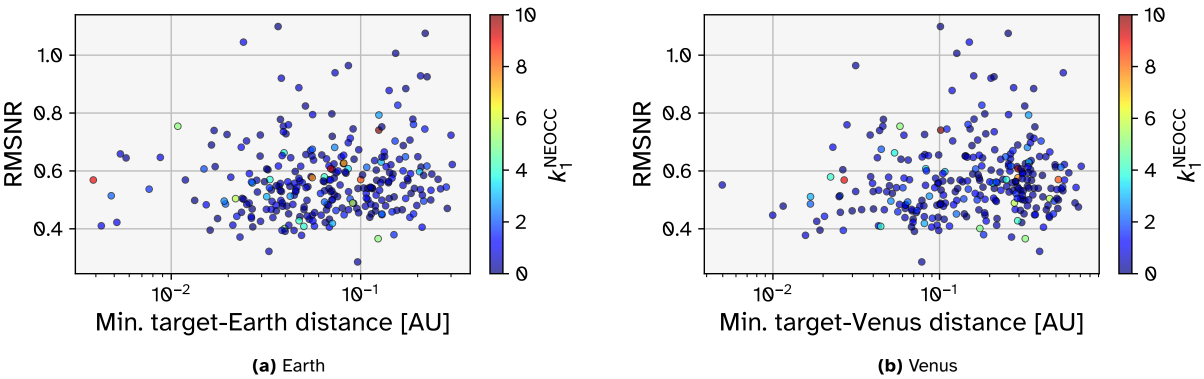


Figure 4.41: Distributions of RMSNR of the Tudat estimations against the minimum distance between the target body and the planet during the propagation. The points are coloured according to their value of k_1^{NEOCC}.

Figure 4.41 was made in order to see if close passes with planets during the propagation period were correlated with worse fits to the observation data and/or estimations of A_2 which are not compatible with those from literature. Such results could occur as a result of the integration settings not being sufficient to handle the fast-changing dynamics of the close pass, or due to insufficiencies in the dynamic model that become more relevant with proximity to a planet (e.g. modelling the planet as a point-mass gravity field rather than using spherical harmonic gravitation terms, or the exclusion of relativistic corrections for the planets). For the inner planets, the Moon, and the three largest bodies in the main asteroid belt, the distance of the closest pass with the target body during the propagation period was recorded. The closest passes occurred for Earth and Venus (and, of course, those of the Moon were very similar to those of Earth); these are plotted in Figure 4.41. Similar plots for the other bodies can be found in Appendix A. However, the figures seem to indicate that close passes are driving factors behind the estimations with poorer fits to the observation data and poorer A_2 estimations. Many of the asteroids which have the closest passes also have quite low RMSNR and also low k_1^{NEOCC}. Occurrences of high RMSNR and high k_1^{NEOCC} can be found across the full range of minimum close pass distances. One limitation of this analysis is that it does not take into account how many close passes occur during the propagation period. It may be that the effects of multiple close passes (with the same planet or with multiple planets) could amplify over time, resulting in worse results even if the minimum close pass distance was not extremely close.

4.3.5. Discussion of Differences

Despite the high degree of overlap between the methods of Fenucci et al. (2024) and the methods used in this research, there are also a number of differences in the inputs and methods which could be contributing factors to the differences observed in the results.

Regarding the 16 main-belt asteroids included as perturbers in the dynamics, the set of asteroids used is not the same. This comes from the use of different sources for the masses of the asteroids (SiMDA for the Tudat estimations and the JPL DE441 ephemerides for the NEOCC estimations) and thus a slightly different list of the top 16 most massive. 14 of the asteroids are the same but 7 Iris and 29 Amphitrite are included in the Tudat estimations while 107 Camilla and 88 Thisbe are included in the NEOCC estimations. For the 14 asteroids in common, there are some significant differences in the masses used, with $M_{\text{NEOCC}}/M_{\text{Tudat}}$ ranging from 0.863 for 65 Cybele to 2.152 for 87 Sylvia. Furthermore, different sources are used for the ephemerides of the main-belt asteroids: the Tudat estimations use ephemerides retrieved from JPL Horizons, while the NEOCC estimations use the JPL DE441 ephemerides.

There are also other differences in the dynamic models used. Fenucci et al. (2024) includes the "relativistic effects of the Sun, the planets, and the Moon were also added to the dynamical model as a first-order Newtonian expansion", while the Tudat estimations only include a relativistic correction for the Sun. This difference may be an important one, as Chesley et al. (2014) noted "the critical importance of the Earth general relativity terms in the dynamical model" for their orbit determination and Yarkovsky estimation of 101955 Bennu. Furthermore, Fenucci et al. (2024) note that for a small set of asteroids, the inclusion of an acceleration due to solar radiation pressure (SRP), along with the estimation of the area-to-mass ratio A/m as a parameter, changed the estimated Yarkovsky parameter from being outside of the acceptable range to being inside the acceptable range.

As mentioned already, Fenucci et al. (2024) makes use of radar observations of asteroids and includes them in the estimation where available, which was the case for 28% of the asteroids in the E-set, including many of the highest SNR_{A_2} estimations. Whilst the inclusion of radar data does not guarantee high SNR_{A_2}, and high SNR_{A_2} can be achieved without radar observations, their inclusion does make a significant difference in some cases. Another potentially significant difference is the treatment of observational outliers. Fenucci et al. (2024) employs an automatic outlier rejection algorithm (developed by Carpino et al. (2003)) which is commonly used in literature. This algorithm is performed iteratively, along with the estimation iterations, and determines if an observation should be rejected (or accepted back if it has been previously rejected) by taking into account both measurement and orbital covariances. In contrast, the Tudat estimations take the relatively blunt approach of removing any observations whose simulated residuals (relative to the JPL Horizons ephemeris) are larger than a particular cutoff value. This may result in some spurious observations being left in for the Tudat estimations, worsening the orbital fit.

Fenucci et al. (2024) does not specify any details about the numerical integration settings employed in their estimations. Figures 4.12 and 4.37 hint that the integration settings of the Tudat estimations may be insufficient in a small number of cases, causing relatively high RMSNR and worse A_2 estimations.

One final difference whose effect is difficult to quantify is the fact that different software is used for performing the estimations. Fenucci et al. (2024) used the proprietary ESA Aegis orbit determination software, whilst the estimations performed for this research were carried out using Tudat, which is open-source. Any differences in the implementation of certain methods are difficult to ascertain since the source code of the ESA Aegis software is not available to inspect.

Because of the limited availability of the specific details on how JPL generates its orbit estimations, it is difficult to pinpoint all of the potential reasons for the differences between the Tudat results and the JPL results. However, it is likely that many of the points mentioned above regarding the reasons for differences in the results also apply to the differences observed with the JPL data: environment inputs, dynamic model, outlier rejection, numerical integration settings, and software implementation differences.

Also, as mentioned in Subsection 4.3.1, JPL makes use of a number of additional high accuracy observation sources which are not available through the MPC for their orbit determinations. These include occultation observations (Dunham et al. 2021), astrometry provided by the Gaia mission (Fenucci et al. 2024), and mission data from space probes (e.g. Cheng et al. 2018).

5 Conclusions & Recommendations

5.1. Conclusions

In Chapter 1, it was identified that the common use of proprietary software in astrodynamics literature impairs the ability to independently reproduce published results. Furthermore, a general lack of transparency regarding precise orbit determination settings means that researchers working on related problems may be spending time independently performing analyses to select appropriate settings, whilst this could be avoided if the results of such analyses were shared along with the results of the research.

In order to address these problems, this thesis aimed to reproduce Yarkovsky estimations from literature using free and open-source software (Tudat), as well as transparently document the selection of relevant orbit determination settings whose details are often omitted from literature. The reproduced results could then be compared to those from literature in order to assess the effectiveness of the chosen settings.

The numerical integration settings were selected in Section 3.3, resulting in the settings specified in Subsection 3.3.3. The dynamic model selection process is detailed in Section 3.4, resulting in the set of accelerations specified in Subsection 3.4.2. The outlier rejection process is explained in Subsection 3.5.1, with the selection of the cutoff value justified in Subsection 3.5.2. All other inputs to the estimation process are listed in detail in Subsection 3.1.3. Based on the selected methods, settings, and results presented in the preceding chapters, some conclusions can be drawn about the successes of the research (Subsection 5.1.1) as well as its limitations (Subsection 5.1.2).

5.1.1. Findings

To evaluate the reproducibility of literature results, 7-dimensional orbit estimations were performed to determine the initial state and Yarkovsky parameter (A_2) for 347 near-Earth asteroids (NEAs). 81% of the estimations were accepted as valid detections of the Yarkovsky effect. 21% were rejected due to low SNR_{A_2}. The rejected 21% still proved to have good fits to the observation data.

The 81% success rate demonstrates that settings relating to the dynamic model, numerical integration, and outlier treatment that were selected and justified in Chapter 3 produce high-quality results compatible with literature (specifically Fenucci et al. (2024) and JPL SBDB) in a large majority of cases. This confirms that robust Yarkovsky effect replication is entirely achievable with Tudat.

The rejected cases demonstrate certain sensitivities of the orbit determination process. The 21% of cases with low SNR_{A_2} are primarily attributed to differences in the observational data sets and observation treatment: the literature data sets include high-precision radar observations, as well as occultation and astrometric data from the Gaia mission for the JPL SBDB estimations. The literature sources analysed also make use of a more sophisticated, statistical outlier rejection process than the one that was employed for in this research. It was shown that not all observation outliers were removed using this outlier rejection process, thereby contaminating some of the results, and in some cases, making them look worse than they actually are.

Comparisons between formal errors and (assumed) true errors of position and velocity demonstrate that the estimated uncertainties are somewhat pessimistic in general (see Figures 4.15 and 4.16). When comparing the formal errors of the Yarkovsky A_2 parameter, however, the estimated uncertainties appeared to be slightly too optimistic overall. This suggests that the assumptions regarding the treatment of observation errors may not be entirely valid. Optimistic formal errors can be a result of unmodelled or uncorrected systematic errors and/or inaccurate observation weight assignment.

The split of results into 81% successes and 21% low SNR_{A_2} underscores the fact that reproducing results from literature is highly feasible, whilst demonstrating that orbit determination accuracy depends sensitively on the appropriate selection of methods and settings. This further reinforces the value of sharing the settings that are found to be appropriate for the given scenario alongside primary scientific results.

Based on these outcomes, the core research questions established in Chapter 1 can be definitively answered:

RQ1: To what extent can results relating to the Yarkovsky effect reported in literature be reproduced via orbit determinations using open-source software (Tudat)?
The results demonstrate that literature-grade Yarkovsky detections can be successfully reproduced using Tudat in the majority of cases (81%). Additionally, with an improved outlier rejection process and the inclusion of additional observation types, this could likely be increased to a much higher level. This demonstrates Tudat's role as a viable open-source alternative to proprietary tools like ESA Aegis.

RQ2: Which specific methodological processes and settings are required to achieve good orbit determinations for the purposes of detecting the Yarkovsky effect?
Achieving orbit determinations that are compatible with literature requires careful selection of settings including the dynamic model, numerical integrator, and outlier rejection strategies. The methods and eventual selections detailed in Chapter 3 proved to be effective for a majority of the bodies on which estimations were performed.

5.1.2. Limitations

A number factors which limited the accuracy or extent of the results and findings of this research are discussed below.

Observations

21% of estimations were rejected on the basis of low SNR_{A_2}. As mentioned above, a contributing factor to this may have been the inclusion of too many outliers in the observation data due to the crude outlier rejection process. Variations in observation noise from different sources are not taken into account with this method, nor are outliers which could be identified through a statistical analysis of the observation residuals. Compared to a more sophisticated statistical algorithm such as the one introduced by Carpino et al. (2003), the method that is used appears to be less effective at identifying and removing true observational outliers, thus limiting the orbital fits that could be achieved.

Only astrometric observations from ground observatories were used in the estimations. Other observation types and sources, such as from radar, astrometric observations from space telescopes, and occultation observations, were not included. While the number of radar observations is a lot lower than the number of astrometric observations (and is in many cases zero), their accuracy can be very high (Desmars et al. 2013), so by not including them, along with space-based astrometry and other observation types, the orbit fit and SNR_{A_2} was limited.

Numerical Integration Settings

Some of the results (e.g. Figures 4.12 and 4.37) indicate that the chosen scheme for numerical integration settings may not be sufficient in a small number of cases, and negatively affected the fit that could be achieved. Specifically, scheme specified that for asteroids with eccentricity $e < 0.6$, the integrator to be used was RK8 and the time step was 6 hours. Although this is sufficient in a large majority of cases, it may not be precise enough for certain dynamical situations such as repeated planetary close passes.

The integrator analysis process only considered fixed-step integrators. The lack of variable-step and other types of integrators in this analysis may mean that there were better options for the types of orbit estimations being performed that were left out. Figure 3.7 shows that the noise floor can be quite high for high-eccentricity asteroids using fixed-step integrators. Considering integrator error may have been a factor for some estimations, and the fact that the estimations were quite slow to perform, this is a limitation of the selection process and chosen settings.

Dynamic Model

The dynamic model used in this research did not include a number of accelerations which may have been relevant in some cases. For example, the general relativistic effects of the planets were not included in the chosen model. Farnocchia et al. (2015b) notes that such effects can be important when close passes with planets occur, and according to Chesley et al. (2014), including at least the Earth's relativistic effect was of "critical importance" when fitting 101955 Bennu's orbit. The importance of including the effect of

Earth's oblateness term is demonstrated by its inclusion in the dynamic model of Farnocchia et al. (2021). Fenucci et al. (2024) noted the importance of including an SRP acceleration for the estimations of a small number of bodies, in order to get a plausible fit for the Yarkovsky effect.

Additionally, only the 16 most massive main-belt asteroids are included in the dynamic model. If the target body experiences a close pass with any another asteroid with a significant mass during the propagation period, this could have a notable effect on the orbital fit.

Yarkovsky Detection

It should be noted that what has actually been detected by the estimations performed in this research is an acceleration in the transverse direction that is proportional to $1/r^2$, with r being the body's heliocentric distance. While the detections are real, in general cannot be said with certainty that the physical mechanisms of the Yarkovsky effect are responsible for it. A similar effect could also be caused by outgassing, for example (Farnocchia et al. 2023).

5.2. Recommendations

Based on the findings and limitations identified and discussed above, a number of recommendations for future work can be made, pertaining to the observation data, the integrator settings, and the dynamic model. These are presented below.

Observations

The outlier rejection procedure used in this research was identified as a limitation to the ability to correctly remove outliers and keep valid observations in the observation data set. An improvement would be to filter based on the normalised residuals instead. This would be better at dealing with different noise levels from different observatories and/or different points in time. An even better solution would be an iterative automatic outlier rejection procedure, such as the algorithm of Carpino et al. (2003). This algorithm has a number of parameters which may need to be tuned from their default values to achieve good results. This algorithm is not yet implemented in Tudat and thus cannot be applied to an orbit estimation. The addition of this functionality could be highly beneficial in terms of identifying and removing observations which negatively affect the outcome of the estimation, and is therefore recommended.

The limited (types of) observations used in the estimations was identified as a limitation. Radar observations are available via the MPC and JPL SBDB and are commonly used in literature for asteroid determination and Yarkovsky detection (e.g. Farnocchia et al. 2013; Greenberg et al. 2020; Dziadura et al. 2023; Fenucci et al. 2024). Although the software infrastructure exists in Tudat, currently, radar observations cannot yet be easily processed and utilised in an orbit estimation within Tudat, limiting the achievable orbit fits for some asteroids. It could be a very valuable addition to Tudat if this functionality was expanded.

Astrometric observations from space telescopes could also help to improve results. These would be more straightforward to implement in Tudat, as the framework already allows for this as long as ephemerides of the space telescopes used are loaded as well. Gaia mission data could also be included for further improved results.

Numerical Integration Settings

Given that the choice and analysis of integrator was identified as a possible contributor to error for a small number of estimations, as well as the fact that the runtimes of the estimations were quite long, it is recommended to conduct another integrator analysis in which other types of integrator are considered. This should include variable-step integrators at a minimum. By properly tuning the parameters of one of these integrators, it may even be possible to use the same integrator settings for all bodies.

Dynamic Model

In order to account for the limitations identified above regarding the dynamic model, it is recommended to conduct another dynamic model selection analysis. This analysis should consider the inclusion of Earth's oblateness term, an SRP acceleration, and the relativistic effects of the planets through the use of the EIH model (see Subsection 2.2.2 for details on these accelerations). In order to counteract the severely increased run time from including many bodies in the EIH formulation, the relativistic terms could be limited to just the inner planets and Jupiter, or perhaps they could be chosen automatically depending on the target body's MOID with each planet.

It is also recommended to conduct a close pass analysis to identify any close encounters between the target bodies and any other large asteroids whose masses are known. If a close encounter within a certain threshold is identified, the large asteroid can be added to the dynamics as a perturber. Such a close pass analysis could be conducted using the ephemeris files for the target bodies and a set of large asteroids with known masses.

Sensitivity Analysis

In order to better understand exactly how the chosen settings affect the outcomes of the estimations and their uncertainties, it is recommended to conduct an extensive sensitivity analysis. This would help to identify the settings which are most critical to the success of the estimations, as well as providing insight as to how settings may need to be adjusted for different estimation scenarios.

This is my full thesis. I need to hand it in within 1.5 hours. Can you please - proofread it and look for typos or other errors, omissions, or inconsistencies - have a look at the References section for any problems - find any abbreviations and symbols that are used in the text that are not yet mentioned in the Nomenclature section - suggest any quick edits that could improve argumentation or readability

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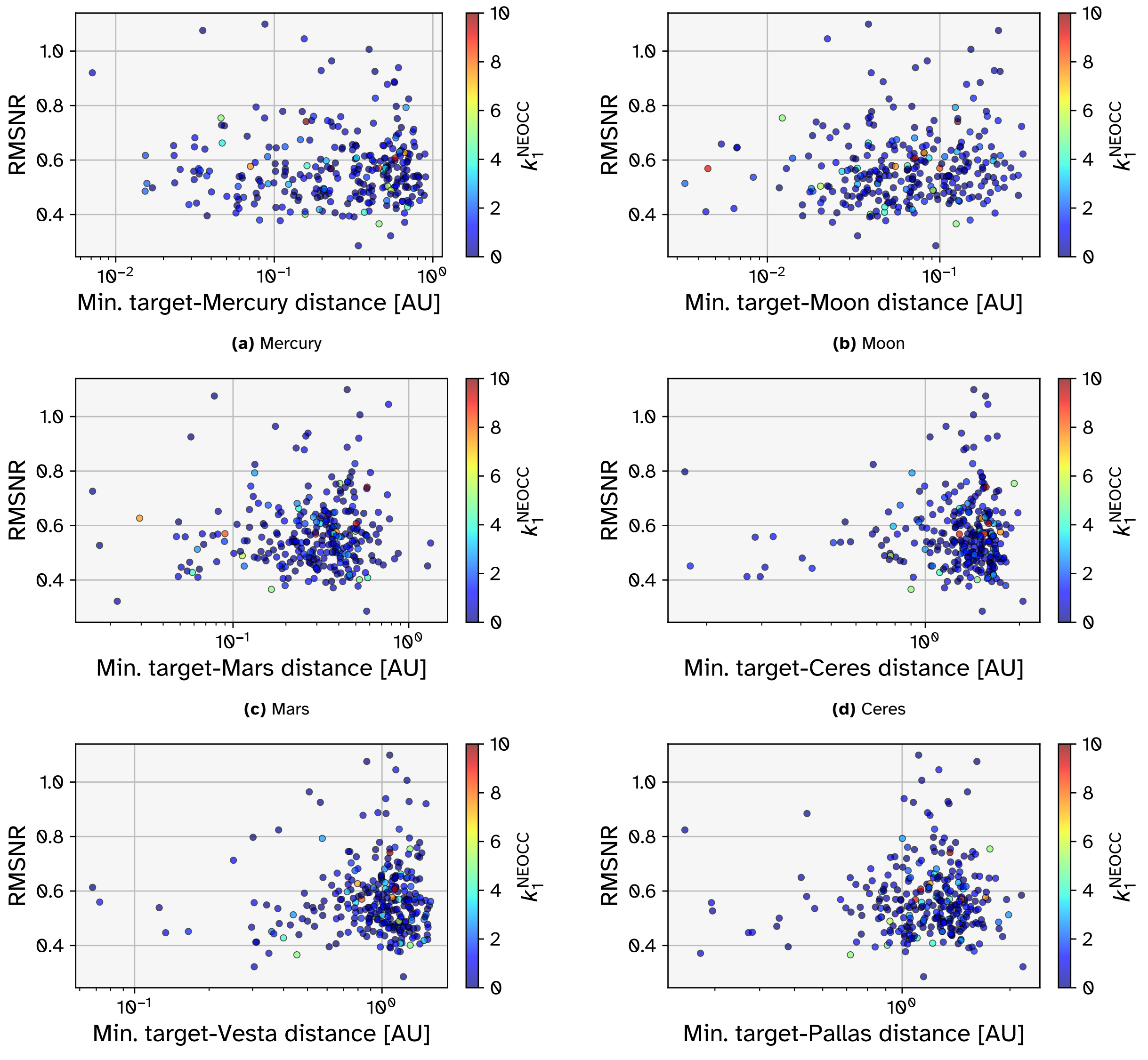
Planetary Close Passes



(a) Mercury

(b) Moon

(c) Mars

(d) Ceres

(e) Vesta

(f) Pallas

Figure A.1: Distributions of RMSNR of the Tudat estimations against the minimum distance between the target body and the inner planets (except Venus and Earth, whose figures were shown in Section 4.3, see Figure 4.41) and the three most massive bodies in the main asteroid belt. The points are coloured according to their value of k_1^{NEOCC}.